Persuasion under Subjective Causal Models

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Abstract

Can a sender exploit the way a receiver interprets information? I study persuasion when the receiver interprets the sender's signal through a subjective causal model, which the sender knows but does not control. The answer depends on whether the receiver regards the signal as a consequence of the state or as one of its causes. In the first case, the sender never gains from a signal that the receiver misinterprets: compared to a standard Bayesian persuasion benchmark, the sender is weakly worse off, and the receiver is never worse off than without information. In the second case, the receiver's posteriors across signal realizations might not average to the prior. The sender can then be strictly better off, and his gain is the receiver's loss: the receiver can be harmed by information that she values.

Keywords: Bayesian persuasion; information design; causal models; directed acyclic graphs; misspecified beliefs.

JEL codes: D82, D83, D91.

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1. Introduction

What we conclude from evidence depends on what we already believe about how the world works. A juror hears testimony and decides whether to convict; a household learns of an interest rate increase and decides how much to spend; a voter learns about a tax reform and decides whether to support it. In each case, the decision depends on a variable that is not observed: whether the defendant is guilty, future inflation, or the effect of the reform on the voter's income. To draw a conclusion about it, the decision maker interprets the available data through a *mental model*: an understanding of how the variables of the environment are related.

In each of these cases, moreover, the information is supplied by a party with an interest in the decision: a prosecutor, a central bank, or a government. If this party knows the mental model, does the mental model constrain the party's ability to persuade the decision maker, or can the party exploit it? The answer depends on whether the decision maker treats the information as a consequence or as a cause of the unobserved variable. In the first case, the party cannot exploit the mental model and can only be constrained by it, while in the second, the party can exploit it at the decision maker's expense.

In this paper, I study how a sender, the interested party, can persuade a receiver, the decision maker, to take the action that the sender prefers when the sender controls the information the receiver observes, which I call the signal, but not the mental model through which the information is interpreted. I focus on a particular class of mental models: *causal models*, which specify, for each variable, its direct causes and effects.[1] Following Spiegler (2016), I represent causal models as directed acyclic graphs (DAGs), in which nodes represent variables and directed edges represent perceived causal relationships: an edge from one variable to another means that the receiver believes that the first is a direct cause of the second. The direct causes of a variable are its *parents*, and its direct effects are its *children*.

The causal model may include variables on which the receiver's payoff does not depend. Consider a juror whose payoff depends on whether the defendant is guilty, but not on whether he had a motive. Suppose a witness testifies that she saw the defendant at the scene. A juror who believes that the testimony is caused by guilt interprets it as evidence of guilt. A juror who instead believes that witnesses are more likely to implicate defendants who they know had a motive interprets the same testimony as evidence that the defendant had a motive, and learns about guilt only to the extent that motive and guilt are related. The same testimony thus leads to different beliefs about guilt, although motive does not enter either juror's payoff.

I formalize this idea in the Bayesian persuasion framework of Kamenica and Gentzkow (2011), hereafter KG11. The state of the world is a vector of state attributes. The receiver's payoff depends on the action taken and on a subset of these attributes; the sender's payoff depends on the action only. The sender commits to a Blackwell experiment, a distribution of a signal conditional on the state, and the receiver observes the signal. Sender and receiver share a prior over the state attributes, so the experiment induces an objective joint distribution over the signal and the state attributes. The receiver, however, interprets the signal through a causal model, a DAG over the signal and the state attributes, which is

[1] People rely on such representations to reason under uncertainty (Sloman, 2005); the experimental evidence is discussed in the literature review.

fixed and known to the sender.

The receiver takes her causal model as given. Following Spiegler (2016), I assume that she forms her beliefs by learning from the data how strong each causal relation is, but not which relations exist. For the juror who believes that testimony is caused by motive, for instance, testimony says nothing more about guilt once the motive is known, whatever the data show. I say that an experiment is *consistent* with the causal model if it creates no relation that the model rules out. At a consistent experiment, the receiver's subjective beliefs coincide with the objective distribution. At any other experiment they do not, and her beliefs about the payoff-relevant state attributes can be wrong too. In that case, I say that the experiment *confuses* the receiver.

Given these beliefs, the receiver applies Bayes rule. She is therefore Bayesian, but with respect to the subjective belief she forms, not to the objective distribution. This introduces a new channel of persuasion. In the KG11 framework, which is the benchmark throughout the paper, the receiver correctly interprets the distribution induced by the sender's experiment. I call her the KG11 receiver. Here, the sender chooses the experiment, while the receiver interprets the signal through a fixed causal model. The sender may therefore attempt to exploit a discrepancy between the objective distribution generated by the experiment and the subjective one that the receiver forms. The question is whether such a discrepancy adds persuasive power, or instead restricts the set of beliefs that can be induced.

The answer depends on how the receiver interprets the signal. I focus on two cases. In the first, the receiver interprets the signal as evidence about the state: it is the consequence of a causal process originating in the state attributes. In the second, the signal is interpreted as a cause of the state attributes. In the language of DAGs, these correspond, respectively, to the signal being a *terminal* node, with no children, and a *root* node, with no parents. These two interpretations generate different persuasion problems.

When the signal is terminal, the receiver treats it as evidence about the state attributes that are its direct causes in the causal model, and revises her beliefs about the remaining state attributes only through them. I establish a *No Confusion Principle*: the sender never benefits from confusing the receiver. Any distribution of the receiver's actions that an arbitrary experiment induces is also induced by a consistent one, so the sender can restrict attention to consistent experiments without loss of generality, in the spirit of a Revelation Principle. I show that the sender solves the KG11 problem under an additional constraint: the signal may depend on the state only through the state attributes that the receiver believes to be its direct causes. Consequently, the sender's value, his highest expected payoff across experiments, is weakly below the KG11 value, his value against the KG11 receiver: when the signal is interpreted as evidence, the causal model can only constrain persuasion.

When the signal is a root, the conclusion can be reversed. The signal is interpreted as one of the causes of the state attributes to which it is connected. Consider the household, which believes that an interest rate increase lowers future inflation and that the rate is unrelated to the other causes of inflation, such as demand. After an increase, the household expects lower inflation and keeps its prior belief about demand. The sender can exploit this structure by choosing an experiment that creates statistical relations that the receiver's causal model rules out. A central bank that raises the rate when demand is high creates such a relation: inflation after an increase is then higher than the household expects. The

household's mistake after an increase need not be offset by an opposite mistake when the rate does not increase. The subjective posteriors then need not average to the prior across the realizations of the signal: the sender can move not only the posterior after each realization, but also the average belief over the state attributes.

I call a difference between the average of the subjective posteriors and the prior a distortion. It is detectable because the receiver holds both beliefs: she knows the prior, so an experiment under which her subjective belief about the state differs from it can be refuted. I say that an experiment satisfies *No Detectable Distortion* (NDD) when there is no such distortion: the subjective posteriors average to the prior. I first restrict the sender to experiments that satisfy NDD, and then drop the restriction. Under NDD, the No Confusion Principle continues to apply and the sender is weakly worse off than in KG11. Without NDD, however, the sender's value can strictly exceed the KG11 value.

Without NDD, the sender's problem is no longer a KG11 problem, but it retains a tractable structure. In a setting with two actions in which the signal has a single child, I show that the sender's problem decomposes into a collection of KG11 problems, one for each value of the child's other causes; in the household example, one for each level of demand. Because the receiver's beliefs about these other causes remain at the prior after every realization of the signal, the sender can design the signal separately for each of their values, and each of the resulting problems is solved by the concavification argument of KG11. The decomposition also identifies the source of the sender's gain. Within each problem, the requirement for persuading the receiver is weaker than in KG11: she evaluates a realization of the signal as if the other causes followed their prior distribution, even when the realization is generated almost exclusively by one of their values. Consequently, the probability with which the sender induces his preferred action can strictly exceed the maximum attainable against the KG11 receiver, and it can equal one even though the receiver does not take that action at the prior. Thus, a causal model in which the signal is evidence can only constrain persuasion, while one in which the signal is a cause can expand it.

I next characterize the causal models that are irrelevant for persuasion, in the sense that the sender faces the same problem as in KG11 and obtains the KG11 value. Irrelevance obtains when the receiver's causal model rules out the same relations between the variables as one in which the signal is caused by exactly the payoff-relevant state attributes; when the signal is terminal or has a single child, this condition is also necessary.

Finally, I study the receiver's welfare. Whenever the No Confusion Principle applies, the receiver's beliefs about the payoff-relevant state attributes are correct at the sender's optimum, so she is never worse off than without information. The causal model then acts as a tax on persuasion: it can reduce the sender's payoff, but it does not expose the receiver to harm. Without NDD, by contrast, the sender's gain over the KG11 benchmark is a transfer from the receiver. In a setting with two actions and a binary payoff-relevant state attribute, I show that each unit of probability of the sender's preferred action beyond the maximum attainable against the KG11 receiver imposes a minimum loss on the receiver, who is then strictly worse off than without information. Moreover, the receiver need not perceive this loss: under a condition on her payoffs, the signal appears weakly beneficial when it is evaluated at her subjective beliefs.

This paper is organized as follows. Section 2 illustrates, in an example of credit ratings, why the re-

ceiver's causal model matters for persuasion. Section 3 introduces the formal framework, and Section 4.1 introduces consistency and No Detectable Distortion. Section 5 studies persuasion under NDD: it establishes the No Confusion Principle and reduces the sender's problem to a constrained Bayesian persuasion problem. Section 6 drops NDD and shows that the sender can then exceed the KG11 benchmark. Section 7 characterizes the causal models under which the sender faces the same problem as in KG11. Section 8 returns to the example, and Section 9 studies the receiver's welfare. Section 10 concludes. Proofs are in the appendices.

Literature. — This paper connects the Bayesian persuasion literature with the literature on causal reasoning with directed acyclic graphs, to study how subjective causal models shape persuasion and optimal information design.

My work contributes to the literature on Bayesian persuasion initiated by Kamenica and Gentzkow (2011) (see Kamenica, 2019, for a review). In the standard framework, the receiver updates against the objective distribution induced by the sender's experiment. I instead consider a receiver who interprets information through a subjective causal model. This departure introduces payoff-irrelevant state attributes into the persuasion problem: although they do not directly enter the receiver's payoff, they can affect how information is interpreted and thus the beliefs on which her choice is based. As a result, causal misspecification can either restrict or expand the sender's persuasive power. Which effect arises depends on the causal role the receiver assigns to the signal and, in the root case, on whether the resulting distortion is detectable from her prior beliefs.

A growing literature studies persuasion when the receiver departs from Bayesian updating. De Clippel and Zhang (2022) characterize optimal persuasion of a receiver who updates according to a non-Bayesian rule. Alonso and Câmara (2016a) study persuasion when sender and receiver hold different priors, allowing the sender to exploit a gap between their beliefs. Levy, Moreno de Barreda, and Razin (2022) consider a receiver who neglects correlation among the signals she observes, and show that the sender can then exceed the KG11 benchmark. Galperti (2019), instead, studies a receiver whose prior rules out some states and a sender who can change her worldview through evidence that her prior deems impossible. My framework is related to these papers in that the receiver's posterior need not coincide with the posterior implied by the sender's experiment and the common prior. The source of the wedge is different, however: in my model, the receiver applies Bayes rule to a distribution constrained by a fixed causal model. The resulting distortion concerns the causal dependence between the signal and the state attributes, rather than the receiver's updating rule, her prior, or the support of the state space.

The causal side of the paper builds on the literature that uses directed acyclic graphs to represent subjective causal reasoning. DAGs are widely used to represent causal structures in statistics and machine learning (Pearl, 2009); Spiegler (2016) introduced them into economic models of subjective causal reasoning and studied the behavioral consequences of common causal misperceptions. Subsequent work studies how misspecified causal models persist or evolve even when agents observe objective data (Spiegler, 2020a; Spiegler, 2020b; Spiegler, 2026). Spiegler (2022), in particular, studies the behavioral consequences of reverse-causality errors. Related work uses causal models to formalize narratives and studies how narratives are selected and evolve, including in political environments (Eliaz and Spiegler, 2020; Eliaz and Spiegler, 2026). Recent experimental work documents that people can hold

different causal interpretations of the same environment and use them to interpret information differently (Charles and Kendall, 2025; Ambuehl and Thysen, 2026; Andre, Haaland, et al., 2026); see Ambuehl, Bhui, and Thysen (2026) for a review. Closest to the role of consistency in my framework, Spiegler (2020b) asks when an agent with a misspecified causal model can be systematically fooled, and characterizes causal models that leave the marginal distribution of a variable undistorted. My notion of No Detectable Distortion imposes an analogous requirement on the marginal distribution over the state attributes, experiment by experiment.

The papers closest to my framework study how persuasion interacts with the receiver's interpretation of information. Aina (2026) studies persuasion through the provision of models for interpreting observable information, while Schwartzstein and Sunderam (2021) and Eliaz, Spiegler, and Thysen (2021) study settings in which the sender supplies the receiver with a model or an interpretation of his messages. Barron and Fries (2025) study how a sender can shape how a receiver interprets objective information by proposing a causal explanation. More recently, Ba et al. (2026) study persuasion when the sender can control both the information the receiver observes and the model through which it is interpreted, while Burkovskaya and Starkov (2026) study a sender who selectively discloses variables and proposes a causal model to persuade the receiver about a causal relationship. In these settings, the sender has some control over the interpretation of information. In my framework, by contrast, the receiver's causal model is exogenous, fixed, and common knowledge. The sender cannot provide, select, or modify the model; he controls only the information. The question is thus whether he can exploit the interaction between an information structure he chooses and a causal model he does not.

My model also differs from much of the literature on subjective causal reasoning in its treatment of the causal model itself. Because the receiver's DAG is taken as given, there is no need for a DAG selection rule, as in Eliaz and Spiegler (2020), and the sender cannot manipulate the receiver by changing her causal model. Moreover, I abstract from personal equilibrium, in which an agent's actions affect the data that later inform her causal model. I instead treat the causal model as fixed and study how it interacts with information chosen strategically by the sender. This distinction is central to the results: causal misspecification can either constrain or expand the sender's persuasive power depending on the causal position of the signal.

2. A Credit Rating Agency Example

Before turning to the formal setup, I illustrate the paper's central mechanism with an example: what a sender can achieve depends not only on *what* he discloses, but also on *how* the receiver interprets it.

Consider a credit rating agency that wants to persuade an investor to lend to a firm. The firm has two attributes: solvency $\omega_1 \in \{s, d\}$, where s denotes solvent and d denotes a defaulter, and current liquidity $\omega_2 \in \{\ell, i\}$, where ℓ denotes liquid and i illiquid. I write $\boldsymbol{\omega} = (\omega_1, \omega_2)$ for the firm's attributes. Suppose that the investor cares only about repayment, so only solvency is payoff-relevant. In particular, the investor lends if and only if her belief that the firm is solvent is at least $\theta = 0.8$. The agency earns a fee of 1 if the investor lends, regardless of the firm's attributes.[2] The agency and

[2] Rating agencies are compensated by the firms they rate and earn their fee when a deal closes.

the investor share the common prior μ_0 of Table 1a, under which liquid firms are more likely to be solvent. Since $\mu_0(\omega_1 = s) = 0.4 < \theta$, the investor does not lend based on the prior alone.

Before observing the firm's attributes, the agency commits to a rating methodology: a (stochastic) rule that specifies how ratings are assigned to firms with different attributes. Formally, the methodology is a Blackwell experiment $\pi \in \mathcal{E}$, where $\pi(x|\boldsymbol{\omega})$ is the probability that a firm with attributes $\boldsymbol{\omega}$ receives rating $x \in X$. The agency chooses the methodology to maximize the ex ante probability $\mathbb{P}$ that the investor lends after observing the rating. Since the agency earns a fee whenever a loan is made, this probability equals its expected payoff. The agency then applies the committed rule, and the investor observes the rating before deciding whether to lend.

I first consider the benchmark of Kamenica and Gentzkow (2011), in which the investor updates her beliefs using the common prior and the rating methodology. I then turn to an investor who interprets the rating through the causal model

$$\text{Rating} \;\rightarrow\; \text{Solvency} \;\leftarrow\; \text{Liquidity}, \tag{R_c}$$

under which the investor treats the rating and liquidity as independent causes of solvency.

$\omega_1 \setminus \omega_2$	ℓ	i
s	0.3	0.1
d	0.1	0.5

(a) Common prior.

$\omega_1 \setminus \omega_2$	ℓ	i
s	1.0	1.0
d	0.75	0.05

(b) An optimal Blackwell experiment $\pi(H|\boldsymbol{\omega})$.

Table 1: Prior and a KG11 optimal Blackwell experiment.

2.1. The KG11 Benchmark

Suppose the investor updates her beliefs using the common prior and the rating methodology. Kamenica and Gentzkow (2011) show that the agency optimally commits to a methodology with two rating realizations, $X = \{H, L\}$, under which the investor lends after observing H but not after L. Since the agency's payoff is the frequency of H, the rating methodology is chosen to issue H as often as possible while keeping the investor's posterior probability of solvency after H at θ. This is achieved by assigning H to every solvent firm and to $1/6$ of defaulters. The resulting value is

$$\mathbb{P}^{\text{BP}} = \mu_0(\omega_1 = s)/\theta = 0.5.$$

One optimal experiment is reported in Table 1b. Any methodology that assigns H to every solvent firm and to one sixth of defaulters is optimal, regardless of the liquidity of the defaulters receiving H, since the investor's posterior after H depends only on the total mass of defaulters receiving that rating.

2.2. The Investor Holds a Causal Model

Now suppose that the investor holds the causal model R_c. One interpretation is that the investor believes ratings to be self-fulfilling: a high rating may give the firm access to cheaper funding, which makes repayment more likely, while a low rating raises its cost of capital and can trigger covenants

that push it toward default. The causal model describes how the investor thinks ratings, solvency and liquidity are related. It need not be the true causal structure.

The causal model pins down one feature of the investor's belief-updating process. Under R_c, the rating and liquidity are two separate causes of solvency, and nothing links them. Thus, the investor treats the rating and liquidity as marginally independent, whatever the rating methodology. Apart from this, I assume that for a given rating and a given liquidity, the belief that the firm is solvent equals the objective frequency that the methodology generates.[3] So, the investor uses the rating to assess solvency within each liquidity group, but she never infers the firm's liquidity from the rating.

Formally, let μ denote the joint distribution of attributes and ratings generated by the prior and the rating methodology. After rating x, the investor believes that the firm is solvent with probability

$$\mu^{R_c}(\omega_1 = s|x) = \sum_{\omega_2 \in \Omega_2} \mu(\omega_1 = s|\omega_2, x)\, \mu_0(\omega_2). \tag{1}$$

This is an average of the solvency rates within the two liquidity groups, with each group weighted at its prior share $\mu_0(\omega_2)$. The benchmark investor computes the same average, but weights each liquidity group at its share $\mu(\omega_2|x)$ among the firms rated x.

The firms that receive a given rating thus have two solvency rates: the *objective* rate $\mu(\omega_1 = s|x)$, the actual frequency of solvent firms among them, and the *subjective* rate $\mu^{R_c}(\omega_1 = s|x)$ that the investor perceives. I call the difference between the two the wedge at rating x:

$$\begin{aligned}\Delta_x &\equiv \mu^{R_c}(\omega_1 = s|x) - \mu(\omega_1 = s|x) \\ &= \big[\mu(\omega_1 = s|\omega_2 = \ell, x) - \mu(\omega_1 = s|\omega_2 = i, x)\big]\, \big[\mu_0(\omega_2 = \ell) - \mu(\omega_2 = \ell|x)\big],\end{aligned}$$

where the first factor is the difference in solvency between the two liquidity groups among firms rated x, and the second factor is the difference between the liquidity weights.

The agency can exploit the wedge in the following way. The investor lends after H when the subjective solvency rate of the firms rated H is at least θ, so their objective rate may fall short of θ by Δ_H. A methodology that creates a positive wedge at H can therefore rate more defaulters H than the benchmark allows, and the investor lends more often.

A methodology determines not only the investor's posterior after each rating, but also the distribution of firm attributes that she perceives: the average of her subjective posteriors across ratings,

$$\mu^{R_c}(\omega_1, \omega_2) = \sum_{x \in X} \mu(x)\, \mu^{R_c}(\omega_1, \omega_2|x) = \sum_{x \in X} \mu(x)\, \mu_0(\omega_2)\, \mu(\omega_1|\omega_2, x), \tag{2}$$

which I refer to as the *subjective prior*. Unlike the objective marginal $\mu(\omega_1, \omega_2)$, the subjective prior need not equal the common prior. I call the difference between the two the *distortion* of the prior.

Under the optimal benchmark methodology of Table 1b, the subjective posterior of solvency after H still clears θ, but the subjective prior differs from the common prior. The reason is that a high rating means the same thing in both liquidity groups: the firms rated H are equally solvent in the two groups,

[3] This belief is the one that best fits the objective frequencies among those consistent with R_c: an investor who estimated her causal model from long-run data on ratings and firms would obtain exactly μ^{R_c} (see Spiegler, 2016).

and every firm rated L is a defaulter. What differs across liquidity groups is how often the high rating is issued: almost all liquid firms receive it, but only one illiquid firm in five. The investor holding R_c ignores this difference, and assumes the high rating to be as frequent in one liquidity group as in the other. The benchmark methodology thus distorts the prior: according to the subjective prior, liquid firms are less solvent and illiquid firms more solvent than under the common prior.

The wedge and the distortion are two faces of the same mistake. The investor assumes that the rating is unrelated to liquidity. Applied to a single rating, the assumption produces the wedge: each liquidity group is weighted at its prior share, rather than at its share among the firms with that rating. Applied across ratings, it produces the distortion: each rating is taken to be as frequent among liquid firms as among illiquid ones. Both vanish if the methodology issues each rating equally often in the two liquidity groups. The benchmark methodology does not, hence the distortion. Yet it creates no wedge, because the firms rated H are equally solvent in both liquidity groups. To create a wedge, the agency must also make liquid and illiquid firms differ in solvency among the firms rated H.

The distortion matters because the investor can detect it. The common prior and the subjective prior are both beliefs that she holds, so she could reject a methodology that, under her causal model, implies a distribution of firm attributes different from the prior. I consider two cases. In the first, the agency is restricted to methodologies that do not distort the prior, $\mu^{R_c}(\omega_1, \omega_2) = \mu_0(\omega_1, \omega_2)$. I call this restriction the *audit* and treat it as a constraint on the agency's feasible set. In the second, the agency is not subject to it. The question is whether the agency can create a wedge without distorting the prior.

2.2.1. The Audit

Suppose first that the audit is imposed. With two ratings, a methodology leaves the prior undistorted only if it issues each rating equally often in both liquidity groups, $\mu(x|\omega_2) = \mu(x)$. The investor's independence assumption then objectively holds, and her posterior after each rating is correct. Hence, there is no wedge to exploit. The firms rated H must therefore be objectively at least θ solvent. The benchmark methodology, however, is not available, since it distorts the prior: the agency must rate liquid and illiquid firms H equally often, although liquid firms are more often solvent.

The optimal methodology under the audit is in Table 2a. In this example, the agency's value is

$$\mathbb{P}^*_{R_c} = 0.25 < \mathbb{P}^{\text{BP}}.$$

In this case, the audit constraint binds: not all solvent firms can be rated H. If every solvent firm is rated H, liquid firms are rated H more often than illiquid ones, whichever way the defaulters rated H are split between the two liquidity groups. Equalizing the two frequencies by rating more illiquid defaulters high would push the posterior of solvency given the high rating below θ. The agency must withhold the high rating from some liquid solvent firms, reducing the frequency of the high rating.

2.2.2. Without the Audit

Suppose that the agency keeps two rating realization, $X = \{H, L\}$, and it is no longer subject to the audit. Nothing then prevents the agency from creating a positive wedge Δ_H: the agency can issue H

$\omega_1 \setminus \omega_2$	ℓ	i
s	$1/3$	1.0
d	0.0	0.1

(a) Audit.

$\omega_1 \setminus \omega_2$	ℓ	i
s	1.0	1.0
d	1.0	0.04

(b) Without audit.

Table 2: Optimal rating methodologies under R_c: with the audit, and without it, among those with two ratings that rate every solvent firm high.

more often to one liquidity group than to the other, and it can let liquid and illiquid firms differ in solvency among the firms rated H.

Consider a methodology that rates every solvent firm H. As in the benchmark, the frequency of H is then $\mu_0(\omega_1 = s)$ divided by the objective solvency of the firms rated H, which the wedge allows to fall to $\theta - \Delta_H$: the agency earns $\mu_0(\omega_1 = s)/(\theta - \Delta_H)$. Thus, the agency wants to maximize the wedge at H, subject to $\mu^{R_c}(\omega_1 = s | x = H) = \theta$.

The agency then has to choose which defaulters receive H. Since the investor evaluates the firms rated H liquidity group by liquidity group, a defaulter rated H lowers the subjective posterior only through the solvency rate of its own liquidity group. Liquid defaulters are cheap: solvent liquid firms are numerous, so even rating every liquid defaulter H leaves the liquid firms rated H three quarters solvent. Illiquid defaulters are expensive: solvent illiquid firms are few, so a few illiquid defaulters dilute the illiquid firms rated H quickly. The agency thus rates every liquid defaulter H, and then illiquid defaulters until the investor's posterior of solvency given the H rating reaches θ (Table 2b). Among the firms rated H, illiquid firms are now the more solvent liquidity group, yet their share among them is smaller than in the prior. The investor, who keeps the prior shares, overweights them.

The firms rated H are then approximately 77% solvent, while the investor's subjective posterior is exactly $\theta = 0.8$. The agency persuades the investor to lend to firms whose objective solvency is below her threshold: $\mathbb{P}_{R_c} = 0.52 > \mathbb{P}^{\text{BP}}$. With two ratings, a wedge always comes with a distortion, so the agency gains only because the audit is not imposed.

2.3. Evidence versus Cause

The investor could instead interpret the rating as evidence of the firm's attributes rather than as a cause of solvency. Consider the two causal models

$$R_a: \text{Liquidity} \rightarrow \text{Solvency} \rightarrow \text{Rating}, \quad R_b: \text{Solvency} \leftarrow \text{Liquidity} \rightarrow \text{Rating}.$$

In R_a, the rating is evidence of solvency, while in R_b it is evidence of liquidity, the cause of solvency.

The wedge can open under an evidence interpretation too: under R_b, the investor reads the rating as evidence of liquidity only and ignores what it says about solvency within each liquidity group, so her posterior can differ from the objective one. What the agency cannot do under R_a or R_b is to distort the prior. When the rating is evidence, the subjective prior equals the common prior whatever the methodology, so the audit is passed automatically and the causal model can only constrain the agency. This is the general pattern. A rating that the investor interprets as evidence can never beat the

benchmark (No Confusion Principle); gains like the one here require the rating to be interpreted as a cause as in R_c (Section 6). I return to this example in Section 8.

3. Setup

Consider two agents: a receiver (she) and a sender (he). The receiver must make a decision under uncertainty. Let $\boldsymbol{\omega} = (\omega_i)_{i=1}^n$ with $n > 1$ be a multivariate state of the world. Each state attribute ω_i belongs to a finite set Ω_i, and the state space is denoted by $\Omega = \times_{i=1}^n \Omega_i$. The receiver and the sender share a common prior $\mu_0 \in \text{int}\,(\Delta(\Omega))$.

The receiver selects an action $a \in A$, where A is finite and $|A| \geq 2$, and obtains a payoff $u(a, \boldsymbol{\omega}_P)$, where $\boldsymbol{\omega}_P = (\omega_i)_{i \in P}$ for a subset $P \subseteq \{1, \ldots, n\}$ of the state attributes, so that not all state attributes are necessarily payoff-relevant. Let $\Omega_P = \times_{i \in P} \Omega_i$. The sender's payoff, $v(a)$, depends only on the receiver's action.

To persuade the receiver to take his desired action, the sender can choose a Blackwell experiment π. A Blackwell experiment π consists of **(i)** a finite realization space X, and **(ii)** a family of conditional probability distributions $(\pi(\cdot|\boldsymbol{\omega}))_{\boldsymbol{\omega} \in \Omega}$ over X. Since the prior μ_0 and the Blackwell experiment π determine an *objective* joint probability distribution over the signal and state spaces, the sender's choice of π is equivalent to the choice of a joint probability distribution $\mu \in \Delta(\Omega \times X)$ where $\mu(\boldsymbol{\omega}, x) = \pi(x|\boldsymbol{\omega})\mu_0(\boldsymbol{\omega})$. Let $\mathcal{E}$ denote the set of all Blackwell experiments with a finite realization space. A realization that occurs with probability zero is never observed, and removing it from X changes neither the probabilities of the other realizations nor the beliefs that the receiver forms at them. Without loss of generality, I therefore describe each experiment by the realizations it generates with positive probability, $\mu(x) > 0$ for all $x \in X$.

The sender commits to an experiment π and communicates it to the receiver. The experiment then generates $x \in X$, observed by the receiver. Finally, the receiver updates her prior about the payoff-relevant state attributes $\boldsymbol{\omega}_P \in \Omega_P$.

The key novelty is how the receiver forms her posterior belief. In Bayesian persuasion à la KG11, the receiver updates her prior on $\boldsymbol{\omega}_P$ by applying Bayes rule to the objective distribution. In my model, the receiver interprets the relation between the signal and the state attributes according to a *subjective* causal model, known to the sender. The receiver does not choose her causal model: it is a primitive of the receiver (see Spiegler, 2016). I formalize causal models through *directed acyclic graphs*.

3.1. Causal Models: Directed Acyclic Graphs

Let $N = \{0, 1, \ldots, n\}$ with $n > 1$ denote the set of *nodes*, where node 0 corresponds to the signal x, so that $\Omega_0 = X$, and each node i for $i \geq 1$ corresponds to the state attribute ω_i. Finally, denote by $N_\omega = N \setminus \{0\}$ the set of nodes corresponding to the state attributes.

Definition 1 : Directed Acyclic Graph (DAG). — *A directed acyclic graph is a pair $R = (N, E)$, where N is a set of nodes and $E \subseteq N \times N$ is a set of directed edges. A directed edge from node i to node j, denoted $i \to_R j$, indicates that i is a direct cause of j, or equivalently, that i influences j.* Acyclicity

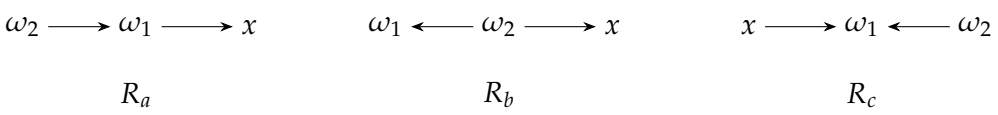


Figure 1: Three causal models. In R_a, ω_2 is the parent of ω_1 and ω_1 is the parent of x; ω_2 is a root and x is terminal. In R_b, ω_2 is a root with two children, ω_1 and x, which are both terminal. In R_c, x and ω_2 are roots and the two parents of ω_1, which is terminal; ω_2 is a co-parent of x.

means that there exists no directed path[4] *that starts and ends at the same node.*

Given two nodes $i, j \in N$, if $i \to_R j$, then i is called a *parent* of j, and j is a *child* of i. Let $\mathrm{pa}(i) = \{j \in N | j \to_R i\}$ be the set of parents of node i in R, and $\mathrm{pa}_\omega(i) = \mathrm{pa}(i) \setminus \{0\}$ the set of parents of i excluding the signal $x = \omega_0$. A node i is a *root* if $\mathrm{pa}(i) = \emptyset$. The set of children of i is $\mathrm{ch}(i) = \{j \in N | i \to_R j\}$, and a node i is *terminal* if $\mathrm{ch}(i) = \emptyset$. The *co-parents* of node i are the other parents of its children. Two nodes are linked if one of them is the parent of the other. Figure 1 illustrates these notions.

In what follows, I restrict attention to the collection of DAGs $\mathcal{R}$ such that each causal model $R \in \mathcal{R}$ is defined over the signal x and all state attributes $(\omega_i)_{i=1}^n$. This restriction reflects the assumption that the receiver is aware of the full set of state attributes that are relevant to her decision-making problem, but she may be misspecified about the causal relationships among them. Omitting a variable from the causal model would amount to assuming it is irrelevant by construction, rather than misinterpreting its role, and would therefore confound causal misspecification with informational unawareness. By working with DAGs defined over the full set of variables, I isolate distortions arising from the causal model while maintaining a common state space for objective and subjective beliefs.

Within $\mathcal{R}$, I further restrict my attention to causal models in which x is either a root or a terminal node, and is connected to at least one state attribute. If x is not connected to any state attribute, the receiver treats it as causally irrelevant, her belief over the state attributes stays at the prior, and the sender's experiment cannot change the receiver's beliefs.

Interpretation. — The sender and the receiver share a common prior μ_0 over the state attributes $(\omega_1, \ldots, \omega_n)$. The signal x is not part of the prior: it has no distribution until the sender chooses the experiment, and its realization is observed only after the experiment is carried out. Ex ante, beliefs are therefore defined only over the state attributes. The receiver nevertheless interprets the realization of x through a causal model defined over $(\omega_1, \ldots, \omega_n, x)$. Before the experiment is chosen, the receiver's causal model restricts her beliefs over the state attributes only through the independence relations it implies among them, and Section 4.1 requires the prior to satisfy these relations.

3.2. Subjective Belief-Formation Process

A directed acyclic graph $R = (N, E)$ encodes a set of conditional independence assumptions known as *local Markov independencies*. These assumptions state that each variable is conditionally independent of its non-descendants given its parents. Intuitively, if node j is not a (direct or indirect) descen-

[4] A directed path is a sequence of edges such that the ending node of each edge in the sequence coincides with the starting node of the next edge in the sequence.

dant of i, then the only source of dependency between i and j can be due to a common parent or ancestor. When conditioning on their parent, the two nodes become independent.

Any prior $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and Blackwell experiment $\pi \in \mathcal{E}$ generate an objective joint probability distribution μ over the signal and state spaces such that $\mu(\boldsymbol{\omega}, x) = \pi(x|\boldsymbol{\omega})\mu_0(\boldsymbol{\omega})$. Given μ, I assume that a DAG $R \in \mathcal{R}$ induces the following subjective belief $\mu^R \in \Delta(\Omega \times X)$:

$$\mu^R(\boldsymbol{\omega}, x) = \mu(x|\boldsymbol{\omega}_{\text{pa}(0)}) \prod_{i \in \text{ch}(0)} \mu(\omega_i|x, \boldsymbol{\omega}_{\text{pa}_\omega(i)}) \prod_{j \in N_\omega \setminus \text{ch}(0)} \mu_0(\omega_j|\boldsymbol{\omega}_{\text{pa}_\omega(j)}), \tag{3}$$

where $\mu(\omega_i|x, \boldsymbol{\omega}_{\text{pa}_\omega(i)}) = \mu(\omega_i, \boldsymbol{\omega}_{\text{pa}_\omega(i)}, x)/\mu(\boldsymbol{\omega}_{\text{pa}_\omega(i)}, x)$ for $i \in \text{ch}(0)$, and a conditional on an empty set of parents is the corresponding marginal: $\mu_0(\omega_j|\boldsymbol{\omega}_{\text{pa}_\omega(j)}) = \mu_0(\omega_j)$ if $\text{pa}_\omega(j) = \varnothing$. If $\mu(x, \boldsymbol{\omega}_{\text{pa}_\omega(i)}) = 0$ for some $i \in \text{ch}(0)$, the experiment carries no information about ω_i at that configuration, and the receiver keeps her prior conditional: $\mu(\omega_i|x, \boldsymbol{\omega}_{\text{pa}_\omega(i)}) := \mu_0(\omega_i|\boldsymbol{\omega}_{\text{pa}_\omega(i)})$. Every factor of (3) is then a probability distribution, and so is μ^R.

There are three components in (3). The first is $\mu(x|\boldsymbol{\omega}_{\text{pa}(0)})$, the probability of x conditional on its parents. Since the prior is a distribution over the state attributes only, the receiver computes this probability directly from π, or indirectly from μ. When x is a root, this factor is $\mu(x)$. When x is terminal, the prior over its parents does not appear in the first two factors: it is part of the third, since the parents of x are state attributes that x does not influence. If the investor of Section 2 regarded the rating as evidence of solvency, $\omega_2 \to \omega_1 \to x$, for instance, (3) would read $\mu^R(\boldsymbol{\omega}, x) = \mu(x|\omega_1)\,\mu_0(\omega_1|\omega_2)\,\mu_0(\omega_2)$, and the prior over solvency, the parent of x, enters through the last two factors. The second factor is the product of the conditional probabilities of state attributes influenced by x. According to the receiver's causal model, x provides information about these state attributes: she believes that there is a link between x and these state attributes, but she does not know the magnitude of this link. Thus, she uses μ to *quantify* the relationships encoded in her subjective causal model. Finally, the last factor is the product of the conditional probabilities of the state attributes not influenced by x. According to the receiver's causal model, x is not a direct cause or effect of these state attributes. For this reason, she relies on her prior to compute these conditional probabilities, rather than on the distribution μ.

This is an assumption on how the receiver forms beliefs. Every factor of (3) is the objective conditional of a variable given its parents in R, since for a state attribute that x does not influence, the prior conditional given its parents coincides with the objective one. Thus μ^R is the distribution that a receiver who fitted her causal model to long-run data generated by μ would estimate, and among the distributions that satisfy the independence relations of R, it is the one that best fits μ (see Spiegler, 2016). I say that the receiver is *causally misspecified* at an experiment, or that the experiment *confuses* her, if the objective distribution μ it induces is different from the subjective one μ^R. This happens exactly when μ does not satisfy the independence relations implied by R (Lemma 2(a)). Even then, her posteriors about the payoff-relevant state attributes can be the objective ones, and her misspecification then does not affect her decision.

Finally, the receiver computes the posterior beliefs of a subset of state attributes $M \subseteq N_\omega$ from μ^R as

$$\mu^R(\boldsymbol{\omega}_M|x) = \frac{\mu^R(\boldsymbol{\omega}_M, x)}{\mu^R(x)} = \frac{\sum_{\Omega_{N_\omega \setminus M}} \mu^R(\boldsymbol{\omega}, x)}{\sum_{\Omega_{N_\omega}} \mu^R(\boldsymbol{\omega}, x)}.$$

I denote by $\mu_x \equiv \mu(\cdot|x)$ and $\mu_x^R \equiv \mu^R(\cdot|x)$ the objective and subjective *payoff-relevant posteriors* on Ω_P.

The model departs from KG11 only in how the receiver forms beliefs. Timing and commitment are the same: the sender commits to π, a realization is observed, and nothing can be manipulated ex post. In KG11 the receiver applies Bayes rule to the objective distribution μ, so payoff-irrelevant state attributes are marginalized out and sender and receiver hold the same posteriors. Here she applies Bayes rule to μ^R. Payoff-irrelevant state attributes then matter whenever her causal model links them to the signal or to a payoff-relevant state attribute, and her posteriors can differ from the sender's even under a common prior and a common experiment. Section 7 characterizes the causal models for which they do not, so that KG11 obtains as a special case.

3.3. Sender's and Receiver's Objectives

After observing a realization $x \in X$ and forming a posterior belief, the receiver chooses an action that maximizes her expected utility. For a belief $q \in \Delta(\Omega_P)$, let $\bar{u}(a, q) \equiv \mathbb{E}_q[u(a, \boldsymbol{\omega}_P)]$ denote her expected utility from action a. She then chooses

$$a \in A^*(\mu_x^R) = \arg\max_{a \in A} \bar{u}(a, \mu_x^R).$$

I assume that for each action $a \in A$, there exists a posterior belief under which a is optimal. Following KG11, let $\hat{a}(\mu_x^R)$ denote the receiver's equilibrium action at belief μ_x^R. If the receiver is indifferent among actions at a given belief, she selects the action that maximizes the sender's utility. The sender's payoff at a receiver's belief over payoff-relevant state attributes $\mu_x^R \in \Delta(\Omega_P)$ is

$$\hat{v}(\mu_x^R) \equiv v(\hat{a}(\mu_x^R)).$$

Taking the receiver's behavior as given, the sender chooses a Blackwell experiment π to maximize the expected utility induced by the receiver's (subjective) posteriors. The *value* of an experiment is the equilibrium expectation of $v(a)$ induced by π, that is

$$V(\pi) \equiv \mathbb{E}_{x \sim \mu}\left[\hat{v}(\mu_x^R)\right].$$

The *gain* from persuasion is the difference between this value and the sender's expected payoff at the prior. The sender *benefits from persuasion* if there exists an experiment with strictly positive gain. An experiment is *optimal* if no other experiment yields a higher value.

4. Consistency and Prior Distortions

I conclude the description of the framework with two definitions that will be used throughout the rest of the analysis: *consistency* and *No Detectable Distortions*.

4.1. Consistency

A DAG R encodes a set of marginal and conditional independence assumptions. Consistency concerns whether the independence relations among variables in $M \subseteq N$ implied by R are satisfied by a probability distribution over M.

Definition 2 : Consistency. — *Let $\mu \in \Delta(\Omega_M)$ and $R = (N, E) \in \mathcal{R}$ with $M \subseteq N$. The pair (R, μ) satisfies* consistency *if every marginal or conditional independence relation among variables in M implied by R also holds in μ.*

I provide conditions to verify consistency in Appendix B.

Applied to the prior, so when $M = N_\omega$, consistency requires the independence relations encoded in the receiver's causal model to hold in the common prior μ_0. I assume that this condition is satisfied: the receiver may hold a misspecified causal model, but not one that implies independence relations that contradict her prior beliefs.

Assumption 1 . — *The receiver's DAG R and the prior $\mu_0 \in \text{int}(\Delta(\Omega))$ satisfy consistency. The set of DAGs R such that (R, μ_0) satisfies consistency is $\mathcal{R}_0 \subseteq \mathcal{R}$.*

Imposing Assumption 1 does not pin down the receiver's causal relations over the state attributes. Several DAGs can be consistent with the same prior, for two reasons. First, this assumption is *one-sided*. If two state attributes are correlated under μ_0, they must be connected in R, either directly or indirectly through other variables. If they are independent, they may still be causally linked in the receiver's model, since she can attribute the absence of correlation to a parametric coincidence. The receiver cannot ignore a correlation present in the prior, but she does not interpret the absence of correlation as evidence against a causal link. Second, consistency places no restriction on the direction of a causal link. For instance, with two state attributes, a DAG containing $\omega_1 \to \omega_2$ and a DAG containing $\omega_2 \to \omega_1$ both satisfy consistency. This is consequential. Although both DAGs reproduce the prior, they may yield different subjective beliefs once x is introduced.

The assumption nonetheless restricts the set of admissible DAGs for a given prior. A DAG that encodes no independence relations is consistent with any prior, so the fully connected DAG always belongs to $\mathcal{R}_0$. Conversely, if the prior satisfies no conditional independence relations among state attributes, consistency forces the receiver's DAG to be fully connected over those attributes whenever x is terminal. In that case, admissible models can differ only in how the signal is connected to the state attributes. Lemma B.2 formalizes these claims and clarifies the qualification needed when x is a root.

The assumption allows me to isolate the effect of how the receiver interprets the signal. The causal model agrees with the prior on the independence relations among the state attributes, so any distortion of her beliefs comes from how she relates the signal to the state. One implication of Assumption 1 is that the receiver does not distort the marginal distribution of the signal. Although the receiver's causal model affects how she interprets a signal realization, it does not affect how likely she believes that realization to be before observing it.

Lemma 1 . — *The receiver's causal model never distorts how likely each realization of the signal is. Formally, fix $\mu_0 \in \text{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$. Then, for any Blackwell experiment $\pi \in \mathcal{E}$, the objective and subjective marginal distributions of the signal coincide:*

$$\mu^R(x) = \mu(x) \quad \textit{for all } x \in X.$$

When the signal is a root in the causal model, the receiver computes the probability of each realization directly from the objective distribution. When it is terminal, she computes it from the experiment and

from the prior, which her causal model reproduces by Assumption 1.

Consistency can also be applied to the joint distribution induced by an experiment. I call $\pi \in \mathcal{E}$ consistent if the objective joint distribution μ it induces satisfies consistency with R. For a given prior μ_0 and a DAG $R \in \mathcal{R}_0$, let $\mathcal{E}_R \subseteq \mathcal{E}$ denote the set of Blackwell experiments consistent with R. Since a distribution that satisfies the independence relations of R factorizes along R, the next result about consistent experiments follows.

Lemma 2(a): Consistent Experiments. — *Fix $\mu_0 \in \text{int}(\Delta(\Omega))$ and a DAG $R \in \mathcal{R}_0$. Then, at any consistent experiment, the receiver is always causally well-specified. Conversely, if the receiver is causally well-specified at an experiment, the experiment is consistent. Thus, $\mu^R = \mu$ if and only if $\pi \in \mathcal{E}_R$.*

An implication of Lemma 2(a) is that the subjective and objective payoff-relevant posteriors coincide at each signal realization whenever the experiment is consistent with the receiver's causal model.

4.2. No Detectable Distortion

Assumption 1 ensures that the receiver's causal model never distorts the marginal distribution of the signal. Her marginal beliefs about the state, however, need not coincide with the common prior: $\mu^R(\omega)$ can differ from $\mu_0(\omega)$. I refer to $\mu^R(\omega)$ as the *subjective prior*, the receiver's ex ante belief about the state induced by her causal interpretation of the experiment.

Since the receiver is Bayesian with respect to her subjective joint beliefs, her subjective posteriors over the state average back to the subjective prior:

$$\sum_{x \in X} \mu^R(x)\mu^R(\omega|x) = \mu^R(\omega).$$

This is a martingale property of her own updating. Under the KG11 benchmark, the subjective prior always coincides with the common prior. This need not be the case when the receiver interprets the experiment through a causal model, in particular whenever the receiver is causally misspecified. Posteriors average to the prior under any model of the receiver, so the sender can raise her belief after some realizations only by lowering it after others. What differs is which prior: the common prior in the KG11 benchmark, which the sender cannot move, and the subjective prior here, which he can. An experiment that distorts the subjective prior in his favor moves the posteriors with it on average.

The example in Section 2 illustrates that when the receiver holds R_c, the sender can beat the KG11 benchmark, for instance by committing to the experiment in Table 2b. The underlying mechanism is indeed the subjective prior distortion. Under this experiment, the subjective prior assigns probability 0.34 to an illiquid defaulter, $\omega = (d, i)$, compared with 0.5 under the common prior. It assigns probability 0.26 to a solvent illiquid firm, $\omega = (s, i)$, against the objective 0.1. The experiment thus shifts probability mass from illiquid defaulters to solvent illiquid firms, making the investor more optimistic about solvency. This is what allows the agency to gain over the benchmark.

Therefore, the sender can manipulate the receiver's subjective prior through the choice of the experiment. However, such a distortion is *detectable*: before observing a signal realization, the receiver could

compare her subjective prior with the common prior, and reject the experiment if they differ. The audit in Section 2 captures this possibility.

Definition 3 : No Detectable Distortion. — *An experiment $\pi \in \mathcal{E}$ satisfies* No Detectable Distortion (NDD) *if the subjective prior coincides with the common prior: $\mu^R(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega})$ for all $\boldsymbol{\omega} \in \Omega$. The set of all such experiments is $\mathcal{E}_{\text{NDD}} \subseteq \mathcal{E}$.*

One class of experiments always satisfies NDD: the consistent experiments.

Lemma 2(b): Consistent Experiments. — *Fix a prior $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and a DAG $R \in \mathcal{R}_0$. Any consistent experiment satisfies NDD. Thus, $\mathcal{E}_R \subseteq \mathcal{E}_{\text{NDD}}$.*

At a consistent experiment, the receiver's subjective joint distribution is the objective one. Therefore, the subjective prior coincides with the common prior. Note that NDD can hold even when the receiver is causally misspecified. NDD requires the receiver's subjective posteriors to average back to the common prior, but it does not require her to interpret every signal realization correctly.

Causal misspecification can create a wedge between the objective posteriors an experiment induces and the subjective posteriors the receiver forms. The sender gains from this wedge if it lets him induce favorable posteriors more often than the KG11 receiver would allow. In the next two sections I show that the sender gains only if the experiment also distorts the receiver's subjective prior. The receiver's causal model can make her subjective posteriors differ from the objective ones. This is not the same as distorting the subjective prior. If the subjective prior is distorted, the receiver is causally misspecified (Lemma 2(b)), but she can be misspecified and still hold the correct prior. I study the two separately.

5. When Do Causal Models Hurt the Sender?

In this section, I study what the sender can gain from misspecification when he is restricted to experiments satisfying NDD. Whether this restriction matters depends on the position of the signal node. If the signal is terminal, every experiment satisfies NDD, so the results characterize everything the sender can achieve. If the signal is a root, some experiments violate NDD, and the results instead describe a sender subject to the audit of Section 2. I study the sender who is not subject to it in the next section.

I show that the sender cannot gain from confusing the receiver if he cannot also manipulate her subjective prior. The reason is a *No Confusion Principle*: whatever the sender achieves with an experiment that the receiver misinterprets, he can also achieve, at the same frequencies, with an experiment she interprets correctly. Thus, confusion alone does not help the sender. His problem can instead be expressed as a constrained Bayesian persuasion problem over the distributions of posteriors induced by consistent experiments. Since these distributions form a subset of those available under KG11, the receiver's causal model can only hurt the sender.

5.1. Prior Distortion and the Position of x

No Detectable Distortion is a property of an experiment. The position of the signal x in the receiver's causal model determines which experiments satisfy it: every experiment in $\mathcal{E}$ when x is terminal, and only some when x is a root. In the first case NDD does not restrict the sender; in the second it can.

Lemma 3 . — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$.*

- **(i)** *If the signal is interpreted as evidence, no experiment can distort the subjective prior: $\mathcal{E}_{\text{NDD}} = \mathcal{E}$.*
- **(ii)** *If the signal is interpreted as a cause, the subjective prior can be distorted. Thus, $\mathcal{E}_{\text{NDD}} \subseteq \mathcal{E}$. In particular, NDD can be expressed as a finite system of moment restrictions on the experiment. Formally, an experiment $\pi \in \mathcal{E}$ satisfies NDD if and only if, for every $\boldsymbol{\omega} \in \Omega$,*

$$\sum_{x \in X} \mu(x) \prod_{i \in \text{ch}(0)} \mu\big(\omega_i | x, \boldsymbol{\omega}_{\text{pa}_\omega(i)}\big) = \frac{\mu_0(\boldsymbol{\omega})}{\prod_{j \in N_\omega \setminus \text{ch}(0)} \mu_0\big(\omega_j | \boldsymbol{\omega}_{\text{pa}_\omega(j)}\big)}.$$

I give an intuition for this result below.

The Signal as Evidence: x is Terminal. — When x is a terminal node in the receiver's causal model, she interprets it as pure evidence: it is the *effect*, not the cause, of the state attributes linked to it. Her subjective factorization then uses the prior conditional for every state attribute, and the experiment enters only through the distribution of x given its parents. Summing over x therefore returns the prior, whatever the experiment: every experiment satisfies NDD. Her beliefs after a realization may still differ from the objective posteriors.

In the credit rating example, $R_a : \omega_2 \to \omega_1 \to x$ and $R_b : \omega_1 \leftarrow \omega_2 \to x$ belong to this class.

The Signal as Cause: x is a Root. — If x is a root node, the interpretation is reversed: the receiver interprets the signal as a cause of some state attributes, rather than as evidence about them or as their effect. In the credit rating example, R_c belongs to this class.

Some DAGs in which x is a root are equivalent to DAGs in which x is terminal: they induce the same subjective beliefs for every experiment $\pi \in \mathcal{E}$. Two DAGs are *Markov equivalent* if they imply the same independence relations, in which case the receiver forms the same beliefs under both (Lemma B.5). For example, $x \to \omega_1 \to \omega_2$ and $\omega_2 \to \omega_1 \to x$ are Markov equivalent. Since x is terminal in the latter DAG, no experiment can distort the subjective prior under either DAG. When I refer to *causal models in which x is a root*, I mean those that are not Markov equivalent to any DAG in which x is terminal. These fall into two kinds: either x is one of two independent causes of a state attribute, as in R_c, or x is a *common cause* of two state attributes that are not linked, as in $\omega_1 \leftarrow x \to \omega_2$. I characterize these cases in Lemma B.6.

In this class of DAGs, the experiment can shift the receiver's subjective prior away from the common prior. In her factorization, the conditional distribution of each child of x, given x and its co-parents, is part of the mechanism that generates the state, and the experiment determines this distribution.

In this section, I restrict the sender to experiments that satisfy NDD. The restriction is in the spirit of Eliaz and Spiegler (2020), who rule out causal models that distort the consequences of the status quo: the distribution of the state is known, so an experiment under which the receiver's causal model implies a different distribution can be refuted. This is the restriction imposed by the audit in Section 2.

Lemma 3**(ii)** shows that NDD can be expressed as a finite system of moment restrictions on the experiment. The number of restrictions does not depend on the number of signal realizations.

In this and the next section, I illustrate the results for causal models in which x has a single child, $\mathrm{ch}(0) = \{i^*\}$, as in R_c. I do not study the common-cause case further, assuming that no variable can account for all the dependence between two state attributes. The NDD condition and the No Confusion Principle hold more generally for all causal models in which x is a root.

When x has a single child, an experiment satisfies NDD whenever the signal is independent of the child's other causes, and the converse holds when different weights on the realizations always produce a different distribution of the child (Corollary A.1). In general, NDD is weaker than this independence: under R_c, a rating that depends on liquidity alone satisfies NDD although it is correlated with liquidity. An experiment can therefore satisfy NDD without being consistent.

5.2. The No Confusion Principle

An experiment $\tilde{\pi} \in \mathcal{E}$ has a *consistent implementation* if there exists a consistent experiment $\pi \in \mathcal{E}_R$ that induces the same subjective payoff-relevant posteriors with the same probabilities: $\mu_x^R = \tilde{\mu}_x^R$ and $\mu(x) = \tilde{\mu}(x)$ for all $x \in X$.

In general, $\tilde{\pi}$ and its consistent implementation π, if it exists, generate different objective joint distributions. They nevertheless induce the same behavior. Under both, the receiver holds the same posteriors with the same probabilities, so she takes the same action after every signal and the sender obtains the same payoff. The difference is that under π she is causally well-specified.

The sender could therefore commit to an experiment that satisfies NDD and still confuses the receiver. The following result shows that he gains nothing from doing so.

Proposition 1 : No Confusion Principle. — *Under* No Detectable Distortion*, the sender gains nothing from confusing the receiver: whatever he achieves with an experiment that confuses her, he can also achieve with one that she interprets correctly. Formally, fix* $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$ *and* $R \in \mathcal{R}_0$*. Every* $\tilde{\pi} \in \mathcal{E}_{\mathrm{NDD}}$ *has a consistent implementation* $\pi \in \mathcal{E}_R$*.*

The intuition is that, under NDD, the receiver's subjective prior under $\tilde{\pi}$ coincides with the common prior. It can therefore be represented as the objective joint distribution of an experiment: $\pi(x|\boldsymbol{\omega}) = \tilde{\mu}^R(x|\boldsymbol{\omega})$. This objective joint distribution factorizes according to R, so π is consistent. By Lemma 2(a), the receiver's subjective joint under π is $\tilde{\mu}^R$ itself: each signal realization occurs with the same probability and induces the same posterior as under $\tilde{\pi}$. Without NDD, this construction need not be feasible, since the receiver's subjective joint may have a different marginal over the state from the common prior.

The No Confusion Principle can be read as a *Revelation Principle* for causal persuasion: under NDD, the sender can restrict attention to the consistent experiments without loss. Since consistent experiments induce a subset of the posterior distributions available under KG11, the receiver's causal model can only weakly hurt the sender. Any gain over the KG11 benchmark must therefore come from distorting the receiver's subjective prior, as in Section 6.

5.3. A Constrained Bayesian Persuasion Problem

By the No Confusion Principle, the sender's problem under NDD is a maximization over consistent experiments. In this subsection, I rewrite it as a Bayesian persuasion problem. As in KG11, the sender

chooses a distribution of posteriors that averages back to the prior, but only among those that a consistent experiment can induce. First, I describe this set and show that it is a subset of the one available in KG11. Second, I show that the sender's value is therefore weakly below the KG11 benchmark, and that he can restrict attention to action recommendations if the signal is terminal or has a single child.

5.3.1. Feasible Posteriors

To characterize the set of attainable posteriors, I exploit the notion of *Bayes consistency*, introduced by Aina (2026). Fix a finite realization space X. A *weighted vector of posteriors* is a pair $\tau = (\alpha_x, \mu_x)_{x \in X}$ where $\alpha \in \text{int}\,(\Delta(X))$ and $\mu_x \in \Delta(\Omega_P)$ for each x. It is *Bayes-consistent* if the prior is the corresponding convex combination of the posteriors, $\sum_{x \in X} \alpha_x \mu_x(\omega_P) = \mu_0(\omega_P)$ for all $\omega_P \in \Omega_P$. The set of Bayes-consistent weighted vectors of posteriors, across all finite realization spaces, is $\mathcal{B}$.

Shmaya and Yariv (2016) and Aina (2026) show that Bayes consistency is the only restriction on the distribution of posteriors under standard Bayesian persuasion. Every Blackwell experiment induces a Bayes-consistent weighted vector, with $\alpha_x = \mu(x)$ and $\mu_x = \mu(\cdot \mid x)$, and conversely, every Bayes-consistent weighted vector can be implemented by a suitable Blackwell experiment. Thus, $\mathcal{B}$ describes the set of posterior distributions available to the sender under KG11.

With a subjective causal model, the set of attainable posterior distributions may be restricted compared to the one under KG11, because the receiver's causal model can constrain how she updates. This restriction is not necessarily tied to NDD: when x is terminal, every experiment satisfies NDD, yet the causal model can still restrict the posterior distributions that the sender can induce. The No Confusion Principle allows us to characterize these distributions by considering consistent experiments only. Every experiment satisfying NDD has a consistent implementation with the same payoff-relevant posteriors and probabilities, while every consistent experiment satisfies NDD by Lemma 2(b). This motivates the following definition.

Definition 4 : R-Bayes Consistency. — *Let $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and $R \in \mathcal{R}_0$. A weighted vector of posteriors $\tau = (\alpha_x, \mu_x)_{x \in X}$ is R-Bayes-consistent if there exists a consistent experiment $\pi \in \mathcal{E}_R$ with realization space X such that $\mu(x) = \alpha_x$ and $\mu^R_x = \mu_x$ for all $x \in X$. Let $\mathcal{B}_R$ denote the set of R-Bayes-consistent weighted vectors of posteriors, across all finite realization spaces.*

Since the receiver is causally well-specified at a consistent experiment (Lemma 2(a)), the posteriors μ_x of an R-Bayes-consistent τ are the objective posteriors of the experiment that induces it. By the No Confusion Principle and Lemma 2(b), $\mathcal{B}_R$ is exactly the set of weighted vectors attainable under NDD. The following lemma compares this set with the one available under standard Bayesian persuasion.

Lemma 4 . — *For every prior $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and DAG $R \in \mathcal{R}_0$, $\mathcal{B}_R \subseteq \mathcal{B}$.*

The inclusion follows because the subjective posteriors induced by a consistent experiment average back to the common prior. The receiver's causal model therefore restricts the sender relative to KG11, not necessarily by ruling out experiments under NDD, but by limiting which weighted vectors of posteriors those experiments can induce.

The restrictions on the sender depend on the position of the signal node. When x is terminal, a consistent experiment depends on the state only through the parents of x (Lemma B.3). When x is a root,

two restrictions apply. The *conditioning restriction* requires the experiment to depend on the state only through the children of x and their other parents. The *independence restriction* requires x to be independent of every state attribute that is not a descendant of x (Lemma B.4).

In the credit rating example, under R_c, the agency can condition the rating on solvency and liquidity, but the rating must be independent of liquidity: $\mathcal{E}_{R_c} = \{\pi \in \mathcal{E} : x \perp\!\!\!\perp_\mu \omega_2\}$. A rating given to every solvent firm and to sufficiently few defaulters to persuade the investor is more frequent among liquid firms, since liquid firms are more often solvent, and thus violates this restriction. The restriction applies to weighted vectors, not to individual posteriors. Any posterior belief about solvency can be induced by a consistent experiment. Under R_c, however, the posterior θ cannot be paired with the posterior 0 in the proportions required by Bayes consistency, which is the optimal split in KG11. Independence requires each rating to occur with the same probability in each liquidity group. The rating that induces the posterior 0 goes only to defaulters, and the liquid group has too few of them. The optimal design under R_c, and its comparison with R_a and R_b, are discussed in Section 8.

5.3.2. The Sender's Problem

By the No Confusion Principle and the definition of $\mathcal{B}_R$, the sender's problem under NDD is a Bayesian persuasion problem over the weighted vectors of posteriors in $\mathcal{B}_R$. The next proposition states this formally. It is the analogue of Proposition 1 in Kamenica and Gentzkow (2011), with one qualification concerning straightforward experiments. An experiment is *straightforward* if its realization space is a subset of A and the receiver takes action x at realization x, that is, $\hat{a}(\mu_x^R) = x$ for all $x \in X$. The KG11 reduction to straightforward experiments pools realizations that induce the same action, but pooling must preserve consistency. This holds when x is terminal, since an experiment that conditions only on $\boldsymbol{\omega}_{\mathrm{pa}(0)}$ still does so after pooling, and when x is a root with a single child, as in R_c (Lemma A.1). When x has several children, pooling can violate the independencies that R imposes among the children given x.

Proposition 2 : Analogue of Proposition 1, KG11. — *Let* $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ *and* $R \in \mathcal{R}_0$.

The following are equivalent:

- **(i)** *There exists a consistent experiment* $\pi \in \mathcal{E}_R$ *that yields sender value* v^*.
- **(ii)** *There exists* $\tau = (\alpha_x, \mu_x)_{x \in X} \in \mathcal{B}_R$ *such that* $\sum_{x \in X} \alpha_x \hat{v}(\mu_x) = v^*$.

If, in addition, x *is terminal or has a single child, both are equivalent to*

- **(iii)** *There exists a* straightforward $\pi \in \mathcal{E}_R$ *that yields sender value* v^*.

The proposition mirrors the KG11 result, with $\mathcal{B}_R$ replacing $\mathcal{B}$. When x is terminal or has a single child, straightforward experiments, whose realizations are action recommendations that the receiver follows, are without loss of generality, and the sender can achieve the optimal value with a design that uses at most $|A|$ realizations. When x has several children, the optimal experiment may need more realizations than actions. In every case the sender has an optimal consistent experiment, and it needs no more realizations than states (Lemma A.2). The reason is that consistency is a condition on each full-state posterior $\mu(\cdot|x) \in \Delta(\Omega)$ taken separately, so the sender chooses posteriors in a closed

subset of $\Delta(\Omega)$ that average back to the prior, as in KG11 with that subset in place of $\Delta(\Omega)$, and Carathéodory's theorem bounds the number of posteriors he needs.

To compare the sender's value with the KG11 benchmark, let

$$V^{\text{BP}} \equiv \sup_{\tau \in \mathcal{B}} \sum_{x \in X} \alpha_x \hat{v}(\mu_x)$$

denote the value of the sender's problem in KG11, the concavification of $\hat{v}$ over all Bayes-consistent weighted vectors of posteriors, evaluated at the prior.

Corollary 1 . — *Under* no detectable distortion*, the receiver's causal model can only hurt the sender relative to the KG11 benchmark.*

Fix $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and $R \in \mathcal{R}_0$. The value of the optimal experiment satisfying NDD is

$$V_R^* \equiv \sup_{\pi \in \mathcal{E}_{\text{NDD}}} V(\pi) = \sup_{\pi \in \mathcal{E}_R} V(\pi) = \sup_{\tau \in \mathcal{B}_R} \sum_{x \in X} \alpha_x \hat{v}(\mu_x).$$

The sender benefits from persuasion if and only if there exists $\tau \in \mathcal{B}_R$ such that $\sum_{x \in X} \alpha_x \hat{v}(\mu_x)$ exceeds his payoff at the prior. Moreover, $V(\pi) \leq V^{\text{BP}}$ for every $\pi \in \mathcal{E}_{\text{NDD}}$, which is all of $\mathcal{E}$ when x is terminal; that is, $V_R^* \leq V^{\text{BP}}$. If x is terminal or has a single child, the value is attained by a straightforward consistent experiment, with at most $|A|$ realizations.[5]

The bound $V_R^* \leq V^{\text{BP}}$ follows from $\mathcal{B}_R \subseteq \mathcal{B}$. Because the causal model restricts the splits of the prior and not each posterior separately, the sender's value is not in general the concavification of $\hat{v}$ over a subset of $\Delta(\Omega_P)$. It is nevertheless a concavification over full-state posteriors. A consistent experiment splits the prior μ_0 into full-state posteriors that all lie in a fixed set, which depends only on R and μ_0, and every such split comes from a consistent experiment (Step 1 in the proof of Lemma A.2). So V_R^* is the concave envelope of $\hat{v}$ over this set, evaluated at μ_0, and the KG11 value is the concave envelope over all of $\Delta(\Omega)$. Here $\hat{v}$ of a full-state posterior is $\hat{v}$ of its marginal on Ω_P.

When x is terminal, the set contains the posteriors that keep the prior distribution of the other state attributes given $\boldsymbol{\omega}_{\text{pa}(0)}$. The sender is then in the position of a KG11 sender who observes only the parents of x. When x is a root with a single child, a posterior in the set is one distribution of the child for each configuration of its other causes. Under R_c it is a pair of solvency rates, one for liquid firms and one for illiquid firms, and the investor's belief about solvency is their average, weighted by the prior shares of the two groups. In these two cases the set is convex, and this is why pooling realizations preserves consistency. With several children it need not be convex, and pooling can fail.

How much the causal model hurts the sender depends on the position of x. When x is terminal, the receiver interprets the signal as news about the parents of x only, and her beliefs about the other state attributes move only through their prior correlation with the parents. The sender's loss compared to the KG11 benchmark depends on how much of P lies outside $\text{pa}(0)$. When x is a root, the signal must also be independent of the state attributes that the receiver does not regard as its effects. Under R_c this is the only constraint, since the rating can depend on both solvency and liquidity. In Section 7, I characterize the causal models under which the sender attains the KG11 benchmark.

[5] As in KG11, the supremum is attained (Lemma A.2).

6. When Can Causal Models Benefit the Sender?

Section 5 shows that the receiver's causal model can only hurt a sender who cannot create detectable distortions. In this section, I study the sender who is free to distort the prior, which is possible only if x is a root. Without NDD, the No Confusion Principle does not apply, and the sender can gain from confusing the receiver.

Throughout this section, I focus on DAGs in which x is a root with a single child, i^*, and in which the set $C \equiv \mathrm{pa}_\omega(i^*)$ of the co-parents of x is non-empty. In Section 6.5, I comment on more actions and on several children.

6.1. Subjective Beliefs

Fix a Blackwell experiment $\pi \in \mathcal{E}$ and a signal realization $x \in X$. Since x is a root node and it has only one child, i^*, (3) becomes[6]

$$\mu^R(\boldsymbol{\omega}|x) = \mu_0(\boldsymbol{\omega}_C)\ \mu(\omega_{i^*}|\boldsymbol{\omega}_C, x)\ \mu_0\big(\boldsymbol{\omega}_{N_\omega\setminus(C\cup\{i^*\})}\ \big|\ \omega_{i^*}, \boldsymbol{\omega}_C\big) \quad \text{for every } \boldsymbol{\omega} \in \Omega, \tag{4}$$

where $\mu(\cdot|\boldsymbol{\omega}_C, x)$ is the objective distribution of ω_{i^*} given the co-parents and the realization, and it is the prior $\mu_0(\cdot|\boldsymbol{\omega}_C)$ when $\mu(\boldsymbol{\omega}_C, x) = 0$. I call a value $\boldsymbol{c} \in \Omega_C$ of the co-parents a *group*, $\mu_0(\boldsymbol{c}) \equiv \mu_0(\boldsymbol{\omega}_C = \boldsymbol{c})$ its *prior share*, and $\mu(\cdot|\boldsymbol{c}, x)$ its *within-group conditional*; I write $\mu_0(\cdot|\boldsymbol{c}) \in \Delta(\Omega_{i^*})$ for the prior distribution of ω_{i^*} in group $\boldsymbol{c}$.

Summing over the state attributes outside C gives $\mu^R(\boldsymbol{\omega}_C|x) = \mu_0(\boldsymbol{\omega}_C)$: the receiver weights each group $\boldsymbol{c}$ by its prior share $\mu_0(\boldsymbol{c})$ at every realization, rather than by its objective share $\mu(\boldsymbol{c}|x)$. Given the group, her belief depends on the experiment only through the within-group conditional, which is the objective one. The two shares coincide when $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_C$, so distorting the prior means letting $\mu(\boldsymbol{c}|x)$ differ from $\mu_0(\boldsymbol{c})$ (Corollary A.1).

6.2. The Design Space

The sender cares about what the receiver believes after each realization, and how often each realization occurs. By (4), the receiver's subjective beliefs, and hence her payoff-relevant posteriors, depend on the experiment only through the within-group conditionals $\mu(\cdot|\boldsymbol{c}, x)$. These conditionals and the probabilities $\mu(x)$ of the realizations depend only on how likely each realization is at each value of ω_{i^*} in each group. The sender can therefore restrict attention to experiments that depend on the state only through ω_{i^*} and $\boldsymbol{\omega}_C$ (Lemma C.1), and I write experiments as $\pi(x|\omega_{i^*}, \boldsymbol{\omega}_C)$ from now on.

Designing experiments in this way shows the structure of the sender's problem. Fix a group $\boldsymbol{c}$. The function $\pi(\cdot|\omega_{i^*}, \boldsymbol{c})$ assigns a distribution over the realizations to each value of ω_{i^*}, so it is an experiment about ω_{i^*} alone, as in KG11. Choosing π is the same as choosing one such experiment for each group, and the choice for one group does not restrict the choice for another. The sender's problem is therefore a collection of KG11 problems, one for each group. In the problem of group $\boldsymbol{c}$, the state is ω_{i^*}, the prior is $\mu_0(\cdot|\boldsymbol{c})$, and the within-group conditional $\mu(\cdot|\boldsymbol{c}, x)$ plays the role of the posterior at x. As in KG11, the only constraint is Bayes plausibility: the within-group conditionals must average

[6] The derivation is in Appendix C.

to $\mu_0(\cdot|\boldsymbol{c})$, whatever the sender does in the other groups (Lemma C.1). In the credit rating example, the agency decides separately how to rate liquid firms and how to rate illiquid firms.

What ties the problems together is the receiver. She sees the realization but not the group, so each x carries one within-group conditional for every group, and her action at x depends on all of them. This is also why the arguments underlying Lemma 4 and Corollary 1 no longer go through: Bayes plausibility holds group by group, not for her subjective posteriors over the full state.

6.3. Two Actions: One Problem per Group

Let $A = \{a_1, a_2\}$, and assume that the receiver's default action at the prior is a_1. The sender's payoff is such that $v(a_2) > v(a_1)$, that is, the sender's preferred action is a_2. At any Blackwell experiment π inducing subjective posteriors over the payoff-relevant state attributes $(\mu_x^R)_{x\in X}$, the sender's payoff is

$$V(\pi) = v(a_1) + [v(a_2) - v(a_1)] \sum_{x:\, \hat{a}(\mu_x^R)=a_2} \mu(x) = v(a_1) + [v(a_2) - v(a_1)]\, \mathbb{P}(\pi),$$

where $\mathbb{P}(\pi)$ is the probability that π persuades the receiver. I call a realization *persuading* if the receiver takes a_2 after it, and *non-persuading* otherwise. Since $v(a_2) > v(a_1)$, the sender's problem is therefore equivalent to maximizing $\mathbb{P}(\pi)$. Let $\mathbb{P}^{\mathrm{BP}}$ be the probability of persuading in the KG11 problem, and $\mathbb{P}_R^* \equiv \sup_{\pi\in\mathcal{E}_{\mathrm{NDD}}} \mathbb{P}(\pi)$ the probability of persuading at the optimal experiment satisfying NDD. In Section 2, $v(a_1) = 0$ and $v(a_2) = 1$, so the agency's value is $\mathbb{P}(\pi)$.

The receiver takes a_2 at x when her expected gain from a_2 over a_1 is non-negative. By (4), this gain is an average across groups: each group contributes the gain at its own within-group conditional, with weight equal to its prior share. A group therefore does not have to persuade the receiver alone, because the other groups help. They help the most when each of them is at its *most favorable conditional*, which puts probability one on a value $\omega_{i^*}^{\boldsymbol{c}'}$ of ω_{i^*} at which the receiver gains the most from a_2 in group $\boldsymbol{c}'$. I call the gain that group $\boldsymbol{c}$ must still deliver in that case the *lowered bar* of group $\boldsymbol{c}$. In the credit rating example, the most favorable conditional of either liquidity group has only solvent firms. At the prior of Table 1a, an investor who lends when she believes the firm solvent with probability at least 0.8 lends after a rating whose liquid firms are all solvent as soon as two thirds of its illiquid firms are solvent.

Define the *capacity* $\kappa_{\boldsymbol{c}}$ of group $\boldsymbol{c}$ as the largest probability, over all experiments, that $\boldsymbol{\omega}_C = \boldsymbol{c}$ and the realization is one at which group $\boldsymbol{c}$ clears its lowered bar:

$$\kappa_{\boldsymbol{c}} \;\equiv\; \max_{\pi\in\mathcal{E}} \sum_{x:\ \mu(\cdot|\boldsymbol{c},x) \text{ clears the bar}} \mu(\boldsymbol{\omega}_C = \boldsymbol{c}, x).$$

Finding the capacity is a KG11 problem within the group: the sender splits the prior $\mu_0(\cdot|\boldsymbol{c})$ into within-group conditionals so as to maximize the probability of those that clear the bar, and the capacity is this maximum times the prior share $\mu_0(\boldsymbol{c})$. In particular, $\kappa_{\boldsymbol{c}} = \mu_0(\boldsymbol{c})$ if the prior $\mu_0(\cdot|\boldsymbol{c})$ itself clears the bar. Section C.2 states the within-group gain, the lowered bar, and the capacity formally.

Proposition 3 : Persuasion without NDD. — *Fix $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$, $R \in \mathcal{R}_0$ as above, and a two-action decision problem with default a_1. Then*

$$\sup_{\pi\in\mathcal{E}} \mathbb{P}(\pi) = \sum_{\boldsymbol{c}\in\Omega_C} \kappa_{\boldsymbol{c}},$$

and the supremum is approached by experiments with $|\Omega_C| + 1$ *realizations: one persuading realization for each group, and one non-persuading realization.*

To see why the capacities bound the probability of persuading, note that at a persuading realization every group must clear its lowered bar, because the bar already counts the most that the other groups can contribute. The probability that $\boldsymbol{\omega}_C = \boldsymbol{c}$ and the realization is persuading is therefore at most $\kappa_{\boldsymbol{c}}$. Summing over the groups,

$$\mathbb{P}(\pi) \leq \sum_{\boldsymbol{c}} \kappa_{\boldsymbol{c}}.$$

To approach the bound, each group $\boldsymbol{c}$ is assigned its own realization $x_{\boldsymbol{c}}$. At $x_{\boldsymbol{c}}$, group $\boldsymbol{c}$ is at a within-group conditional that clears its bar, as in the problem that defines its capacity, while every other group $\boldsymbol{c}' \neq \boldsymbol{c}$ assigns an arbitrarily small probability ε to $x_{\boldsymbol{c}}$ through the state $\omega_{i^*}^{\boldsymbol{c}'}$ that maximizes its within-group gain. Thus, at $x_{\boldsymbol{c}}$, group $\boldsymbol{c}$ clears its lowered bar while every other group is arbitrarily close to its most favorable conditional, which the receiver weights by the group's prior share. Group $\boldsymbol{c}$ gives up only a probability of order ε to the other groups' realizations, losing an arbitrarily small part of its capacity.

6.4. The Value of Persuasion: the Binary Case

Suppose now that ω_{i^*} is the only payoff-relevant state attribute and that it is binary. Let $i^* = 1$, $P = \{1\}$, and $\Omega_1 = \{\omega_1^-, \omega_1^+\}$, and let a_2 be the receiver's preferred action when ω_1^+ is certain. The receiver's gain from a_2 is then affine and increasing in her belief that $\omega_1 = \omega_1^+$, so there is a threshold θ such that she takes a_2 if and only if her belief that $\omega_1 = \omega_1^+$ is at least θ. I assume $\theta \in (\mu_0(\omega_1^+), 1)$, so that a_1 is the default action.

In the binary case, $\mathbb{P}^{\mathrm{BP}} = \mu_0(\omega_1^+)/\theta$: as in Section 2.1, the optimal KG11 experiment induces the posterior θ with probability $\mu_0(\omega_1^+)/\theta$ and the posterior 0 otherwise.

In the binary case, the lowered bar and the capacity have closed forms. By (4), the receiver's belief that $\omega_1 = \omega_1^+$ after realization x is

$$\mu_x^R(\omega_1^+) = \sum_{\boldsymbol{c} \in \Omega_C} \mu_0(\boldsymbol{c}) \mu(\omega_1^+ | \boldsymbol{c}, x),$$

and the receiver takes a_2 at x if and only if this belief is at least θ. With every other group at its most favorable conditional, which assigns probability one to ω_1^+, the other groups contribute $1 - \mu_0(\boldsymbol{c})$ to this belief, so she takes a_2 if and only if $\mu_0(\boldsymbol{c})\mu(\omega_1^+|\boldsymbol{c}, x) + 1 - \mu_0(\boldsymbol{c}) \geq \theta$, that is, if and only if

$$\mu(\omega_1^+ | \boldsymbol{c}, x) \geq t_{\boldsymbol{c}}(\theta) \equiv \max\Big\{0, \frac{\theta - (1 - \mu_0(\boldsymbol{c}))}{\mu_0(\boldsymbol{c})}\Big\}, \tag{5}$$

so that $t_{\boldsymbol{c}}(\theta)$ is the lowered bar of group $\boldsymbol{c}$. When $t_{\boldsymbol{c}}(\theta)$ is positive, it is below θ because the other groups contribute positively to the receiver's belief; when $\theta \leq 1 - \mu_0(\boldsymbol{c})$, it is zero because the other groups alone can persuade the receiver.

Within group $\boldsymbol{c}$, the sender thus faces a KG11 problem with prior $\mu_0(\omega_1^+|\boldsymbol{c})$ and the lowered bar $t_{\boldsymbol{c}}(\theta)$ in place of θ. If $\mu_0(\omega_1^+|\boldsymbol{c}) \geq t_{\boldsymbol{c}}(\theta)$, the capacity is the entire prior share, $\kappa_{\boldsymbol{c}} = \mu_0(\boldsymbol{c})$. If $\mu_0(\omega_1^+|\boldsymbol{c}) < t_{\boldsymbol{c}}(\theta)$, the sender optimally induces the two within-group conditionals $t_{\boldsymbol{c}}(\theta)$ and 0,

placing as much mass as possible on the former. Hence, the capacity is $\kappa_{\boldsymbol{c}} = \mu_0(\omega_1^+, \boldsymbol{c})/t_{\boldsymbol{c}}(\theta)$. When $\mu_0(\omega_1^+|\boldsymbol{c}) < t_{\boldsymbol{c}}(\theta)$, I say that group $\boldsymbol{c}$ *binds*. This condition can also be expressed as

$$\theta > \bar{\theta}_{\boldsymbol{c}} \equiv 1 - \mu_0(\omega_1^-, \boldsymbol{c}).$$

The right-hand side is the receiver's belief that $\omega_1 = \omega_1^+$ at a realization generated by all states in group $\boldsymbol{c}$ and only states with $\omega_1 = \omega_1^+$ outside group $\boldsymbol{c}$. The next proposition sums the capacities and compares the result with $\mathbb{P}^{\text{BP}}$.

Proposition 4 . — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$, $R \in \mathcal{R}_0$ as above with $P = \{1\}$ and $\Omega_1 = \{\omega_1^-, \omega_1^+\}$, and $\theta \in (\mu_0(\omega_1^+), 1)$. Then*

$$\sup_{\pi \in \mathcal{E}} \mathbb{P}(\pi) = \overline{\mathbb{P}}(\theta) \equiv \sum_{\boldsymbol{c} \in \Omega_C} \min\Big\{\mu_0(\boldsymbol{c}),\ \frac{\mu_0(\omega_1^+, \boldsymbol{c})}{t_{\boldsymbol{c}}(\theta)}\Big\}, \tag{6}$$

with the convention that the second term is $\mu_0(\boldsymbol{c})$ when $t_{\boldsymbol{c}}(\theta) = 0$. Moreover:

- **(i)** *$\overline{\mathbb{P}}(\theta) = 1$ if and only if $\theta \leq \min_{\boldsymbol{c} \in \Omega_C} \bar{\theta}_{\boldsymbol{c}}$, and this bound is strictly above $\mu_0(\omega_1^+)$. The supremum is attained if and only if the inequality is strict, and then by an experiment with $|\Omega_C|$ realizations.*
- **(ii)** *If $\theta \geq \max_{\boldsymbol{c} \in \Omega_C} \mu_0(\omega_1^+|\boldsymbol{c})$, then $\overline{\mathbb{P}}(\theta) > \mathbb{P}^{\text{BP}}$.*

The supremum is approached by the experiment described in Proposition 3; in the credit rating example, it has one persuading rating for each liquidity group. Part **(i)** is the region in which no group binds: every group's prior conditional clears its lowered bar, so every realization can persuade the receiver, and the value 1 is strictly above $\mathbb{P}^{\text{BP}} < 1$. Part **(ii)** is the region in which the receiver's threshold is at least the prior conditional of every group. There, each group's share $\mu_0(\omega_1^+, \boldsymbol{c})/\theta$ of $\mathbb{P}^{\text{BP}}$ is at most its prior share $\mu_0(\boldsymbol{c})$, and its lowered bar is below θ, so its capacity is at least as large.

When neither condition holds, the comparison depends on the prior. A group whose prior conditional exceeds θ contributes only its prior share $\mu_0(\boldsymbol{c})$ to $\overline{\mathbb{P}}(\theta)$, which is less than its share $\mu_0(\omega_1^+, \boldsymbol{c})/\theta$ of $\mathbb{P}^{\text{BP}}$. Hence, a group that is already favorable under its prior can limit the sender's gain from distortion. Section 8.2 illustrates this intermediate region in the credit rating example, where the liquid group has the higher prior conditional.

The number of realizations. — Unlike in Section 5, the sender cannot restrict attention to straightforward experiments: two realizations that each persuade the receiver need not persuade her when they are pooled. At the realization $x_{\boldsymbol{c}}$ of the construction above, group $\boldsymbol{c}$ is at its lowered bar and the other groups help. Once the realizations are pooled, every group is close to the conditional it has at its own realization, which is its lowered bar if the group binds, and no group is left to help.

When every group binds, that is, when $\theta > \max_{\boldsymbol{c}} \bar{\theta}_{\boldsymbol{c}}$, all $|\Omega_C| + 1$ realizations of the construction in Proposition 3 are needed. The reason is that at a persuading realization at most one group can be close to its lowered bar, so each group attains its capacity only at a realization of its own. Since not every realization persuades when every group binds, with $|\Omega_C|$ realizations some group has no realization of its own: at every persuading realization it is strictly above its lowered bar, and it therefore contributes less than its capacity. With only two groups, as in the credit rating example, two ratings fall short as

soon as one group binds (Section 8.2).

Replicating the Bayesian Persuasion Outcome. — The binary case also shows when an experiment that satisfies NDD attains $\mathbb{P}^{BP}$. The Bayesian persuasion outcome splits $\mu_0(\omega_1^+)$ between the posteriors θ, with probability $\mu_0(\omega_1^+)/\theta$, and 0. I say that π *replicates* it if the receiver's subjective posterior $\mu_x^R(\omega_1^+)$ equals θ on a set of realizations of total probability $\mu_0(\omega_1^+)/\theta$ and 0 at every other realization. Let

$$\tilde{\theta} \equiv \frac{\mu_0(\omega_1^+)}{\max_{\boldsymbol{c}} \mu_0(\omega_1^+|\boldsymbol{c})}.$$

The value of $\tilde{\theta}$ identifies when replication is compatible with NDD. Some consistent experiment replicates the Bayesian persuasion outcome if and only if $\theta \leq \tilde{\theta}$ (Lemma C.3). By the No Confusion Principle, an experiment that satisfies NDD and replicates the outcome has a consistent implementation that does too. Thus, when $\theta > \tilde{\theta}$, no experiment satisfying NDD can replicate the Bayesian persuasion outcome.

6.5. Extensions

The results of this section are stated for two actions and a single child. With more actions, each action is triggered within a group at its own lowered bar, so the sender again faces one KG11 problem per group, with a step payoff in place of a single bar. With several children, what matters is their causal relation to one another. Children on no causal path to a payoff-relevant state attribute are summed out of the belief about those attributes and play no role. Children that are pairwise linked and share their other parents enter the receiver's belief as a single vector-valued child, and I expect the results above to extend to this case with the vector in place of ω_{i^*}. Children that are not linked pose a different problem: the receiver computes the posterior of each child correctly and multiplies the posteriors, rather than reweighting the groups. I do not pursue these extensions here.

7. Irrelevance of Causal Models

Finally, I characterize the conditions under which a causal model is *irrelevant*, meaning that it leaves the sender with the same persuasion problem as in Kamenica and Gentzkow (2011). The definition of irrelevance distinguishes two requirements: one on the receiver's beliefs and one on the experiments consistent with her causal model.

Definition 5 : Irrelevance. — *Fix $\mu_0 \in \text{int}\,(\Delta(\Omega))$. A causal model $R \in \mathcal{R}_0$ is*

- **(i)** belief-irrelevant *if for every $\pi \in \mathcal{E}$ and every $x \in X$, $\mu_x^R(\boldsymbol{\omega}_P) = \mu_x(\boldsymbol{\omega}_P)$;*
- **(ii)** design-irrelevant *if $\mathcal{E}_R \subseteq \{\pi \in \mathcal{E} : \pi(x|\boldsymbol{\omega}) = \pi(x|\boldsymbol{\omega}_P)$ for all $\boldsymbol{\omega} \in \Omega\}$.*

R is irrelevant *if it is both belief-irrelevant and design-irrelevant.*

Belief-irrelevance is a property of the causal model. It requires the receiver's subjective payoff-relevant posteriors to coincide with the objective ones after every signal realization. A causal model is belief-irrelevant not only when it is correct, but whenever its distortions are confined to dimensions that

do not affect the receiver's decision problem: she may be wrong about how the payoff-irrelevant state attributes covary with the signal.

The implication for the sender's value is immediate. Under belief-irrelevance, the receiver holds the same payoff-relevant posterior as the KG11 receiver after every signal realization, so she takes the same action: $\hat{v}(\mu_x^R) = \hat{v}(\mu_x)$ for all $x \in X$. Since the sender and the receiver also agree on how likely each realization is (Lemma 1), *every* experiment yields the payoff it would yield with the KG11 receiver:

$$V(\pi) = \sum_{x \in X} \mu(x)\, \hat{v}(\mu_x^R) = \sum_{x \in X} \mu(x)\, \hat{v}(\mu_x).$$

Since this holds for every experiment, it holds for the optimal one: the sender attains the KG11 value.

Design-irrelevance pertains to the sender's choice set: consistent experiments depend on the state only through its payoff-relevant attributes. Thus, $\boldsymbol{\omega}_P$ is sufficient for the sender's design problem, and the attributes in $N_\omega \setminus P$ are then not only payoff-irrelevant; they also play no role in the sender's choice.

Neither notion implies the other. Belief-irrelevance without design-irrelevance arises when the receiver's causal model is fully connected and $P \subsetneq N_\omega$ (Remark D.1). The receiver forms the same beliefs as the KG11 receiver under every experiment, but $\mathcal{E}_R = \mathcal{E}$: the sender is not restricted to experiments that depend only on the payoff-relevant attributes.

Conversely, design-irrelevance need not imply belief-irrelevance. Let x be terminal with only some of the payoff-relevant state attributes as parents, $\mathrm{pa}(0) \subsetneq P$. Every consistent experiment depends on $\boldsymbol{\omega}_{\mathrm{pa}(0)}$ only (Lemma B.3), so design-irrelevance holds. But the receiver reads the signal as evidence about $\boldsymbol{\omega}_{\mathrm{pa}(0)}$ only, and revises her belief about the other payoff-relevant attributes through the prior, whatever the experiment. Some experiment therefore induces $\mu_x^R \neq \mu_x$ (Lemma D.1).

7.1. Conditions for Irrelevance

To establish irrelevance, it is enough to look at a Markov-equivalent model in which x is terminal.

Proposition 5 . — *Let $R \in \mathcal{R}_0$. Then R is irrelevant if it is Markov equivalent to a DAG in which x is terminal and $\mathrm{pa}(0) = P$. The converse holds if x is terminal or has a single child.*

The reason is in two steps. First, a causal model inherits irrelevance from any model that is Markov equivalent to it, since Markov-equivalent causal models induce the same beliefs and the same consistent experiments (Lemma B.5). Second, when x is terminal the receiver applies Bayes rule once, with the likelihood $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})$. If $\mathrm{pa}(0) = P$, this is the likelihood the KG11 receiver uses to update her belief about $\boldsymbol{\omega}_P$, so her payoff-relevant posteriors are the objective ones, and an experiment is consistent exactly when it depends on the state through $\boldsymbol{\omega}_P$ only (Lemma D.1).

When x is terminal, R is Markov equivalent to such a terminal model if and only if $\mathrm{pa}(0) = P$. When x is a root, it is if and only if the children of x are exactly the payoff-relevant state attributes, every two of them are linked, and none of them has a payoff-irrelevant parent (Lemma D.2). When x is a root with a single child i^*, this leaves one model only: $x \to \omega_{i^*}$ with no other parent of ω_{i^*} and $P = \{i^*\}$. The converse says that, when x is terminal or has a single child, no other causal model is irrelevant. When x has several children, the condition is sufficient.

The consequences for persuasion follow from the reduction to the terminal model. The sender can induce every Bayes-consistent vector of posteriors by committing to an experiment designed on the reduced state space Ω_P.

Corollary 2 . — *Under the condition of Proposition 5, the sender faces the KG11 problem on Ω_P: an experiment is consistent if and only if it depends on the state through $\boldsymbol{\omega}_P$ only, and $\mathcal{B}_R = \mathcal{B}$.*

Table 3 summarizes when belief-irrelevance and design-irrelevance arise in the DAGs with x as a terminal node depending on the relation between P and $\mathrm{pa}(0)$. In the top row, the sender obtains the KG11 value, because belief-irrelevance (hence also irrelevance) implies that $\mathcal{B}_R = \mathcal{B}$ (Lemma D.1). What differs between the two columns is just the design space, which is restricted to Ω_P when $\mathrm{pa}(0) = P$. In the bottom row, the sender is strictly worse off than in KG11, independently of the design space chosen, because the receiver discards some payoff-relevant state attributes when processing the signal.

8. A Credit Rating Agency Example — Continued

I return to the credit rating agency of Section 2 with two changes. The threshold θ is no longer fixed at 0.8, and the investor may hold any of the following three causal models:

$$\text{Liquidity} \rightarrow \text{Solvency} \rightarrow \text{Rating} \qquad (R_a)$$

$$\text{Solvency} \leftarrow \text{Liquidity} \rightarrow \text{Rating} \qquad (R_b)$$

$$\text{Rating} \rightarrow \text{Solvency} \leftarrow \text{Liquidity}. \qquad (R_c)$$

I assume that the prior μ_0 has full support and that liquid firms are more likely to be solvent, $\mu_0(\omega_1 = s|\omega_2 = \ell) > \mu_0(\omega_1 = s|\omega_2 = i)$. The investor's lending threshold is $\theta \in (\mu_0(\omega_1 = s), 1)$, so the investor does not lend without further information. She lends if and only if her belief that the firm is solvent is at least θ, and the agency's value is the probability $\mathbb{P}$ that she does.

I first impose No Detectable Distortion (NDD), the audit introduced in Section 2. By Lemma 3**(i)**, the audit is immaterial under R_a and R_b, where the rating is evidence, while under R_c it can be binding. By the No Confusion Principle and Proposition 2, I can then restrict attention to consistent methodologies with binary rating realizations, $X = \{H, L\}$, such that the investor lends after H and not after L. I then drop the audit and characterize what the agency gains under R_c from the wedge between subjective and objective solvency.

8.1. With the Audit

Under R_a, the agency faces the KG11 problem since the causal model is irrelevant, by Proposition 5.

Lemma 5(a): $\mathrm{R_a}$. — *The optimal methodology rates high all solvent firms and a fraction of defaulters,*

	$\mathbf{pa(0) \subseteq P}$	$\mathbf{pa(0) \not\subseteq P}$
$\boldsymbol{P \subseteq pa(0)}$	irrelevant	belief-irrelevant only
$\boldsymbol{P \not\subseteq pa(0)}$	design-irrelevant only	neither

Table 3: Irrelevance when x is terminal, depending on the relation between P and $\mathrm{pa}(0)$.

so that $\mu_H^{R_a}(\omega_1 = s) = \theta$ *and* $\mu_L^{R_a}(\omega_1 = s) = 0$. *The investor lends with probability* $\mathbb{P}^*_{R_a} = \mathbb{P}^{\text{BP}}$.

To compare R_b and R_c with the benchmark, note that the objective posteriors over solvency average back to the prior. Hence the frequency of the high rating is pinned down by the objective posteriors it induces:

$$\mathbb{P}(\pi) = \mu(x = H) = \frac{\mu_0(\omega_1 = s) - \mu_L(\omega_1 = s)}{\mu_H(\omega_1 = s) - \mu_L(\omega_1 = s)}.$$

The objective posteriors determine how often the high rating can be issued, while the investor's subjective posterior after H determines whether it persuades. Under the audit, by the No Confusion Principle, the objective and subjective posteriors coincide.

The agency maximizes the size of the high-rated pool, which is determined by two posteriors. On the one hand, $\mu_H(\omega_1 = s)$ is the solvency that the high rating must carry to persuade the investor, which has to be at least θ. On the other hand, $\mu_L(\omega_1 = s)$ is the solvency assigned to the non-persuasive rating, which the agency would like to minimize.

A causal model can reduce the probability of lending in two ways. It can impose a *ceiling* on the posterior after the high rating, so that the investor cannot be persuaded once her threshold exceeds it, or a *floor* on the posterior after the low rating, so that some solvent firms stay in the non-persuasive pool even when the high-rated pool reaches her threshold. R_b does both; R_c with the audit only the second.

Evidence of liquidity. — Under R_b, a consistent methodology conditions the rating only on liquidity (Lemma B.3).

Lemma 5(b): R_b. — *If* $\theta > \mu_0(\omega_1 = s|\omega_2 = \ell)$, *persuasion is not possible. Otherwise, the optimal methodology rates high all liquid firms and a fraction of the illiquid ones, so that* $\mu_H^{R_b}(\omega_1 = s) = \theta$ *and* $\mu_L^{R_b}(\omega_1 = s) = \mu_0(\omega_1 = s|\omega_2 = i)$, *and* $\mathbb{P}^*_{R_b} < \mathbb{P}^{\text{BP}}$.

The investor interprets the rating as information about liquidity alone, so the agency can choose how much weight each liquidity group receives under each rating, but not the solvency rate within either group. The most favorable information a rating can convey is that the firm is liquid, so the posterior after the high rating cannot exceed $\mu_0(\omega_1 = s|\omega_2 = \ell)$. This is the *ceiling*. Similarly, the posterior after the low rating cannot fall below $\mu_0(\omega_1 = s|\omega_2 = i)$, which is the *floor*. Both bounds arise because the rating reveals liquidity, while the prior determines how liquidity predicts solvency.

Unlike the benchmark value $\mu_0(\omega_1 = s)/\theta$, which is positive for every $\theta < 1$, the value under R_b drops to zero once the threshold exceeds the ceiling. At the prior of Table 1a, liquid firms are solvent 75% of the time, so the agency's value is zero at $\theta = 0.8$.

The shortfall from the KG11 benchmark is due to the causal model, not to the audit: it is R_b itself that puts a ceiling and a floor on the posteriors a rating can induce.

A cause of solvency. — Under R_c, the investor never revises her view of liquidity from the rating. The agency can condition the rating on both attributes, but the audit requires the two liquidity

groups to receive each rating at the same rate (Corollary A.1**(ii)**). In other words, the high rating cannot act as a liquidity screen. The threshold $\tilde{\theta}$ of Lemma C.3 is $\mu_0(\omega_1 = s)/\mu_0(\omega_1 = s|\omega_2 = \ell)$ here.

Lemma 5(c): $\mathrm{R_c}$. — *If $\theta \leq \tilde{\theta}$, the optimal methodology replicates the Bayesian persuasion outcome, $\mathbb{P}^*_{R_c} = \mathbb{P}^{\mathrm{BP}}$. Otherwise, the audit binds and the optimal methodology sets $\mu_H^{R_c}(\omega_1 = s) = \theta$ but leaves $\mu_L^{R_c}(\omega_1 = s) > 0$, and $\mathbb{P}^*_{R_c} < \mathbb{P}^{\mathrm{BP}}$.*

Under R_c there are two regimes, separated by $\tilde{\theta}$, the point at which two requirements on the *size* of the high-rated pool become incompatible. The first requirement is already present in the benchmark. A high-rated pool of size $\mu(x = H)$ with solvency θ uses $\theta\mu(x = H)$ units of solvent mass, of which only $\mu_0(\omega_1 = s)$ exist. Hence $\mu(x = H) \leq \mu_0(\omega_1 = s)/\theta$. This upper bound falls as θ rises.

The second requirement comes from the audit. To keep $\mu_L(\omega_1 = s) = 0$, the agency must rate every solvent firm high. Since a fraction $\mu_0(\omega_1 = s|\omega_2 = \ell)$ of liquid firms is solvent, at least that fraction of liquid firms must receive the high rating. The audit requires the same rating rate among illiquid firms, so $\mu(x = H) \geq \mu_0(\omega_1 = s|\omega_2 = \ell)$. This lower bound is independent of θ.

The Bayesian persuasion outcome is feasible exactly when these two bounds are compatible: $\mu_0(\omega_1 = s|\omega_2 = \ell) \leq \mu_0(\omega_1 = s)/\theta$, or equivalently $\theta \leq \tilde{\theta}$. When $\theta \leq \tilde{\theta}$, the agency can rate every solvent firm high and allocate defaulters across the two liquidity groups so that the common rating rate holds. The audit is then slack. Above $\tilde{\theta}$, the agency can still bring the high-rated pool to the investor's threshold, but it can no longer keep every solvent firm in that pool.

The firms left out are solvent *liquid* firms. Once all solvent illiquid firms have been rated high, every additional illiquid firm assigned the high rating is a defaulter. A pool of size $\mu(x = H)$ then contains at most $\mu(x = H)\mu_0(\omega_2 = \ell)$ solvent liquid firms, its entire liquid part, and at most $\mu_0(\omega_1 = s, \omega_2 = i)$ solvent illiquid firms, while it must contain $\theta\mu(x = H)$ solvent firms in total. The size of the high-rated pool is therefore limited by the stock of solvent illiquid firms, $\mu(x = H) \leq \mu_0(\omega_1 = s, \omega_2 = i)/\theta - \mu_0(\omega_2 = \ell)$. The scarce input is no longer solvency in the population, as in the benchmark, but solvency *in the illiquid group*.

At the prior of Table 1a and $\theta = 0.8$, $\tilde{\theta} \approx 0.53$, so the audit binds. The agency's value is $\mathbb{P}^*_{R_c} = 0.25$, half the benchmark, and the optimal methodology is in Table 2a.

Unlike under R_b, persuasion remains possible at every threshold: the audit shrinks the persuading pool gradually, whereas the ceiling of R_b shuts persuasion down.

8.2. Dropping the Audit

Suppose now that the audit is dropped. Nothing changes under R_a or R_b, since every methodology already satisfies NDD. Under R_c, the agency can now separate the two margins that coincide under the audit: the subjective posterior determines whether a rating persuades, while the objective posterior determines how much solvent mass is needed to support it.

Recall the wedge of Section 2, Δ_x, the subjective minus the objective solvency of the firms rated x: the product of the difference in solvency between liquid and illiquid firms among them and the difference between the prior share of liquid firms and their share among them. The wedge is positive at x when the

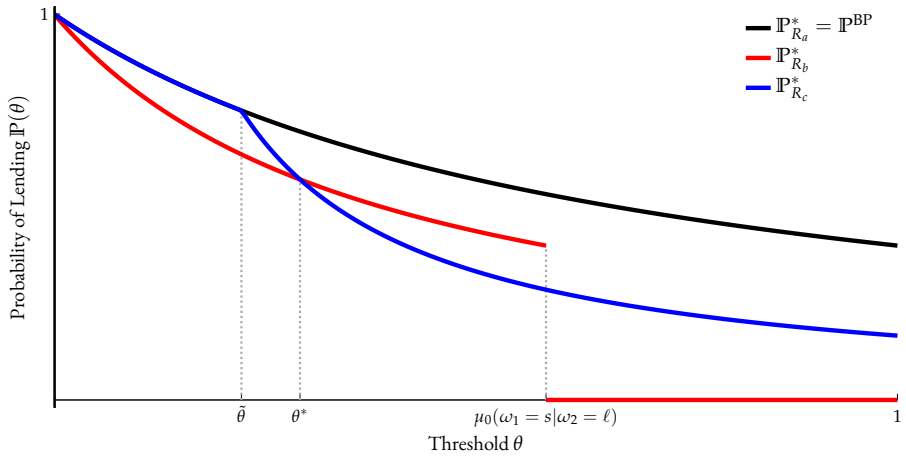


Figure 2: Probability of lending as a function of the threshold under the audit, at the prior of Table 1a. Under R_b the value drops to zero at $\theta = \mu_0(\omega_1 = s|\omega_2 = \ell)$. Under R_c it coincides with the benchmark up to $\tilde{\theta}$ and falls below it thereafter. The ranking of the two models reverses at θ^*, where $\mathbb{P}^*_{R_b} = \mathbb{P}^*_{R_c}$, and again at $\mu_0(\omega_1 = s|\omega_2 = \ell)$.

liquidity group that is more solvent at the rating has a smaller share of the rating than its prior weight, so that the investor overweights it, and it is larger the larger the difference in solvency rate between the two liquidity groups. At every persuading rating the agency brings the investor's subjective posterior to θ, so the rating's objective solvency is $\theta - \Delta_x$, and the larger the wedge, the fewer truly solvent firms the rating needs. The agency therefore wants to maximize the wedge Δ_x at every persuading rating.

Two ratings. — Suppose first that the agency keeps two ratings, as in Section 2. To see when it can persuade the investor after both ratings, consider a rating that every firm of liquidity group $c \in \Omega_2$ receives, including the group's defaulters; I say that such a rating is targeted at group c. The threshold $\bar{\theta}_c = 1 - \mu_0(\omega_1 = d, \omega_2 = c)$ of Section 6.4, above which group c binds, has a second reading here: it is the highest posterior the investor can obtain from a rating targeted at group c. It is attained when the rating goes only to solvent firms in the other group. By (1), the investor evaluates group c at its prior solvency and the other group as entirely solvent. For example, for $c = \ell$ her posterior is

$$\mu_0(\omega_2 = \ell)\mu_0(\omega_1 = s \mid \omega_2 = \ell) + \mu_0(\omega_2 = i) = \bar{\theta}_\ell.$$

Hence a rating targeted at group c can persuade if and only if $\theta \leq \bar{\theta}_c$.

The agency can use two ratings, one targeted at each liquidity group. Each rating is assigned to the firms in its target group and to a small share of solvent firms from the other group, so that the investor's posteriors after the two ratings can be made arbitrarily close to $\bar{\theta}_\ell$ and $\bar{\theta}_i$. Thus, both ratings can persuade when $\theta < \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$. Once $\theta \geq \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, at most one of the two ratings can persuade. The reason is that above $\bar{\theta}_c$ a rating persuades only if group c is more solvent at that rating than in the population, since even an entirely solvent other group contributes no more than its prior weight to the investor's posterior; with two ratings, every firm of group c receives one of them, so the group cannot be more solvent than in the population at both. The agency is then left with a single persuading rating and faces the problem of Section 2 at a general threshold: which defaulters to assign the persuading rating, and how many.

Lemma 5(d): R_c, no audit, two ratings. — *Let R_c be the investor's causal model and the audit not be imposed. With $\overline{\mathbb{P}}(\theta)$ as in Proposition 4, denote by $\overline{\mathbb{P}}_{\text{bin}}(\theta)$ the largest probability that the investor lends over the methodologies with two rating realizations $X = \{H, L\}$.*

- **(i)** *If $\theta < \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, the investor lends after both ratings of some methodology, so two ratings attain the supremum over all methodologies: $\overline{\mathbb{P}}_{\text{bin}}(\theta) = \overline{\mathbb{P}}(\theta) = 1$.*
- **(ii)** *If $\theta \geq \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, the investor lends after at most one rating, and two ratings fall short of the supremum over all methodologies: $\overline{\mathbb{P}}_{\text{bin}}(\theta) < \overline{\mathbb{P}}(\theta)$. The supremum $\overline{\mathbb{P}}_{\text{bin}}(\theta)$ is attained by a methodology that rates high all solvent firms, all or none of the defaulters of one liquidity group, and a fraction of the defaulters of the other group, so that $\mu_H^{R_c}(\omega_1 = s) = \theta$.*
- **(iii)** *$\overline{\mathbb{P}}_{\text{bin}}(\theta) < \mathbb{P}^{\text{BP}}$ if and only if $\theta \in \big(\tilde{\theta}, \mu_0(\omega_1 = s \mid \omega_2 = \ell)\big)$, and $\overline{\mathbb{P}}_{\text{bin}}(\theta) = \mathbb{P}^{\text{BP}}$ only if θ is one of these two thresholds.*

Part **(iii)** is the interval in which the audit would reduce the agency's value while R_b still allows persuasion. At the prior of Table 1a and $\theta = 0.8$, the best two-rating methodology is the one in Table 2b, with value 0.52, compared with the KG11 value 0.5.

More than two ratings. — With more ratings, the agency can assign a separate persuading rating to each liquidity group. A rating x_c meant for group $c \in \Omega_2$ goes to the firms of that group and to a small share of the solvent firms of the other group. Under R_c, the investor weights the other group by its prior share even when only a few of its firms receive x_c, and if those firms are solvent they contribute the whole share to her posterior. The firms of group c then need only reach the lowered bar of Section 6.4: for the illiquid group, $t_i(\theta) = \max\{0,\ (\theta - \mu_0(\omega_2 = \ell))/\mu_0(\omega_2 = i)\}$, and symmetrically $t_\ell(\theta)$ for the liquid group.

Group c binds when its bar exceeds its prior solvency $\mu_0(\omega_1 = s|\omega_2 = c)$, that is, when $\theta > \bar{\theta}_c$. Its capacity is its full prior share $\mu_0(\omega_2 = c)$ if it does not bind, and its stock of solvent firms divided by its bar, $\mu_0(\omega_1 = s, \omega_2 = c)/t_c(\theta)$, if it does. With the liquidity groups as the groups of Section 6.3, Proposition 4 gives the supremum $\overline{\mathbb{P}}(\theta)$ of the probability that the investor lends over all methodologies: the sum of the two capacities. It is approached by the construction of Proposition 3, the ratings x_c above plus a third rating for the remaining defaulters, and it equals one exactly when no group binds, $\theta \leq \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, with the supremum attained only if the inequality is strict.

The third rating is what lets the agency separate the two group-specific persuasion problems, instead of forcing them into a single rating with a common liquidity composition.

At the prior of Table 1a and $\theta = 0.8$, the illiquid group binds and the liquid group does not. The liquid group contributes its full prior share and the illiquid group its capacity, for a supremum $\overline{\mathbb{P}}(0.8) = 0.55$, against the KG11 value 0.5.

8.3. Comparing Causal Models

The three causal models rank differently depending on whether the audit is imposed. Figure 2 illustrates the ranking with the audit at the prior of Table 1a. There is no global ranking between R_b and R_c. They yield the same value at $\theta^* = 0.575$. For lower thresholds, R_c is less restrictive than R_b; on

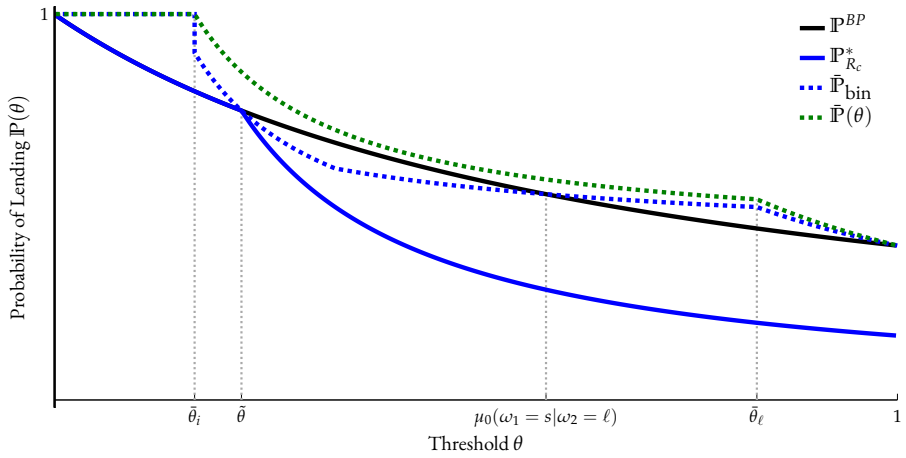


Figure 3: The agency's value under R_c as a function of the threshold θ, at the prior of Table 1a. With the audit, the value coincides with the benchmark up to $\tilde{\theta}$ and falls below it thereafter. Without the audit, the supremum over all methodologies, $\overline{\mathbb{P}}(\theta)$, exceeds the benchmark at every threshold and equals one up to $\bar{\theta}_i$, below which no liquidity group binds. The best methodology with two ratings, with value $\overline{\mathbb{P}}_{\text{bin}}(\theta)$, reaches the supremum only below $\bar{\theta}_i$; it exceeds the benchmark outside $[\tilde{\theta}, \mu_0(\omega_1 = s|\omega_2 = \ell)]$ and falls below it in between (Lemma 5(d)).

the intermediate range, R_b is less restrictive; and once θ exceeds $\mu_0(\omega_1 = s \mid \omega_2 = \ell)$, persuasion under R_b becomes impossible while R_c still yields a positive value. The benchmark model R_a remains optimal throughout.

Without the audit, only R_c changes: its value approaches $\overline{\mathbb{P}}(\theta)$. Whenever $\overline{\mathbb{P}}(\theta) > \mathbb{P}^{\text{BP}}$, R_c is then better for the agency than R_a, and R_b remains the most restrictive of the three models. At the prior of Table 1a, this holds at every threshold (Figure 3), including the range between $\min\{\bar{\theta}_\ell, \bar{\theta}_i\}$ and $\mu_0(\omega_1 = s \mid \omega_2 = \ell)$, where the illiquid group binds while the liquid group is still more solvent than the threshold and the comparison depends on the prior. There the liquid group contributes only its prior share, which is less than its share of the benchmark, and whether the gain of the illiquid group makes up for it depends on how much solvent mass that group has.

The audit thus matters only under R_c, where it takes away the one instrument that the investor does not check: the share of liquid firms among the firms that receive a rating.

9. Welfare

So far the receiver's causal model has entered the analysis as a constraint on the sender, or, without NDD, as a lever for him. In this section I turn to the receiver's welfare. When her posteriors about the payoff-relevant state attributes are correct, as they are at the sender's optimum whenever the No Confusion Principle applies, her causal model acts as a *tax on the sender*: it restricts what he can induce and leaves her no worse off than without information. When NDD fails, the sender's gain over the KG11 benchmark is not created by the causal model but *transferred from the receiver*, at a rate that I characterize. Proofs are in Appendix E.

The receiver chooses her action on the basis of the subjective posterior she forms through her causal

model, but her payoff is determined by the objective one. Write

$$\hat{u}(q) \equiv \max_{a \in A} \bar{u}(a, q)$$

for her indirect utility at a belief $q \in \Delta(\Omega_P)$, and $\hat{u}(\mu_0)$ for its value at the prior marginal on Ω_P. The function $\hat{u}$ is convex and piecewise linear, and its kinks are the beliefs at which the optimal action switches; at a kink the receiver is indifferent between two actions. In the credit rating example a belief is the probability that the firm is solvent, and the kinks are the thresholds separating consecutive actions.

Definition 6 : Receiver's Welfare. — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$, $R \in \mathcal{R}_0$ and $\pi \in \mathcal{E}$. The receiver's* welfare *and the* payoff of the KG11 receiver *are*

$$U(\pi) \equiv \sum_{x \in X} \mu(x)\, \bar{u}\big(\hat{a}(\mu_x^R), \mu_x\big) \qquad U^{\mathrm{BP}}(\pi) \equiv \sum_{x \in X} \mu(x)\, \hat{u}(\mu_x).$$

$U(\pi)$ is the receiver's expected payoff: at each realization she takes the action that is optimal at her subjective posterior, and the payoff is evaluated at the objective one. $U^{\mathrm{BP}}(\pi)$ is the payoff that a receiver with a fully connected causal model obtains from the same experiment. Since $\hat{a}(\mu_x)$ maximizes the expected payoff at μ_x, $U(\pi) \leq U^{\mathrm{BP}}(\pi)$, with equality exactly when the distortion in the receiver's posterior never moves her to an action that is worse at the objective posterior. Since the objective posteriors average to the prior and $\hat{u}$ is convex, $U^{\mathrm{BP}}(\pi) \geq \hat{u}(\mu_0)$. Thus the receiver's net gain from the signal is the difference between an information gain and a misspecification loss,

$$U(\pi) - \hat{u}(\mu_0) = \underbrace{\Big[U^{\mathrm{BP}}(\pi) - \hat{u}(\mu_0)\Big]}_{\text{information gain} \geq 0} - \underbrace{\Big[U^{\mathrm{BP}}(\pi) - U(\pi)\Big]}_{\text{misspecification loss} \geq 0}. \tag{7}$$

In KG11 the loss is zero and information never hurts the receiver. Here it can be strictly positive, and information can make her worse off.

9.1. Correct Posteriors: a Tax on the Sender

I first consider experiments at which the receiver's subjective and objective posteriors over the payoff-relevant state attributes coincide. This is weaker than being causally well-specified: her subjective joint belief may differ from the objective one, as long as the difference does not reach Ω_P.

Let $A = \{a_1, a_2\}$, with a_1 the receiver's default action and a_2 the action the sender prefers. I say that *obedience binds* at an experiment if, at every realization at which the receiver takes a_2, she is indifferent between the two actions at her objective posterior.

Proposition 6 . — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$, and let $\pi \in \mathcal{E}$ satisfy $\mu_x^R(\omega_P) = \mu_x(\omega_P)$ for every $x \in X$.*

- **(i)** $U(\pi) = U^{\mathrm{BP}}(\pi) \geq \hat{u}(\mu_0)$.
- **(ii)** *If there are two actions and obedience binds at π, then $U(\pi) = \hat{u}(\mu_0)$.*

Proposition 6**(i)** states that as long as the payoff-relevant posteriors are correct, the receiver weakly gains from information, whether or not her causal model is. Two receivers whose beliefs about Ω differ but agree on Ω_P take the same action and obtain the same payoff. In this sense, her misspecification never reaches her decision. Part **(ii)** concerns experiments at which the sender leaves the receiver no

surplus. In the KG11 problem obedience binds at every optimum. When the sender optimizes over $\mathcal{B}_R$, whether it binds depends on the feasible set. In the credit rating example it binds at the agency's optimum under each of the three causal models (Lemma 5(a), 5(b) and 5(c)).

The proposition applies at the sender's optimum whenever the No Confusion Principle holds. By Corollary 1 the optimal value on $\mathcal{E}_{\text{NDD}}$ is attained at a consistent experiment, and at a consistent experiment the payoff-relevant posteriors are correct by Lemma 2(a). With two actions, if obedience binds at that optimum, Proposition 6**(ii)** says that the receiver's welfare there does not depend on R. The reason is that $U(\pi) = \sum_x \mu(x)\hat{u}(\mu_x)$ is an average of the convex function $\hat{u}$ over posteriors whose mean is the prior, so it exceeds $\hat{u}(\mu_0)$ only if the posteriors straddle a kink. With two actions there is one kink; binding obedience places every persuading posterior on it, and every other posterior lies where the receiver strictly prefers a_1. All posteriors thus lie on the linear piece of $\hat{u}$ on which it equals her expected payoff from a_1, and averaging returns her default payoff. The causal model changes how often she is persuaded, which is the sender's concern, but not what she expects to earn.

With more than two actions the receiver can gain strictly from information even if her payoff-relevant posteriors are correct and the sender gives her no more than he has to. Let the payoff-relevant state attribute be binary, so that a kink of $\hat{u}$ is a probability of ω_1^+. She gains strictly if and only if the experiment induces a posterior above the lowest kink (Lemma E.1 in Appendix E). The number of actions alone does not decide this: the sender must want to move a posterior above the lowest kink, and he must be able to.

What he is able to do depends on his feasible set, and so on the receiver's causal model. In the credit rating example with three actions, the agency can induce every posterior under R_a, and under R_b only the posteriors between the two prior conditionals $\mu_0(\omega_1 = s|\omega_2 = i)$ and $\mu_0(\omega_1 = s|\omega_2 = \ell)$. If the top kink lies above $\mu_0(\omega_1 = s|\omega_2 = \ell)$, it is out of reach under R_b: the agency can at most use the lower kink, and the investor is left at her default payoff. Under R_a she gains whenever the agency's payoff makes the top kink worth using.

9.2. Without NDD: a Transfer from the Receiver

When $\mu_x^R(\boldsymbol{\omega}_P) \neq \mu_x(\boldsymbol{\omega}_P)$ at some realization, the misspecification loss in (7) can be strictly positive and the receiver can end up below her no-information payoff. I bound how far she can fall in the setting of Section 6.4: two actions with a_1 the default, a binary payoff-relevant state attribute $\Omega_1 = \{\omega_1^-, \omega_1^+\}$, and a threshold $\theta \in (\mu_0(\omega_1^+), 1)$ above which the receiver takes a_2. Since her gain from a_2 over a_1 is affine in her belief, it equals $K\,(q(\omega_1^+) - \theta)$ at every belief q, for some $K > 0$; the constant $K\theta$ is what she loses by taking a_2 when ω_1^- is certain. The probability $\mathbb{P}(\pi)$ that π persuades her and $\mathbb{P}^{\text{BP}} = \mu_0(\omega_1^+)/\theta$ are those of Section 6.4.

The receiver's expected payoff from a_1 is affine in her belief, and the objective posteriors average to the prior, so its average over the objective posteriors is her default payoff $\hat{u}(\mu_0)$. Her net gain from the signal is therefore the gain from a_2, evaluated at the objective posterior and summed over the realizations that persuade her:

$$U(\pi) - \hat{u}(\mu_0) = K \sum_{x:\, \mu_x^R(\omega_1^+) \geq \theta} \mu(x)\,\big(\mu_x(\omega_1^+) - \theta\big). \tag{8}$$

The identity holds at every experiment. Which realizations persuade the receiver is determined by her subjective posteriors, while the weights and the posteriors in the sum are objective: a realization that persuades her raises her welfare if her objective posterior is above θ and lowers it if it is below.

The payoff that the receiver herself expects from π is computed at her subjective posteriors. I call it her perceived payoff and write $U^{\mathrm{R}}(\pi) \equiv \sum_{x \in X} \mu(x)\, \hat{u}(\mu_x^R)$.

Proposition 7 : Transfer from the receiver. — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$, let $A = \{a_1, a_2\}$ with a_1 the default action, and let $\Omega_P = \Omega_1$ be binary. For every $\pi \in \mathcal{E}$:*

(i) *The receiver's net gain from the signal is*

$$U(\pi) - \hat{u}(\mu_0) = K\theta \left[\mathbb{P}^{\mathrm{BP}} - \mathbb{P}(\pi)\right] - K \sum_{x:\, \mu_x^R(\omega_1^+) < \theta} \mu(x)\, \mu_x(\omega_1^+). \tag{9}$$

In particular, if $\mathbb{P}(\pi) > \mathbb{P}^{\mathrm{BP}}$, then $\hat{u}(\mu_0) - U(\pi) \geq K\theta \left[\mathbb{P}(\pi) - \mathbb{P}^{\mathrm{BP}}\right] > 0$.

(ii) *If $\mathbb{P}(\pi) > \mathbb{P}^{\mathrm{BP}}$, then her subjective prior satisfies $\mu^R(\omega_1^+) > \mu_0(\omega_1^+)$ and*

$$U^{\mathrm{R}}(\pi) - \hat{u}(\mu_0) \;\geq\; \left[u(a_1, \omega_1^+) - u(a_1, \omega_1^-)\right] \left[\mu^R(\omega_1^+) - \mu_0(\omega_1^+)\right], \tag{10}$$

with equality if and only if $\mu_x^R(\omega_1^+) = \theta$ at every realization at which she takes a_2. Hence, if $u(a_1, \omega_1^+) \geq u(a_1, \omega_1^-)$, then $U^{\mathrm{R}}(\pi) \geq \hat{u}(\mu_0) > U(\pi)$.

By Corollary 1, $\mathbb{P}(\pi) \leq \mathbb{P}^{\mathrm{BP}}$ for every $\pi \in \mathcal{E}_{\mathrm{NDD}}$ (Section 6.4): the sender beats the KG11 benchmark only at an experiment that violates NDD, and his gain over the benchmark is then financed by the receiver: each unit of persuasion above $\mathbb{P}^{\mathrm{BP}}$ costs her at least $K\theta$. The mechanism is the one of Section 6. The mass of states with $\omega_1 = \omega_1^+$ is fixed by the prior, so persuading more often than $\mathbb{P}^{\mathrm{BP}}$ means that the persuading realizations have, on average, an objective posterior below θ. The receiver takes a_2 on them because her subjective posterior is at θ, and she pays the difference. The last term in (9) is the probability that $\omega_1 = \omega_1^+$ and she keeps the default. It is zero, and the bound holds with equality, whenever the realizations at which she keeps the default are generated only by states with $\omega_1 = \omega_1^-$, which is the case at the experiments of Section 6 that approach $\overline{\mathbb{P}}(\theta)$. Under the assumptions of Proposition 4, her loss along these experiments therefore approaches $K\theta\,[\overline{\mathbb{P}}(\theta) - \mu_0(\omega_1^+)/\theta]$, and if $\theta \leq \min_c \bar{\theta}_c$ it can be brought arbitrarily close to $K\,(\theta - \mu_0(\omega_1^+))$, her loss from taking a_2 at the prior.

In the credit rating example at $\theta = 0.8$, the best methodology with two ratings of Figure 3 (Lemma 5(d)) persuades with probability 0.52 against $\mathbb{P}^{\mathrm{BP}} = 0.5$, and its low rating carries only defaulters, so the investor's loss is exactly $0.8 \times 0.02\, K = 0.016\, K$. Along the ratings that approach the bound $\overline{\mathbb{P}}(0.8) = 0.55$, her loss approaches $0.04\, K$. Below $\theta = 1/2$ she lends after every rating, and her loss is $K(\theta - 0.4)$: she behaves as if she had been told nothing and lent anyway.

By part **(ii)**, the receiver cannot detect the transfer. When the sender persuades her more often than $\mathbb{P}^{\mathrm{BP}}$, her subjective posteriors are on average too optimistic about ω_1^+, and her perceived payoff is at or above her no-information payoff whenever the default pays weakly more when $\omega_1 = \omega_1^+$, while her actual payoff is below it. In the credit rating example, where refraining pays 0 in both states and her subjective posterior sits at θ on every high rating, she values a rating that harms her at exactly zero: (10) holds with equality and its right-hand side is zero. If refraining paid more when the firm is solvent, she

would be willing to pay for such a rating.

When the No Confusion Principle holds, the receiver acts on correct posteriors about the payoff-relevant state attributes, and her misspecification only restricts what the sender can induce. When NDD fails, the sender's gain over the KG11 benchmark is a transfer from the receiver, which she cannot detect. The receiver is harmed only at experiments that violate NDD and persuade her more often than $\mathbb{P}^{BP}$.

10. Conclusion

This paper studies a persuasion problem in which the receiver interprets information through a subjective causal model. The sender chooses the experiment, but not the directed acyclic graph through which the receiver relates the signal to the state attributes: by assumption, she applies Bayes rule to a joint distribution that combines the independence relations of her causal model with the objective conditional distributions, and this distribution need not be the objective one. As a result, objective and subjective beliefs may differ, and the sender could persuade not only through what the receiver learns but also through how she interprets it. Whether he can do so depends on where she places the signal in her causal model.

The first main result is a Revelation Principle for persuasion under subjective causal models: the No Confusion Principle. When the sender cannot distort the receiver's prior, automatically so when the signal is a terminal node of the receiver's DAG, and under No Detectable Distortion (NDD) when the signal is a root node, any outcome attainable through an arbitrary experiment can also be attained by an experiment consistent with the receiver's causal model. The sender gains only from the information he provides, never from confusing the receiver about it. Optimal persuasion can then be studied as a constrained version of Bayesian persuasion, in which feasible experiments must respect the receiver's causal model, and the sender's value is at most the KG11 benchmark.

Building on this result, the paper characterizes how the structure of the receiver's causal model shapes the set of feasible posterior beliefs. When the signal is terminal, it can depend on the state only through the state attributes that the receiver regards as its direct causes. When the signal is a root, it can depend on the state only through its children and their other parents, and it must be independent of the state attributes that are not its descendants. Kamenica and Gentzkow (2011) is a special case, recovered when the receiver's causal model is observationally equivalent (Markov equivalent) to one in which the signal is caused by exactly the payoff-relevant state attributes: the causal model is then irrelevant, and the sender faces the same problem as in KG11. When the signal is terminal or has a single child, this condition is also necessary for irrelevance. Irrelevance therefore does not require the causal model to be correct, but only that its distortions do not reach the payoff-relevant state attributes.

These conclusions rest on prior distortion being ruled out. The second main result concerns what the sender can achieve when he is free to violate NDD. When the signal is a root and its child has other parents, the receiver weights each configuration of these co-parents by its prior probability at every signal realization, whatever the experiment. An experiment that makes the signal and the co-parents dependent therefore moves not only her posteriors but also the belief to which they average. With two actions, the sender's problem separates into one KG11 problem for each configuration of the co-

parents, at a threshold that the other configurations lower, and his value is the sum of their values. This value can strictly exceed the KG11 benchmark; it is in general approached but not attained, and approaching it can require more signal realizations than actions. The receiver's causal model therefore disciplines the sender under NDD and, without NDD, allows him to obtain more than the KG11 value, although even then it can leave him below the benchmark.

The two cases also differ in who is hurt by the causal model. Whenever the No Confusion Principle applies, the receiver's posteriors over the payoff-relevant state attributes are correct at the sender's optimum, she is never worse off than without information, and any loss relative to the KG11 benchmark falls on the sender. Without NDD, in the two-action problem, the sender's gain over the KG11 benchmark is a transfer from the receiver: she can end up below her no-information payoff, and she cannot detect it, since the payoff she expects from the experiment need not be below the one she expects without information. NDD thus separates persuasion that leaves the receiver unharmed from persuasion that hurts her.

Several questions are left for future work. First, in this paper I restrict attention to causal models in which the signal is a root or a terminal node; a signal with both parents and children would combine the two mechanisms. Second, NDD is imposed here as a restriction on the sender, motivated by a receiver who would find an experiment that distorts her prior implausible. A model in which she compares the distribution over state attributes that she forms with her prior, and reacts when they differ, would make the boundary between the two cases endogenous. Finally, I assume that the sender knows the receiver's causal model. How the sender should optimall design the experiment when he is uncertain about the receiver's causal model, or when he addresses receivers with different causal models, is an open question.

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A. Appendix: Proofs

To ease the notation, I denote with $\boldsymbol{\omega}_M = (\omega_i)_{i\in M}$ and $\boldsymbol{\omega}_{-M} = (\omega_i)_{i\in N_\omega\setminus M}$. Similarly, let $\Omega_M = \times_{i\in M}\Omega_i$ and $\Omega_{-M} = \times_{i\in N_\omega\setminus M}\Omega_i$. Finally $\sum_{\Omega_M}\cdot = \sum_{\boldsymbol{\omega}_M\in\Omega_M}\cdot$.

Throughout, conditional probabilities on null events follow the prior completion of Section 3: if $\mu(x,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = 0$, then $\mu(\omega_i|x,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = \mu_0(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$. Every factor of (3) is then a probability kernel, so a joint written as the right-hand side of (3) factorizes along R in the sense of (Lauritzen, 1996, Theorem 3.27) and is consistent with R.

Proof of Lemma 1

Fix $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$, $R \in \mathcal{R}_0$, and $\pi \in \mathcal{E}$, and let $\mu(\boldsymbol{\omega},x) = \pi(x|\boldsymbol{\omega})\mu_0(\boldsymbol{\omega})$ be the induced objective joint distribution. The objective marginal distribution of x is given by

$$\mu(x) = \sum_{\Omega}\pi(x|\boldsymbol{\omega})\mu_0(\boldsymbol{\omega}) = \sum_{\Omega_{\mathrm{pa}(0)}}\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\mu_0(\boldsymbol{\omega}_{\mathrm{pa}(0)}),$$

where $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) = \sum_{\Omega_{-\mathrm{pa}(0)}}\pi(x|\boldsymbol{\omega})\,\mu_0(\boldsymbol{\omega}_{-\mathrm{pa}(0)}|\boldsymbol{\omega}_{\mathrm{pa}(0)})$ is the objective likelihood of x given the parents of x, well-defined because μ_0 has full support. By (3), the subjective joint distribution is

$$\mu^R(\boldsymbol{\omega},x) = \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\prod_{i\in N_\omega}k_i(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}(i)}),$$

where $k_i(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}(i)}) = \mu_0(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ if $i \notin \mathrm{ch}(0)$, and $k_i(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}(i)}) = \mu(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)},x)$ otherwise.

I consider separately the cases in which x is a terminal node and x is a root node.

x is terminal. Since $\mathrm{ch}(0) = \emptyset$, k_j is the prior conditional $\mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ and $\mathrm{pa}(j) = \mathrm{pa}_\omega(j)$ for all $j \in N_\omega$. Because x has no children, every path between two state attributes that passes through x has x as a collider and is blocked by any conditioning set of state attributes, so the conditional independencies that R entails among the state attributes are exactly those of the DAG $R|_{N_\omega}$ obtained from R by deleting x. Since (R,μ_0) satisfies consistency, μ_0 is Markov with respect to $R|_{N_\omega}$ and factorizes as $\mu_0(\boldsymbol{\omega}) = \prod_{j\in N_\omega}\mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ (Lauritzen, 1996, Theorem 3.27). It follows that $\mu^R(\boldsymbol{\omega},x) = \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\,\mu_0(\boldsymbol{\omega})$, and

$$\mu^R(x) = \sum_{\Omega}\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\,\mu_0(\boldsymbol{\omega}) = \sum_{\Omega_{\mathrm{pa}(0)}}\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\,\mu_0(\boldsymbol{\omega}_{\mathrm{pa}(0)}) = \mu(x).$$

x is a root. Since $\mathrm{pa}(0) = \emptyset$, $\mu(x|\,\mathrm{pa}(0)) = \mu(x)$, and

$$\mu^R(x) = \mu(x)\sum_{\Omega}\prod_{i\in N_\omega}k_i(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}(i)}).$$

List N_ω in a causal order of R and sum the attributes out from the last to the first. When ω_i is summed out, every child of i has already been summed out, so ω_i appears only in its own kernel $k_i(\cdot|\boldsymbol{\omega}_{\mathrm{pa}(i)})$, which is a probability distribution on Ω_i: for $i \notin \mathrm{ch}(0)$ it is a prior conditional, and for $i \in \mathrm{ch}(0)$ it is $\mu(\cdot|x,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$, a probability distribution by the convention adopted in (3). Each summation returns one, so $\mu^R(x) = \mu(x)$ for all $x \in X$. This case does not use Assumption 1: when x is a root,

the receiver takes the distribution of the signal directly from the experiment, and $\mu^R(x) = \mu(x)$ for every $R \in \mathcal{R}$. ■

Remark A.1 . — *Under NDD, the receiver's posteriors about the payoff-relevant state attributes average back to the prior. Formally, for $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$, and $\pi \in \mathcal{E}_{\mathrm{NDD}}$*

$$\sum_{x \in X} \mu(x)\, \mu_x^R(\boldsymbol{\omega}_P) = \mu_0(\boldsymbol{\omega}_P) \qquad \textit{for all } \boldsymbol{\omega}_P \in \Omega_P.$$

Proof of Remark A.1

Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$, $R \in \mathcal{R}_0$, and let $\pi \in \mathcal{E}_{\mathrm{NDD}}$. Then, $\mu^R(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega})$ for all $\boldsymbol{\omega} \in \Omega$, and

$$\sum_{x \in X} \mu^R(x)\, \mu^R(\boldsymbol{\omega}|x) = \mu^R(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega}),$$

where $\mu^R(\boldsymbol{\omega}|x) = \mu^R(\boldsymbol{\omega}, x)/\mu^R(x)$ is well-defined because $\mu^R(x) = \mu(x) > 0$ by Lemma 1. Finally, fix $\boldsymbol{\omega}_P \in \Omega_P$ and sum both sides over $\boldsymbol{\omega}_{N_\omega \setminus P} \in \Omega_{N_\omega \setminus P}$:

$$\sum_{x \in X} \mu^R(x)\, \mu_x^R(\boldsymbol{\omega}_P) = \mu^R(\boldsymbol{\omega}_P) = \mu_0(\boldsymbol{\omega}_P).$$

Since $\mu^R(x) = \mu(x)$ for all x by Lemma 1,

$$\sum_{x \in X} \mu(x)\, \mu_x^R(\boldsymbol{\omega}_P) = \mu_0(\boldsymbol{\omega}_P).$$ ■

Proof of Lemma 2(a) and of 2(b)

(a) Sufficiency. Let $\pi \in \mathcal{E}_R$. I show that $\mu^R = \mu$. Write $\omega_0 = x$. For $j \in N_\omega \setminus \mathrm{ch}(0)$, $\mathrm{pa}(j) = \mathrm{pa}_\omega(j)$ contains only state attributes, and the objective marginal of μ over Ω is μ_0, so $\mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) = \mu(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$. Hence at every $(\boldsymbol{\omega}, x)$ at which $\mu(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) > 0$ for all $i \in \mathrm{ch}(0)$, (3) can be written as

$$\mu^R(\boldsymbol{\omega}, x) = \prod_{i \in N} \mu(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}(i)}). \tag{A.1}$$

Consistency of (R, μ) means that μ satisfies the Markov property with respect to R, hence factorizes as the right-hand side (Lauritzen, 1996, Theorem 3.27) at every configuration at which all conditioning events are positive, while $\mu(\boldsymbol{\omega}, x) = 0$ at any other configuration, since $\mu(\boldsymbol{\omega}, x) \leq \mu(\boldsymbol{\omega}_{\mathrm{pa}(i)})$ for every i. Hence $\mu^R(\boldsymbol{\omega}, x) = \mu(\boldsymbol{\omega}, x)$ wherever all conditioning events are positive, and $\mu^R(\boldsymbol{\omega}, x) \geq 0 = \mu(\boldsymbol{\omega}, x)$ elsewhere, so $\mu^R \geq \mu$ pointwise. Both are probability distributions on $\Omega \times X$: μ by construction, and μ^R because (3) is a product of one probability kernel per node of R, which sums to one when the nodes are summed out in reverse causal order. Two probability distributions ordered pointwise are equal. Thus $\mu^R = \mu$. In particular, $\mu_x^R(\boldsymbol{\omega}_P) = \mu^R(\boldsymbol{\omega}_P, x)/\mu^R(x) = \mu(\boldsymbol{\omega}_P, x)/\mu(x) = \mu_x(\boldsymbol{\omega}_P)$, since $\mu(x) > 0$ for $\pi \in \mathcal{E}$.

Necessity. Suppose $\mu^R = \mu$. I show that $\pi \in \mathcal{E}_R$. Every factor of (3) is a probability kernel for one node of R and depends on the other variables only through the parents of that node in R: $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})$ for x; $\mu(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ for $i \in \mathrm{ch}(0)$, whose parents are x and $\mathrm{pa}_\omega(i)$; and $\mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ for $j \in N_\omega \setminus \mathrm{ch}(0)$. This is also true at a configuration with $\mu(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = 0$, where the factor of i is the prior conditional $\mu_0(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$, since which of the two expressions applies depends only on

$(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$. Hence μ^R factorizes along R for every $\pi \in \mathcal{E}$, and therefore satisfies every independence relation implied by R (Lauritzen, 1996, Theorem 3.27). Since $\mu = \mu^R$, so does μ: (R, μ) satisfies consistency, that is, $\pi \in \mathcal{E}_R$.

(b) Let $\pi \in \mathcal{E}_R$. By sufficiency in 2(a), $\mu^R = \mu$, and the marginal of μ over the state is the prior. Hence $\mu^R(\boldsymbol{\omega}) = \mu(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega})$ for all $\boldsymbol{\omega} \in \Omega$, so $\pi \in \mathcal{E}_{\mathrm{NDD}}$. ■

Proof of Lemma 3

(i) Since $\mathrm{ch}(0) = \varnothing$, (3) reads

$$\mu^R(\boldsymbol{\omega}, x) = \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) \prod_{j \in N_\omega} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}). \tag{A.2}$$

Every factor is well-defined and is a probability distribution in its first argument: the prior conditionals because μ_0 has full support, and $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) = \mu(x, \boldsymbol{\omega}_{\mathrm{pa}(0)})/\mu_0(\boldsymbol{\omega}_{\mathrm{pa}(0)})$.

Let $R|_{N_\omega}$ be the DAG obtained from R by deleting x. Because x has no children, every path between two state attributes that passes through x has x as a collider; since x has no descendants, neither x nor a descendant of x belongs to any conditioning set $M \subseteq N_\omega$, so every such path is blocked by every M. Paths that avoid x are the same in R and in $R|_{N_\omega}$, and their blocking status is the same because the descendant sets of their colliders differ at most by $x \notin M$. Hence the conditional independencies that R entails among state attributes are exactly those of $R|_{N_\omega}$.

By Assumption 1, μ_0 satisfies every independence that $R|_{N_\omega}$ entails, that is, it is (globally) Markov with respect to $R|_{N_\omega}$. Since x is the child of no node, the parent set of each $j \in N_\omega$ in $R|_{N_\omega}$ is $\mathrm{pa}(j) = \mathrm{pa}_\omega(j)$, and the Markov property yields the factorization $\mu_0(\boldsymbol{\omega}) = \prod_{j \in N_\omega} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ (Lauritzen, 1996, Theorem 3.27).

Substituting into (A.2) gives $\mu^R(\boldsymbol{\omega}, x) = \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\, \mu_0(\boldsymbol{\omega})$. Summing over x and using the fact that $\sum_{x \in X} \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) = 1$ gives $\mu^R(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega})$. The equality $\mathcal{E}_{\mathrm{NDD}} = \mathcal{E}$ then follows from the definition of No Detectable Distortion.

(ii) Since x is a root, summing (3) over X gives the receiver's subjective prior:

$$\mu^R(\boldsymbol{\omega}) = \left[\sum_{x \in X} \mu(x) \prod_{i \in \mathrm{ch}(0)} \mu\big(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}\big) \right] \cdot \prod_{j \in N_\omega \setminus \mathrm{ch}(0)} \mu_0\big(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}\big). \tag{A.3}$$

NDD requires $\mu^R(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega})$ for all $\boldsymbol{\omega}$. Since μ_0 has full support, every factor $\mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ is strictly positive, and dividing (A.3) by their product gives the condition in the statement. For the single-child case, no path between two state attributes passes through x, since x is linked only to its child ω_{i^*}. The independencies that R entails among state attributes are therefore those of the DAG $R|_{N_\omega}$ obtained by deleting x, which again coincides with the latent projection $\Pi(R)$ because a single child leaves no bidirected edge, so by Assumption 1 μ_0 is Markov with respect to $R|_{N_\omega}$ and factorizes as $\mu_0(\boldsymbol{\omega}) = \prod_{j \in N_\omega} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$. Substituting into the right-hand side of that condition gives (A.4). ■

The following corollary relates NDD to an independence restriction when x has a single child, as in

R_c. In that case the condition of Lemma 3**(ii)** reads

$$\sum_{x\in X} \mu(x)\,\mu\big(\omega_{i^*}|x,\,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}\big) = \mu_0\big(\omega_{i^*}|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}\big) \qquad \forall\,(\omega_{i^*},\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}). \tag{A.4}$$

Corollary A.1 . — *When the signal has a single effect, the receiver treats the signal and the other causes of that effect as independent. A signal that is independent of them therefore never distorts her prior. The converse holds when different weights on the realizations always produce a different distribution of the effect. Her prior is then undistorted only if the weights she uses are the true ones, and they are only if the signal is independent of the other causes.*

Fix $R \in \mathcal{R}_0$ with $\mathrm{ch}(0) = \{i^*\}$, and let $\pi \in \mathcal{E}$.

- **(i)** *If $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$, then $\pi \in \mathcal{E}_{NDD}$.*
- **(ii)** *Suppose that $\{\mu(\cdot|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)},x)\}_{x\in X}$ are affinely independent for every $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$. Then π satisfies NDD if and only if $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$.*

In the objective joint distribution, the distribution of ω_{i^*} given x's co-parents is the average across signal realizations of the conditionals $\mu(\cdot|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)},x)$ weighted by $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$. The receiver, who treats x and $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$ as independent, weights them by $\mu(x)$ instead. NDD requires the two averages to coincide at every $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$. If $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$, the weights are equal and NDD holds. If the child conditionals are affinely independent, equal averages require equal weights, so NDD implies $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$; affine independence is possible only if $|X| \le |\Omega_{i^*}|$. If two realizations induce the same child conditional at every $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$, moving probability between them at one $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$ changes neither average but makes $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$ differ from $\mu(x)$: NDD holds and x is correlated with $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$. Under R_c, a rating that depends on liquidity alone is the extreme case: within each liquidity group every realization induces the prior conditional of solvency, so NDD holds although the rating is correlated with liquidity. NDD is thus weaker than independence.

Proof of Corollary A.1

Since the objective marginal over Ω is μ_0, for every $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$

$$\mu_0(\omega_{i^*}|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) = \mu(\omega_{i^*}|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) = \sum_{x\in X} \mu(x|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})\,\mu(\omega_{i^*}|x,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}),$$

so (A.4) is equivalent to

$$\sum_{x\in X} \underbrace{\big[\mu(x) - \mu(x|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})\big]}_{c_x}\,\mu(\cdot|x,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) = 0 \in \mathbb{R}^{\Omega_{i^*}} \qquad \forall\,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}. \tag{A.5}$$

(i) If $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$, then $c_x = 0$ for all $x \in X$.

(ii) Let $p_x = \mu(\cdot|x,\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$, so (A.5) becomes $\sum_x c_x p_x = 0$ with $\sum_x c_x = 0$. Affine independence implies that (A.5) holds only if $c_x = 0$ for all $x \in X$, that is $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$. Given that p_x lies in the $(|\Omega_{i^*}|-1)$-dimensional simplex, this is possible only if $|X| \le |\Omega_{i^*}|$. The converse is **(i)**.

■

Proof of Proposition 1

Let $\tilde{\mu}$ be the objective joint induced by $\tilde{\pi}$. In both cases, I construct π whose objective joint equals the receiver's subjective joint under $\tilde{\pi}$, $\mu = \tilde{\mu}^R$; the conclusion then follows from Lemma 2(a), since $\mu^R = \mu = \tilde{\mu}^R$ gives $\mu^R_x = \tilde{\mu}^R_x$, and $\mu(x) = \tilde{\mu}^R(x) = \tilde{\mu}(x) > 0$ by Lemma 1 applied to $\tilde{\pi}$, so that $\pi \in \mathcal{E}$ and the realization space is unchanged.

x **is terminal.** Define

$$\pi(x|\boldsymbol{\omega}) := \tilde{\mu}(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) = \sum_{\boldsymbol{\omega}_{-\mathrm{pa}(0)}} \tilde{\pi}(x|\boldsymbol{\omega})\, \mu_0(\boldsymbol{\omega}_{-\mathrm{pa}(0)}|\boldsymbol{\omega}_{\mathrm{pa}(0)}),$$

which is well-defined because μ_0 has full support, and is a probability distribution on X for every $\boldsymbol{\omega}$. Its objective joint is $\mu(\boldsymbol{\omega}, x) = \tilde{\mu}(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\mu_0(\boldsymbol{\omega}) = \tilde{\mu}^R(\boldsymbol{\omega}, x)$ because, by the proof of Lemma 3**(i)**,

$$\tilde{\mu}^R(\boldsymbol{\omega}, x) = \tilde{\mu}(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) \prod_{j\in N_\omega} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) \qquad \text{and} \qquad \mu_0(\boldsymbol{\omega}) = \prod_{j\in N_\omega} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}).$$

Hence $\mu(\boldsymbol{\omega}, x) = \tilde{\mu}(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) \prod_{j\in N_\omega} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ is a factorization of μ along R: the factor attached to node 0 is a kernel of $\boldsymbol{\omega}_{\mathrm{pa}(0)}$, and the factor attached to each $j \in N_\omega$ is a kernel of $\boldsymbol{\omega}_{\mathrm{pa}(j)}$, since $\mathrm{pa}(j) = \mathrm{pa}_\omega(j)$ when x is terminal. A distribution that factorizes along a DAG satisfies all the conditional independencies the DAG entails (Lauritzen, 1996, Theorem 3.27), so (R, μ) is consistent and $\pi \in \mathcal{E}_R$.

x **is a root.** Define, for every $\boldsymbol{\omega} \in \Omega$,

$$Q(\boldsymbol{\omega}) := \frac{\mu_0(\boldsymbol{\omega})}{\prod_{j\in N_\omega\setminus \mathrm{ch}(0)} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})} > 0, \qquad \pi(x|\boldsymbol{\omega}) := \frac{\tilde{\mu}(x) \prod_{i\in\mathrm{ch}(0)} \tilde{\mu}(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})}{Q(\boldsymbol{\omega})},$$

where, by the prior completion of (3), a factor $\tilde{\mu}(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ with $\tilde{\mu}(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = 0$ is the prior conditional $\mu_0(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$. The numerator is non-negative and, by Lemma 3**(ii)**, whose factors carry the same completion, $\tilde{\pi} \in \mathcal{E}_{\mathrm{NDD}}$ means exactly that it sums to $Q(\boldsymbol{\omega})$ over x; hence $\pi(\cdot|\boldsymbol{\omega}) \in \Delta(X)$ for every $\boldsymbol{\omega}$. The objective joint induced by π is

$$\mu(\boldsymbol{\omega}, x) = \pi(x|\boldsymbol{\omega})\mu_0(\boldsymbol{\omega}) = \tilde{\mu}(x) \prod_{i\in\mathrm{ch}(0)} \tilde{\mu}(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) \prod_{j\in N_\omega\setminus\mathrm{ch}(0)} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) = \tilde{\mu}^R(\boldsymbol{\omega}, x),$$

by (3) for a root x. The display has one factor per node of R, each conditioning on the node's parents: $\tilde{\mu}(x)$ for node 0 (with $\mathrm{pa}(0) = \emptyset$), $\tilde{\mu}(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ for $i \in \mathrm{ch}(0)$ (with $\mathrm{pa}(i) = \{0\} \cup \mathrm{pa}_\omega(i)$), and $\mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ for $j \notin \mathrm{ch}(0)$ (with $\mathrm{pa}(j) = \mathrm{pa}_\omega(j)$). Each factor is a probability distribution on its coordinate: the first and third by construction, the second because it is $\tilde{\mu}(\cdot|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ when $\tilde{\mu}(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) > 0$ and the prior conditional $\mu_0(\cdot|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ otherwise. Hence μ factorizes along R (Lauritzen, 1996, Theorem 3.27), (R, μ) is consistent, and $\pi \in \mathcal{E}_R$. ■

Proof of Lemma 4

Let $\tau \in \mathcal{B}_R$ be induced by $\pi \in \mathcal{E}_R$. By definition of $\mathcal{B}_R$, $\alpha_x = \mu(x)$ and $\mu_x = \mu^R_x$ for all $x \in X$, and by Lemma 2(a), $\mu^R_x = \mu(\cdot|x)$. Hence τ is the weighted vector of objective posteriors of π, and since the objective joint has the prior as its marginal over the state, $\sum_x \alpha_x \mu_x(\boldsymbol{\omega}_P) = \sum_x \mu(x)\, \mu(\boldsymbol{\omega}_P|x) = \mu(\boldsymbol{\omega}_P) = \mu_0(\boldsymbol{\omega}_P)$ for all $\boldsymbol{\omega}_P \in \Omega_P$. ■

Proof of Corollary 1

$\mathcal{E}_R \subseteq \mathcal{E}_{\text{NDD}}$ by Lemma 2(a), and every $\tilde{\pi} \in \mathcal{E}_{\text{NDD}}$ has a consistent implementation in $\mathcal{E}_R$ with the same marginal over X and the same subjective posteriors, hence the same value, by the No Confusion Principle; this gives the first equality. The second is **(i)** $\Leftrightarrow$ **(ii)** in Proposition 2. The uninformative experiment belongs to $\mathcal{E}_R$ and yields the sender's payoff at the prior, which gives the claim on the benefit from persuasion. The bound follows from the first two equalities and $\mathcal{B}_R \subseteq \mathcal{B}$ (Lemma 4), and $\mathcal{E}_{\text{NDD}} = \mathcal{E}$ for a terminal x by Lemma 3**(i)**. The last claim is Proposition 2**(iii)**. ■

Intermediate Result for Proof of Proposition 2

The reduction to straightforward experiments pools realizations, and pooling must preserve consistency. A *garbling* of π is a kernel $\xi\colon X \to \Delta(Y)$, and $\pi^{\xi}(y|\boldsymbol{\omega}) = \sum_x \xi(y|x)\pi(x|\boldsymbol{\omega})$; I take Y to be the set of realizations that π^{ξ} reaches with positive probability, so that $\pi^{\xi} \in \mathcal{E}$.

Lemma A.1 . — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$, and suppose that either x is terminal or x is a root with a single child. If $\pi \in \mathcal{E}_R$, then $\pi^{\xi} \in \mathcal{E}_R$ for every garbling ξ.*

Proof of Lemma A.1

Let $\pi \in \mathcal{E}_R$ with objective joint μ, $\xi\colon X \to \Delta(Y)$ be a garbling, and μ^{ξ} be the objective joint of π^{ξ}.

x is terminal. By Lemma B.3, $\pi(x|\boldsymbol{\omega}) = \pi(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})$, so $\pi^{\xi}(y|\boldsymbol{\omega}) = \sum_x \xi(y|x)\pi(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})$ depends on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_{\mathrm{pa}(0)}$, and $\pi^{\xi} \in \mathcal{E}_R$ by the same result.

x is a root with $\mathrm{ch}(0) = \{i^*\}$. Let $\mu^{+}(\boldsymbol{\omega}, x, y) = \mu(\boldsymbol{\omega}, x)\xi(y|x)$, whose $(\boldsymbol{\omega}, y)$-marginal is μ^{ξ}, and let R^{+} be R augmented with a node y and the single edge $x \to y$. Since (R, μ) is consistent, μ factorizes along R, so μ^{+} factorizes along R^{+} and satisfies every independence that R^{+} entails. I show that μ^{ξ} satisfies the local Markov condition of R with y in the role of node 0; the global Markov property, hence consistency, then follows (Lauritzen, 1996, Theorem 3.27). Write $\mathrm{nd}(i)$ for the non-descendants of i in R, excluding i, and note that in R^{+} the node x is linked to ω_{i^*} and y alone, and y to x alone. Since x is a root, it is never a collider on any path, so conditioning on its descendant y never opens a path through x.

Node 0. Required: $y \perp\!\!\!\perp \boldsymbol{\omega}_{\mathrm{nd}(0)}$. Every path from y starts $y \leftarrow x \to \omega_{i^*}$ and continues as a path from x in R; it either ends at a descendant of x or contains a node at which the direction changes from $\to$ to $\leftarrow$, a collider, which is unconditioned. Hence y is d-separated from $\boldsymbol{\omega}_{\mathrm{nd}(0)}$ in R^{+}.

Nodes $j \in N_\omega \setminus \{i^*\}$. Required: $\omega_j \perp\!\!\!\perp \boldsymbol{\omega}_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)} | \boldsymbol{\omega}_{\mathrm{pa}(j)}$ and, if $x \in \mathrm{nd}(j)$, $\omega_j \perp\!\!\!\perp y | \boldsymbol{\omega}_{\mathrm{pa}(j)}$, where $\mathrm{pa}(j) \subseteq N_\omega$. A path between state attributes cannot pass through y, which is linked only to x, so it is a path of R and is blocked by $\boldsymbol{\omega}_{\mathrm{pa}(j)}$ because R entails the same statement. A path from ω_j to y ends with $x \to y$, and its initial segment is a path from ω_j to x in R; since R entails $\omega_j \perp\!\!\!\perp x | \boldsymbol{\omega}_{\mathrm{pa}(j)}$, that segment is blocked at an interior node, which remains interior with the same collider status in the extended path.

Node i^*. Required: $\omega_{i^*} \perp\!\!\!\perp \boldsymbol{\omega}_{\mathrm{nd}(i^*)\setminus\mathrm{pa}_\omega(i^*)} | (y, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$. A path from ω_{i^*} through x can only continue to y, where it ends; so any path from ω_{i^*} to a state attribute avoids x and y, and is a path of R. In R it is blocked given $(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$; since x is not on the path, it is blocked by $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$ alone,

either at a conditioned non-collider or at an unconditioned collider none of whose descendants is conditioned. Replacing x by y in the conditioning set does not alter either fact, since y is a descendant of x only, and x is not on the path.

Thus μ^ξ satisfies the local Markov condition of R with y as node 0, and $\pi^\xi \in \mathcal{E}_R$. ■

Proof of Proposition 2

Throughout, for $\pi \in \mathcal{E}_R$ the value is $V(\pi) = \sum_{x\in X} \mu(x)\, \hat{v}(\mu_x^R)$ and $\mu_x^R = \mu_x$ by Lemma 2(a).

(i) $\Leftrightarrow$ **(ii)** An experiment $\pi \in \mathcal{E}_R$ with value v^* induces $\tau = (\mu(x), \mu_x^R)_{x\in X} \in \mathcal{B}_R$ with $\sum_x \alpha_x \hat{v}(\mu_x) = V(\pi) = v^*$. Conversely, the consistent experiment that induces any $\tau \in \mathcal{B}_R$ has value $\sum_x \alpha_x \hat{v}(\mu_x)$.

(iii) $\Rightarrow$ **(i)** is trivial.

(i) $\Rightarrow$ **(iii)** under the additional hypothesis. Let $\pi \in \mathcal{E}_R$ attain v^*. For each $a \in A$ set $X^a = \{x \in X : \hat{a}(\mu_x^R) = a\}$, let $A' = \{a \in A : X^a \neq \emptyset\}$, and define the garbling $\xi(a|x) = \mathbf{1}\{x \in X^a\}$ with realization space A'; set $\pi' = \pi^\xi$. By Lemma A.1, $\pi' \in \mathcal{E}_R$. Realization a has objective probability $\mu'(a) = \sum_{x\in X^a} \mu(x) > 0$ and carries the posterior $\mu^R_{\pi',a} = \sum_{x\in X^a} \frac{\mu(x)}{\mu'(a)}\, \mu_x^R$, a convex combination of posteriors at each of which a is optimal for the receiver; by linearity of expected utility in the posterior, a is optimal at $\mu^R_{\pi',a}$, and by the tie-breaking rule $\hat{a}(\mu^R_{\pi',a}) = a$. Hence π' is straightforward, and

$$V(\pi') = \sum_{a\in A'} \mu'(a)\, \hat{v}(\mu^R_{\pi',a}) = \sum_{a\in A'} \sum_{x\in X^a} \mu(x)\, v(a) = \sum_{x\in X} \mu(x)\, \hat{v}(\mu_x^R) = V(\pi) = v^*.$$

■

Lemma A.2 : Number of realizations. — *The sender has an optimal consistent experiment. It needs no more realizations than states, and no more realizations than actions when the signal is evidence or has a single effect. Formally, fix* $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ *and* $R \in \mathcal{R}_0$. *The supremum of* $V(\pi)$ *over* $\mathcal{E}_R$ *is attained, and by an experiment with at most* $|\Omega|$ *realizations. If* x *is terminal or has a single child, it is attained by an experiment with at most* $\min\{|A|, |\Omega|\}$ *realizations.*

Proof of Lemma A.2

Throughout, for $\pi \in \mathcal{E}$ with objective joint μ write $\alpha_x = \mu(x) > 0$ and $\nu_x = \mu(\cdot|x) \in \Delta(\Omega)$ for the full-state posterior at x, so that $\mu(\omega, x) = \alpha_x \nu_x(\omega)$ and $\sum_{x\in X} \alpha_x \nu_x = \mu_0$. For $\nu \in \Delta(\Omega)$, let $\nu^P \in \Delta(\Omega_P)$ denote its marginal on the payoff-relevant attributes. The proof has three steps. Consistency is a condition on each ν_x taken separately (Step 1). The sender then chooses weights on posteriors in a fixed set, and the argument behind Carathéodory's theorem shows that at most $|\Omega|$ posteriors are ever needed (Step 2); compactness gives attainment (Step 3).

Step 1: *Consistency is a condition on each posterior separately.* I claim that there is a closed set $S_R \subseteq \Delta(\Omega)$ such that, for every $\pi \in \mathcal{E}$, $\pi \in \mathcal{E}_R$ if and only if $\nu_x \in S_R$ for every $x \in X$. Conversely, given a finite X, weights $\alpha \in \Delta(X)$ with $\alpha_x > 0$ for every x, and a family $(\nu_x)_{x\in X}$ in S_R with $\sum_x \alpha_x \nu_x = \mu_0$, the kernel

$$\pi(x|\omega) = \frac{\alpha_x\, \nu_x(\omega)}{\mu_0(\omega)} \tag{A.6}$$

is well defined because μ_0 has full support, is a probability distribution on X for every $\boldsymbol{\omega}$ because $\sum_x \alpha_x \nu_x(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega})$, and induces the objective joint $\mu(\boldsymbol{\omega}, x) = \alpha_x \nu_x(\boldsymbol{\omega})$. Hence $\pi \in \mathcal{E}$ with $\mu(x) = \alpha_x$ and $\mu(\cdot|x) = \nu_x$, and $\pi \in \mathcal{E}_R$ by the claim.

x **terminal.** Let $S_R = \{\nu \in \Delta(\Omega) : \nu(\boldsymbol{\omega}) = \nu(\boldsymbol{\omega}_{\mathrm{pa}(0)})\, \mu_0(\boldsymbol{\omega}_{-\mathrm{pa}(0)}|\boldsymbol{\omega}_{\mathrm{pa}(0)}) \ \forall \boldsymbol{\omega} \in \Omega\}$, a closed set since it is cut out by linear equations. A distribution ν belongs to S_R if and only if $\nu(\boldsymbol{\omega})/\mu_0(\boldsymbol{\omega})$ depends on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_{\mathrm{pa}(0)}$: if $\nu(\boldsymbol{\omega}) = g(\boldsymbol{\omega}_{\mathrm{pa}(0)})\mu_0(\boldsymbol{\omega})$, summing over $\boldsymbol{\omega}_{-\mathrm{pa}(0)}$ gives $\nu(\boldsymbol{\omega}_{\mathrm{pa}(0)}) = g(\boldsymbol{\omega}_{\mathrm{pa}(0)})\mu_0(\boldsymbol{\omega}_{\mathrm{pa}(0)})$, hence $\nu(\boldsymbol{\omega}) = \nu(\boldsymbol{\omega}_{\mathrm{pa}(0)})\mu_0(\boldsymbol{\omega})/\mu_0(\boldsymbol{\omega}_{\mathrm{pa}(0)})$; conversely, if $\nu \in S_R$ then $\nu(\boldsymbol{\omega})/\mu_0(\boldsymbol{\omega}) = \nu(\boldsymbol{\omega}_{\mathrm{pa}(0)})/\mu_0(\boldsymbol{\omega}_{\mathrm{pa}(0)})$. Since $\pi(x|\boldsymbol{\omega}) = \alpha_x \nu_x(\boldsymbol{\omega})/\mu_0(\boldsymbol{\omega})$, the experiment depends on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_{\mathrm{pa}(0)}$ if and only if $\nu_x \in S_R$ for every x, and by Lemma B.3 the former is $\pi \in \mathcal{E}_R$.

x **is a root.** Let S_R be the set of $\nu \in \Delta(\Omega)$ of the form

$$\nu(\boldsymbol{\omega}) = \prod_{i \in \mathrm{ch}(0)} k_i(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) \prod_{j \in N_\omega \setminus \mathrm{ch}(0)} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) \qquad \forall \boldsymbol{\omega} \in \Omega, \tag{A.7}$$

for some family of probability kernels $k_i \colon \Omega_{\mathrm{pa}_\omega(i)} \to \Delta(\Omega_i), i \in \mathrm{ch}(0)$. The families of kernels form a product of simplices, a compact set, and (A.7) maps it continuously onto S_R; hence S_R is compact, in particular closed. Let $\pi \in \mathcal{E}_R$. Then (R, μ) is consistent, so μ factorizes along R (Lauritzen, 1996, Theorem 3.27):

$$\mu(\boldsymbol{\omega}, x) = \mu(x) \prod_{i \in \mathrm{ch}(0)} \mu(\omega_i|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) \prod_{j \in N_\omega \setminus \mathrm{ch}(0)} \mu(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) \qquad \forall (\boldsymbol{\omega}, x),$$

since the parents of 0, of $i \in \mathrm{ch}(0)$ and of $j \in N_\omega \setminus \mathrm{ch}(0)$ are $\emptyset$, $\{0\} \cup \mathrm{pa}_\omega(i)$ and $\mathrm{pa}_\omega(j)$ respectively. The identity holds whatever value a conditional on a null event is given, in particular under the prior completion: if $\mu(\boldsymbol{\omega}, x) = 0$, the first node in a topological order of R whose configuration, jointly with its predecessors, has probability zero has a parents' configuration of positive probability, and by the local Markov property its conditional given its parents equals its conditional given all its predecessors, which is zero. Since the Ω-marginal of μ is μ_0, $\mu(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) = \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$; dividing by $\mu(x) = \alpha_x > 0$ gives (A.7) for ν_x with $k_i(\cdot|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = \mu(\cdot|x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$, so $\nu_x \in S_R$. Conversely, if $\nu_x \in S_R$ for every x, with kernels $(k_i^x)_{i \in \mathrm{ch}(0)}$, then

$$\mu(\boldsymbol{\omega}, x) = \alpha_x \nu_x(\boldsymbol{\omega}) = \alpha_x \prod_{i \in \mathrm{ch}(0)} k_i^x(\omega_i|\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) \prod_{j \in N_\omega \setminus \mathrm{ch}(0)} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$$

is a product of probability kernels, one per node of R, each conditioning on the parents of its node. Hence μ factorizes along R, satisfies every independence that R entails (Lauritzen, 1996, Theorem 3.27), and $\pi \in \mathcal{E}_R$.

The set S_R is convex when x is terminal and when x is a root with a single child i^*, since (A.7) is then linear in the single kernel k_{i^*} and maps the convex set of kernels onto S_R. It is not convex in general when x has several children: for the common-cause DAG $\omega_1 \leftarrow x \rightarrow \omega_2$, S_R is the set of product distributions on $\Omega_1 \times \Omega_2$.

Step 2: *No experiment needs more than* $|\Omega|$ *realizations.* For $\pi \in \mathcal{E}_R$, $\mu_x^R = \mu_x = \nu_x^P$ by Lemma 2(a), so $V(\pi) = \sum_x \alpha_x \hat{v}(\nu_x^P)$. Fix $\pi \in \mathcal{E}_R$ with realizations $X = \{1, \dots, m\}$, so that $\nu_1, \dots, \nu_m \in S_R$ by Step 1, and let $\Lambda = \{\lambda \in \mathbb{R}_+^m : \sum_k \lambda_k \nu_k = \mu_0\}$. Summing the $|\Omega|$ equa-

tions that define Λ gives $\sum_k \lambda_k = 1$, so Λ is a compact subset of the unit simplex, and $\alpha \in \Lambda$. The function $\lambda \mapsto \sum_k \lambda_k \hat{v}(\nu_k^P)$ is linear, hence attains its maximum over Λ; among its maximizers, let λ^* have the fewest positive coordinates. I claim that λ^* has at most $|\Omega|$ positive coordinates. Otherwise the vectors ν_k with $\lambda_k^* > 0$, more than $|\Omega|$ of them in $\mathbb{R}^{|\Omega|}$, are linearly dependent: some $\delta \neq 0$ with $\delta_k = 0$ whenever $\lambda_k^* = 0$ satisfies $\sum_k \delta_k \nu_k = 0$, and summing the coordinates of this identity gives $\sum_k \delta_k = 0$, so δ has coordinates of both signs. Replacing δ by $-\delta$ if necessary, $\sum_k \delta_k \hat{v}(\nu_k^P) \geq 0$. Then $\lambda^* + s\delta \in \Lambda$ for $s \in [0, \bar{s}]$, where $\bar{s} > 0$ is the first s at which a coordinate of $\lambda^* + s\delta$ reaches zero, and the objective does not decrease along the way; so $\lambda^* + \bar{s}\delta$ is a maximizer with fewer positive coordinates than λ^*, a contradiction. Dropping the zero coordinates of λ^*, Step 1 gives an experiment $\pi' \in \mathcal{E}_R$ with at most $|\Omega|$ realizations and $V(\pi') = \sum_k \lambda_k^* \hat{v}(\nu_k^P) \geq V(\pi)$; if only one coordinate is positive, split the single realization into two of equal weight, which changes neither consistency nor the value, and $|\Omega| \geq 2$.

Step 3: *Attainment.* Let

$$\mathcal{K} = \left\{ (\lambda, \nu_1, \ldots, \nu_{|\Omega|}) \in \Delta(\{1, \ldots, |\Omega|\}) \times S_R^{|\Omega|} : \sum_k \lambda_k \nu_k = \mu_0 \right\}$$

and $f(\lambda, \nu) = \sum_k \lambda_k \hat{v}(\nu_k^P)$ on $\mathcal{K}$. Every point of $\mathcal{K}$, with its zero weights dropped, is an experiment in $\mathcal{E}_R$ with value $f(\lambda, \nu)$ (Step 1), and every $\pi \in \mathcal{E}_R$ is dominated by one with at most $|\Omega|$ realizations (Step 2), which is a point of $\mathcal{K}$ once its weight vector is padded with zeros and one of its posteriors is repeated in the unused slots. Hence $V_R^* = \sup_{\pi \in \mathcal{E}_R} V(\pi) = \sup_{\mathcal{K}} f$, where V_R^* is the value defined in Corollary 1, equal to the supremum over $\mathcal{E}_R$ by Proposition 1. The set $\mathcal{K}$ is a closed subset of a compact set, since S_R is closed, hence compact.

The function f is upper semicontinuous: $\hat{v}(\nu) = \max\{v(a) : a \in A^*(\nu)\}$ is bounded and upper semicontinuous because A^* is upper hemicontinuous, and for each k the term $\lambda_k \hat{v}(\nu_k^P)$ is upper semicontinuous, since along a convergent sequence it is bounded by λ_k times a bound on $|v|$ when the limit weight is zero, and inherits the upper semicontinuity of $\hat{v}$ when it is positive. An upper semicontinuous function attains its supremum on a compact set, so V_R^* is attained, by an experiment with at most $|\Omega|$ realizations.

Finally, if x is terminal or has a single child, garble the experiment just obtained by pooling realizations that recommend the same action, as in the proof of Proposition 2, (i) $\Rightarrow$ (ii). By Lemma A.1 the pooled experiment is in $\mathcal{E}_R$, it has the same value, and it has at most $\min\{|A|, |X|\} \leq \min\{|A|, |\Omega|\}$ realizations. ■

Proof of Lemma 5(a)

In what follows, I use y^z to denote the fact that the variable y takes value z.

The Markov boundary of x in R_a is $\mathrm{MB}_{R_a}(0) = \{1\}$. By Lemma B.3, the agency designs the rating on solvency alone, $(\pi(\cdot|\omega_1))_{\omega_1 \in \Omega_1}$, and objective and subjective posteriors coincide

$$\mu_x^{R_a}(\omega_1^s) = \frac{\pi(x|\omega_1^s)\mu_0(\omega_1^s)}{\mu(x)}.$$

The investor lends after $x = H$ only if $\mu_H^{R_a}(\omega_1^s) \geq \theta$, that is, only if $\pi(x^H|\omega_1^s)\, \mu_0(\omega_1^s) \geq \theta\, \mu(x^H)$.

Hence, for any experiment that persuades,

$$\mathbb{P}(\pi) = \mu(x^H) \leq \frac{\pi(x^H|\omega_1^s)\,\mu_0(\omega_1^s)}{\theta} \leq \frac{\mu_0(\omega_1^s)}{\theta} = \mathbb{P}^{\text{BP}},$$

with equality if and only if $\pi(x^H|\omega_1^s) = 1$.

Set $\pi(x^H|\omega_1^s) = 1$ and choose $\pi(x^H|\omega_1^d)$ so that $\mu_H^{R_a}(\omega_1^s) = \theta$, namely

$$\pi(x^H|\omega_1^d) = \frac{\mu_0(\omega_1^s)}{\mu_0(\omega_1^d)} \cdot \frac{1-\theta}{\theta} \in (0,1),$$

where the inclusion follows from $\theta \in (\mu_0(\omega_1^s), 1)$. This experiment is feasible and meets both equality conditions, so $\mathbb{P}^*_{R_a} = \mathbb{P}^{\text{BP}}$. Since every solvent firm is rated $x = H$, $\mu_L^{R_a}(\omega_1^s) = 0$.

Remark A.2 : Non-uniqueness of the optimal experiment. — *Under R_a, the sender's payoff depends on $\pi(\cdot|\omega_1)$ only, so there is a continuum of optimal Blackwell experiments: any π satisfying $\pi(x^H|\omega_1^s, \omega_2) = 1$ for all $\omega_2 \in \Omega_2$ and*

$$\sum_{\tilde{\omega}_2 \in \Omega_2} \pi(x^H|\omega_1^d, \tilde{\omega}_2)\,\mu_0(\tilde{\omega}_2|\omega_1^d) = \frac{\mu_0(\omega_1^s)}{\mu_0(\omega_1^d)} \cdot \frac{1-\theta}{\theta}$$

achieves the optimum. Equivalently, any $(\pi(x^H|\omega_1^d, \omega_2^\ell),\ \pi(x^H|\omega_1^d, \omega_2^i)) \in [0,1]^2$ satisfying

$$\pi(x^H|\omega_1^d, \omega_2^i) = \frac{1}{\mu_0(\omega_1^d, \omega_2^i)} \left(\frac{1-\theta}{\theta}\,\mu_0(\omega_1^s) - \pi(x^H|\omega_1^d, \omega_2^\ell)\,\mu_0(\omega_1^d, \omega_2^\ell) \right)$$

is optimal. This family is non-empty for $\pi(x^H|\omega_1^d, \omega_2^\ell) \in \left[\frac{\mu_0(\omega_1^s)}{\theta\,\mu_0(\omega_1^d, \omega_2^\ell)} - \frac{1-\mu_0(\omega_1^d, \omega_2^\ell)}{\mu_0(\omega_1^d, \omega_2^\ell)},\ \frac{1-\theta}{\theta} \frac{\mu_0(\omega_1^s)}{\mu_0(\omega_1^d, \omega_2^\ell)} \right]$. The implied marginal rating rate for liquidity type ω_2 is

$$\mu(x^H|\omega_2) = \mu_0(\omega_1^s|\omega_2) + \pi(x^H|\omega_1^d, \omega_2)\,\mu_0(\omega_1^d|\omega_2).$$

These rates need not coincide across liquidity groups under R_a, which is what distinguishes $\mathcal{E}_{R_a}$ and $\mathcal{E}_{R_c}$. ■

Proof of Lemma 5(b)

In what follows, I use y^z to denote the fact that the variable y takes value z.

The Markov boundary of x in R_b is $\text{MB}_{R_b}(0) = \{2\}$, so by Lemma B.3 the agency designs the rating on liquidity alone, $(\pi(\cdot|\omega_2))_{\omega_2 \in \Omega_2}$, and subjective and objective posteriors coincide. To ease notation, I denote the probability that an x-rated firm is liquid as $q_x \equiv \mu(\omega_2^\ell|x) = \pi(x|\omega_2^\ell)\,\mu_0(\omega_2^\ell)/\mu(x)$. The investor's posterior on solvency is

$$\mu_x^{R_b}(\omega_1^s) = q_x\,\mu_0(\omega_1^s|\omega_2^\ell) + (1-q_x)\,\mu_0(\omega_1^s|\omega_2^i), \tag{A.8}$$

a convex combination of the two prior conditional solvency rates: the rating moves her belief about solvency only by moving her belief about liquidity.

Since $q_x \leq 1$, (A.8) gives $\mu_H^{R_b}(\omega_1^s) \leq \mu_0(\omega_1^s|\omega_2^\ell)$ for every experiment. If $\theta > \mu_0(\omega_1^s|\omega_2^\ell)$, the investor never lends and persuasion is impossible.

Let $\theta \leq \mu_0(\omega_1^s|\omega_2^\ell)$. By (A.8), the investor lends after $x = H$ if and only if

$$q_H \;\geq\; \bar{q} \equiv \frac{\theta - \mu_0(\omega_1^s|\omega_2^i)}{\mu_0(\omega_1^s|\omega_2^\ell) - \mu_0(\omega_1^s|\omega_2^i)} \in (\mu_0(\omega_2^\ell), 1],$$

since $\theta \in (\mu_0(\omega_1^s), \mu_0(\omega_1^s|\omega_2^\ell)]$. The agency thus faces the problem of Lemma 5(a) with liquidity in place of solvency and $\bar{q}$ in place of θ, and the argument is the same. For any experiment that persuades,

$$\mathbb{P}(\pi) = \mu(x^H) \leq \frac{\pi(x^H|\omega_2^\ell)\,\mu_0(\omega_2^\ell)}{\bar{q}} \leq \frac{\mu_0(\omega_2^\ell)}{\bar{q}},$$

with equality if and only if $\pi(x^H|\omega_2^\ell) = 1$, for which $q_H = \bar{q}$ and $\mu_H^{R_b}(\omega_1^s) = \theta$. Both are attained by $\pi(x^H|\omega_2^\ell) = 1$ and

$$\pi(x^H|\omega_2^i) = \frac{1-\bar{q}}{\bar{q}} \cdot \frac{\mu_0(\omega_2^\ell)}{\mu_0(\omega_2^i)} = \frac{\mu_0(\omega_2^\ell)}{\mu_0(\omega_2^i)} \cdot \frac{\mu_0(\omega_1^s|\omega_2^\ell) - \theta}{\theta - \mu_0(\omega_1^s|\omega_2^i)} \in [0, 1),$$

where the inclusion follows from $\bar{q} \in (\mu_0(\omega_2^\ell), 1]$. Since every liquid firm is rated $x = H$, a low-rated firm is illiquid for sure, $q_L = 0$, and (A.8) gives $\mu_L^{R_b}(\omega_1^s) = \mu_0(\omega_1^s|\omega_2^i)$. The value is

$$\mathbb{P}^*_{R_b} = \frac{\mu_0(\omega_2^\ell)}{\bar{q}} = \frac{\mu_0(\omega_1^s) - \mu_0(\omega_1^s|\omega_2^i)}{\theta - \mu_0(\omega_1^s|\omega_2^i)}.$$

Finally, subtracting $\mathbb{P}^*_{R_b}$ from $\mathbb{P}^{\text{BP}} = \mu_0(\omega_1^s)/\theta$,

$$\mathbb{P}^{\text{BP}} - \mathbb{P}^*_{R_b} = \frac{\mu_0(\omega_1^s|\omega_2^i)\,\big(\theta - \mu_0(\omega_1^s)\big)}{\theta\,\big(\theta - \mu_0(\omega_1^s|\omega_2^i)\big)} > 0,$$

since $\theta > \mu_0(\omega_1^s) > \mu_0(\omega_1^s|\omega_2^i)$. ■

Proof of Lemma 5(c)

In what follows, I use y^z to denote the fact that the variable y takes value z.

Since R_c encodes $x \perp\!\!\!\perp_{R_c} \omega_2$, any $\pi \in \mathcal{E}_{R_c}$ must induce μ such that

$$\mu(x|\omega_2^\ell) = \mu(x|\omega_2^i) = \mu(x) \quad \text{for all } x \in X, \tag{A.9}$$

where $\mu(x|\omega_2) = \sum_{\tilde{\omega}_1 \in \Omega_1} \pi(x|\tilde{\omega}_1, \omega_2)\,\mu_0(\tilde{\omega}_1|\omega_2)$. Under (A.9), $\mu(\omega_2|x) = \mu_0(\omega_2)$, so subjective and objective posteriors coincide and the investor lends after $x = H$ if and only if

$$\theta\,\mu(x^H) \;\leq\; \pi(x^H|\omega_1^s, \omega_2^\ell)\,\mu_0(\omega_1^s, \omega_2^\ell) + \pi(x^H|\omega_1^s, \omega_2^i)\,\mu_0(\omega_1^s, \omega_2^i), \tag{A.10}$$

The agency maximizes $\mu(x^H)$ subject to (A.9) and (A.10), which binds at any optimum. Thus, $\mu_H^{R_c}(\omega_1^s) = \theta$.

The right-hand side of (A.10) is at most $\mu_0(\omega_1^s)$, so $\mu(x^H) \leq \mu_0(\omega_1^s)/\theta = \mathbb{P}^{\text{BP}}$, with equality if and only if every solvent firm is rated $x = H$. That in turn requires $\mu(x^H|\omega_2) \geq \mu_0(\omega_1^s|\omega_2)$ for each group, hence by (A.9) $\mu(x^H) \geq \mu_0(\omega_1^s|\omega_2^\ell)$. The Bayesian persuasion outcome is therefore attainable only if $\mu_0(\omega_1^s|\omega_2^\ell) \leq \mu_0(\omega_1^s)/\theta$, that is, $\theta \leq \tilde{\theta}$.

Case $\theta \leq \tilde{\theta}$. Let

$$\pi(x^H|\omega_1^s, \omega_2) = 1, \qquad \pi(x^H|\omega_1^d, \omega_2) = \frac{\mu_0(\omega_1^s)/\theta - \mu_0(\omega_1^s|\omega_2)}{\mu_0(\omega_1^d|\omega_2)} \qquad \text{for all } \omega_2 \in \Omega_2.$$

The second probability is nonnegative because $\theta \leq \tilde{\theta}$ and $\mu_0(\omega_1^s|\omega_2^\ell) > \mu_0(\omega_1^s|\omega_2^i)$, and at most one because $\mu_0(\omega_1^s)/\theta < 1$. By construction $\mu(x^H|\omega_2) = \mu_0(\omega_1^s)/\theta$ for both groups, so (A.9) holds, and every solvent firm is rated x^H, so (A.10) binds and $\mu_L^{R_c}(\omega_1^s) = 0$. Hence $\mathbb{P}^*_{R_c} = \mathbb{P}^{\text{BP}}$.

Case $\theta > \tilde{\theta}$. Now some solvent liquid firms must be left out, and the binding limit is not the stock of solvent firms but the liquid group's share of the pool. By (A.9), the pool contains $\mu(x^H)\,\mu_0(\omega_2^\ell)$ liquid firms, so the first term on the right of (A.10) is at most $\mu(x^H)\,\mu_0(\omega_2^\ell)$; the second is at most $\mu_0(\omega_1^s, \omega_2^i)$. Since $\tilde{\theta} = \mu_0(\omega_2^\ell) + \mu_0(\omega_1^s, \omega_2^i)/\mu_0(\omega_1^s|\omega_2^\ell) > \mu_0(\omega_2^\ell)$, we have $\theta > \mu_0(\omega_2^\ell)$ and (A.10) gives

$$\mu(x^H) \;\leq\; \bar{\mu} \;\equiv\; \frac{\mu_0(\omega_1^s, \omega_2^i)}{\theta - \mu_0(\omega_2^\ell)},$$

with equality if and only if every liquid firm in the pool is solvent and every solvent illiquid firm is in the pool. Both hold under

$$\pi(x^H|\omega_1^s, \omega_2^\ell) = \frac{\bar{\mu}}{\mu_0(\omega_1^s|\omega_2^\ell)}, \qquad \pi(x^H|\omega_1^d, \omega_2^\ell) = 0,$$

$$\pi(x^H|\omega_1^s, \omega_2^i) = 1, \qquad \pi(x^H|\omega_1^d, \omega_2^i) = \frac{\mu_0(\omega_1^s|\omega_2^i)\,(1-\theta)}{\mu_0(\omega_1^d|\omega_2^i)\,(\theta - \mu_0(\omega_2^\ell))},$$

which gives $\mu(x^H|\omega_2) = \bar{\mu}$ for both groups, so (A.9) holds. The two free probabilities lie in $[0,1]$: $\bar{\mu} \leq \mu_0(\omega_1^s|\omega_2^\ell)$ is $\theta \geq \tilde{\theta}$, and $\pi(x^H|\omega_1^d, \omega_2^i) \in [0,1]$ is $\mu_0(\omega_1^s|\omega_2^i) \leq \bar{\mu} \leq 1$, which follows from $\theta \leq 1$ and $\bar{\mu} < \mu_0(\omega_1^s)/\theta < 1$. Hence $\mathbb{P}^*_{R_c} = \bar{\mu}$.

Finally, $\bar{\mu} < \mathbb{P}^{\text{BP}}$: at π the pool holds $\bar{\mu}\,\mu_0(\omega_2^\ell) < \mu_0(\omega_1^s, \omega_2^\ell)$ solvent liquid firms, the inequality being $\theta > \tilde{\theta}$, so $\theta\bar{\mu} < \mu_0(\omega_1^s)$. Since objective posteriors average to the prior,

$$\mu_L^{R_c}(\omega_1^s) = \frac{\mu_0(\omega_1^s) - \theta\bar{\mu}}{1 - \bar{\mu}} > 0. \qquad \blacksquare$$

Proof of Lemma 5(d)

First, to derive the closed form of $\overline{\mathbb{P}}_{\text{bin}}(\theta)$ let

$$\mathbb{P}_c^{\text{all}}(\theta) \equiv \mu_0(\omega_2 = c) + \frac{\mu_0(\omega_2 = \bar{c})\,\mu_0(\omega_1 = s, \omega_2 = \bar{c})}{\theta - \mu_0(\omega_1 = s, \omega_2 = c)} \qquad \text{for } \theta \leq \bar{\theta}_c,$$

$$\mathbb{P}_c^{\text{none}}(\theta) \equiv \mu_0(\omega_1 = s, \omega_2 = c) + \frac{\mu_0(\omega_1 = s, \omega_2 = \bar{c})}{t_{\bar{c}}(\theta)} \qquad \text{for } \theta \geq \bar{\theta}_{\bar{c}}.$$

where $c \in \Omega_2$ is a liquidity group and $\bar{c}$ denotes the other group. Let $c^* = \arg\max_{c \in \Omega_2} \bar{\theta}_c$.

These are the probabilities with which two methodologies persuade the investor. Both rate high every solvent firm and, of group $\bar{c}$, as many defaulters as bring the investor's posterior after the high rating to θ. The first rates high every defaulter of group c, and the second none. The conditions on θ are those under which the methodology exists. For $\theta \geq \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, the largest probability with which a methodology with two ratings persuades the investor is

$$\overline{\mathbb{P}}_{\text{bin}}(\theta) = \begin{cases} \max\{\mathbb{P}_{c^*}^{\text{all}}(\theta),\, \mathbb{P}_{c^*}^{\text{none}}(\theta)\} & \text{if } \min\{\bar{\theta}_\ell, \bar{\theta}_i\} \leq \theta \leq \max\{\bar{\theta}_\ell, \bar{\theta}_i\}, \\ \max\{\mathbb{P}_\ell^{\text{none}}(\theta),\, \mathbb{P}_i^{\text{none}}(\theta)\} & \text{if } \theta \geq \max\{\bar{\theta}_\ell, \bar{\theta}_i\}. \end{cases} \tag{A.11}$$

I write $q_c \equiv \mu_0(s|c)$. By assumption, $q_\ell > q_i$ and $\mu_0(\ell)\, q_\ell + \mu_0(i)\, q_i = \mu_0(s) < \theta$. As in Section C.3, $p_x(c)$ is the solvency of the firms of group c that receive rating x, with $p_x(c) = q_c$ when no firm of group c receives x. The investor lends at x if and only if $\mu_0(\ell)\, p_x(\ell) + \mu_0(i)\, p_x(i) \geq \theta$. Recall that $\bar{\theta}_c = 1 - \mu_0(d,c) = \mu_0(s,c) + \mu_0(\bar{c})$.

Part (i). If the investor lends after both ratings of a methodology, its value is one, so that $\overline{\mathbb{P}}(\theta) = 1$ and the supremum over all methodologies is attained. By Proposition 4(i), this is the case if and only if $\theta < 1 - \max_c \mu_0(d,c) = \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, and a methodology with two ratings then attains it.

Part (ii). Let $\theta \geq \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$. A methodology after whose ratings the investor never lends has value zero. Otherwise, let H be the rating after which she lends and L the other one. For $p \in [0,1]$, let

$$M_c(p) \equiv \min\Big\{\frac{\mu_0(s,c)}{p},\ \frac{\mu_0(d,c)}{1-p}\Big\},$$

with $M_c(0) = \mu_0(d,c)$ and $M_c(1) = \mu_0(s,c)$. This is the largest mass of firms of group c whose solvency is p, since a mass g of them consists of $g\, p$ solvent firms and $g\,(1-p)$ defaulters. The function M_c is continuous, strictly increasing on $[0, q_c]$ and strictly decreasing on $[q_c, 1]$, and $M_c(q_c) = \mu_0(c)$. The firms of group c that receive H have mass at most $M_c(p_H(c))$, also when there are none. Hence the value of the methodology is at most

$$F(\theta) \equiv \max\Big\{\sum_{c\in\Omega_2} M_c(p_H(c)) :\ (p_H(\ell), p_H(i)) \in [0,1]^2,\ \sum_{c\in\Omega_2} \mu_0(c)\, p_H(c) \geq \theta\Big\}.$$

Let $(p_H(\ell), p_H(i))$ be a maximizer. It has three properties.

1. $\mu_0(\ell)\, p_H(\ell) + \mu_0(i)\, p_H(i) = \theta$. Indeed, since $\mu_0(\ell)\, q_\ell + \mu_0(i)\, q_i < \theta$, some group has $p_H(c) > q_c$, and if the inequality were strict, lowering $p_H(c)$ would raise $M_c(p_H(c))$.
2. $p_H(c) \geq q_c$ for both groups. If $p_H(c) < q_c$ for some group, then $p_H(\bar{c}) > q_{\bar{c}}$, and raising $p_H(c)$ while lowering $p_H(\bar{c})$, so that the constraint still holds with equality, would raise both $M_c(p_H(c))$ and $M_{\bar{c}}(p_H(\bar{c}))$.
3. $p_H(c) \in \{q_c, 1\}$ for some group. The points with the first two properties form a segment, on which $M_\ell(p_H(\ell)) + M_i(p_H(i)) = \mu_0(s,\ell)/p_H(\ell) + \mu_0(s,i)/p_H(i)$ is strictly convex. Thus, the maximizer is an endpoint of this segment, and at an endpoint some $p_H(c)$ is q_c or 1.

If $p_H(c) = q_c$, then $p_H(\bar{c}) = (\theta - \mu_0(s,c))/\mu_0(\bar{c})$, which exceeds $q_{\bar{c}}$ because $\theta > \mu_0(s)$, and which is at most one if and only if $\theta \leq \bar{\theta}_c$. The value is $\mu_0(c) + \mu_0(s,\bar{c})/p_H(\bar{c}) = \mathbb{P}_c^{\text{all}}(\theta)$. If $p_H(c) = 1$, then $p_H(\bar{c}) = (\theta - \mu_0(c))/\mu_0(\bar{c})$, which is below one, and which is at least $q_{\bar{c}}$ if and only if $\theta \geq \bar{\theta}_{\bar{c}}$; it is then equal to $t_{\bar{c}}(\theta)$. The value is $\mu_0(s,c) + \mu_0(s,\bar{c})/t_{\bar{c}}(\theta) = \mathbb{P}_c^{\text{none}}(\theta)$. Hence $F(\theta)$ is the right-hand side of (A.11).

It remains to show that $F(\theta)$ is attained. Take a maximizer $(p_H(\ell), p_H(i))$. Let H go to every solvent firm and, in each group, to a mass $\mu_0(s,c)\,(1-p_H(c))/p_H(c)$ of defaulters, which is at most $\mu_0(d,c)$ because $p_H(c) \geq q_c$, and let L go to the other defaulters. The firms of group c that receive H have solvency $p_H(c)$ and mass $M_c(p_H(c))$, so the investor's posterior after H is θ, and H has probability $F(\theta)$. Some defaulter receives L: otherwise $p_H(c) = q_c$ for both groups, and her posterior after H would be $\mu_0(s) < \theta$. By the first claim of this part, she does not lend after L, so the value of the methodology is $F(\theta)$. If $p_H(c) = q_c$, every defaulter of group c is rated high, and if $p_H(c) = 1$,

none is.

Finally, by Proposition 4**(i)**, no methodology attains $\overline{\mathbb{P}}(\theta)$, whereas $\overline{\mathbb{P}}_{\text{bin}}(\theta)$ is attained, so $\overline{\mathbb{P}}_{\text{bin}}(\theta) < \overline{\mathbb{P}}(\theta)$.

Part (iii). If $\theta < \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$, then $\overline{\mathbb{P}}_{\text{bin}}(\theta) = 1 > \mathbb{P}^{\text{BP}}$. Let $\theta \geq \min\{\bar{\theta}_\ell, \bar{\theta}_i\}$. Recalling that $\mathbb{P}^{\text{BP}} = \mu_0(s)/\theta$, the agency's gain (or loss) is

$$\mathbb{P}_c^{\text{all}}(\theta) - \mathbb{P}^{\text{BP}} = \frac{\mu_0(c)\,(\theta - q_c)\,(\theta - \mu_0(s))}{\theta\,(\theta - \mu_0(s,c))},$$

$$\mathbb{P}_c^{\text{none}}(\theta) - \mathbb{P}^{\text{BP}} = \frac{\mu_0(c)\,(1-\theta)\,(\mu_0(s) - q_c\,\theta)}{\theta\,(\theta - \mu_0(c))},$$

where the denominators are positive wherever the values are defined. Since $q_i < \mu_0(s) < \theta < 1$, both differences are positive for $c = i$. For $c = \ell$, the first has the sign of $\theta - q_\ell$, and the second has the sign of $\tilde{\theta} - \theta$, because $\mu_0(s) = q_\ell\,\tilde{\theta}$. Hence $\overline{\mathbb{P}}_{\text{bin}}(\theta) \leq \mathbb{P}^{\text{BP}}$ if and only if neither value is defined for $c = i$, that is, $\bar{\theta}_i < \theta < \bar{\theta}_\ell$, and $\tilde{\theta} \leq \theta \leq q_\ell$; both values are then defined for $c = \ell$. The first condition follows from the second, because $\tilde{\theta} = \mu_0(\ell) + \mu_0(s,i)/q_\ell > \mu_0(\ell) + \mu_0(s,i) = \bar{\theta}_i$ and $\bar{\theta}_\ell - q_\ell = \mu_0(i)\,(1 - q_\ell) > 0$. The inequality $\overline{\mathbb{P}}_{\text{bin}}(\theta) \leq \mathbb{P}^{\text{BP}}$ is strict if and only if $\theta \in (\tilde{\theta}, q_\ell)$. ■

B. Appendix: Consistency

B.1. Consistency

The notion of consistency is weaker than *faithfulness*, which requires every statistical independence in μ to be encoded in the DAG, letting the absence of correlation rule out causal links. Consistency imposes only the converse: the DAG must not assert independencies that μ contradicts.

Consistency between a DAG $R \in \mathcal{R}$ and a joint distribution μ can be verified using *d-separation*. D-separation identifies whether two variables become independent when conditioning on a third set of variables by analyzing how paths between variables are blocked.

A node i *blocks* a path if one of the following occurs. First, if the path contains a segment of the form $k \to i \to j$ or $k \leftarrow i \to j$, then conditioning on i removes any dependence between k and j. Second, if the path contains a *collider* $k \to m \leftarrow j$, then k and j are independent unless we condition on m or its descendants, which instead creates a dependence. Thus, if $i \neq m$ and i is not a descendant of m, conditioning on i preserves the independence between k and j. Formally:

Definition B.1 : D-Separation. — *Let $N_1, N_2, N_3 \subseteq N$ be pairwise disjoint. The sets N_1 and N_2 are d-separated given N_3, denoted $(d\text{-}sep_R(N_1; N_2|N_3))$, if N_3 blocks every path between any $j \in N_1$ and any $k \in N_2$.*

I write $\boldsymbol{\omega}_{N_1} \perp\!\!\!\perp_R \boldsymbol{\omega}_{N_2} \mid \boldsymbol{\omega}_{N_3}$ when N_1 and N_2 are d-separated given N_3 in R, and say that R implies this independence relation.

D-separation is the operational content of "implied by R" in the definition of consistency: the conditional independencies a DAG entails are exactly those obtained by d-separation (Pearl, 2009, Theorem 1.2.4). Consistency therefore reduces to a graphical test.

Lemma B.1 . — *Let $\mu \in \Delta(\Omega_M)$ and $R = (N, E)$ with $M \subseteq N$. Then (R, μ) satisfies consistency if and only if, for all pairwise disjoint $N_1, N_2, N_3 \subseteq M$,*

$$d\text{-}sep_R(N_1; N_2 \mid N_3) \quad \Longrightarrow \quad \boldsymbol{\omega}_{N_1} \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{N_2} \mid \boldsymbol{\omega}_{N_3}.$$

The converse implication is not required: μ may satisfy independencies that R does not entail. When $M = N$, the condition is the global Markov property of μ with respect to R, and it holds if and only if μ factorizes along R (Lauritzen, 1996, Theorem 3.27).

Example: Consistency. — *Let $N = \{0, 1, 2\}$ and $M = \{1, 2\}$. Let $\mu \in \operatorname{int}(\Delta(\Omega_M))$ be such that $\omega_1 \not\perp\!\!\!\perp_\mu \omega_2$. Consider R' : $\omega_1 \leftarrow \omega_0 \to \omega_2$, where ω_0 is a common cause of ω_1 and ω_2. In this DAG, conditioning on ω_0 blocks the path between ω_1 and ω_2, implying $\omega_1 \perp\!\!\!\perp_{R'} \omega_2|\omega_0$, but $\omega_1 \not\perp\!\!\!\perp_{R'} \omega_2$. R' is consistent with μ' because the (unconditional) dependence between ω_1 and ω_2 is encoded in R'. Furthermore, the conditional independence between ω_1 and ω_2 given ω_0 arises only due to the inclusion of ω_0. Since $\omega_0 \notin \operatorname{supp}(\mu)$, this does not contradict consistency between μ and R'. Let now the causal model be such that R'' : $\omega_1 \to \omega_0 \leftarrow \omega_2$. In this case, $\omega_1 \perp\!\!\!\perp_{R''} \omega_2$, but $\omega_1 \not\perp\!\!\!\perp_{R''} \omega_2|\omega_0$. Thus, (R'', μ) violates consistency, as ω_1 and ω_2 are independent according to R'' but not to μ, and dependency arises only because of the inclusion of ω_0.*

B.2. Prior-free Consistency

Because consistency only constrains the independencies that R *entails* among the state attributes, a causal model entailing none of them is consistent with every prior. This identifies the maximal causal model referred to in Section 4.1 and Section 7.

Lemma B.2 : Prior-free consistency. — *Let $R \in \mathcal{R}$, and let the* latent projection $\Pi(R)$ *of R onto N_ω be the graph over N_ω that keeps the edges of R among state attributes and, when x is a root, adds a bidirected edge $\omega_i \leftrightarrow \omega_j$ between every two children i, j of x. It implies exactly the independence relations that R implies among the state attributes. Call a graph over N_ω* fully connected *if every two state attributes are joined by an edge of either kind.*

- **(i)** *If R restricted to N_ω is fully connected, then $R \in \mathcal{R}_0$ for every $\mu_0 \in \operatorname{int}(\Delta(\Omega))$.*
- **(ii)** *$R \in \mathcal{R}_0$ for every $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ if and only if $\Pi(R)$ is fully connected.*
- **(iii)** *If x is terminal, $\Pi(R) = R|_{N_\omega}$, so* **(i)** *and* **(ii)** *coincide: $R \in \mathcal{R}_0$ for all priors if and only if $R|_{N_\omega}$ is fully connected.*

Two aspects of this lemma are worth noticing. First, the converse in **(ii)** is stated for the latent projection and not for $R|_{N_\omega}$. If x is a root with two children i, j that are not linked to each other, marginalizing x leaves the bidirected edge $\omega_i \leftrightarrow \omega_j$, so no independence between ω_i and ω_j is entailed and $R \in \mathcal{R}_0$ for every prior, even though $R|_{N_\omega}$ is not fully connected. This is the same phenomenon as in the example above, and it is the source of the asymmetry between the terminal and the root regime in Section 7. Second, let μ_0 be a prior not entailing any conditional independence relation among state attributes. Reading **(iii)** in the other direction, for such a prior and terminal x every $R \in \mathcal{R}_0$ is fully connected over the state attributes: all the variation across admissible causal models then lies in how x is linked to the state attributes. This is the sense in which the conditions of Proposition 5 are minimal.

B.3. Consistent Experiments

The characterization of $\mathcal{E}_R$ is determined by the structure of R, which determines the relevant conditioning variables. The key concept is the *Markov boundary* of x in R: a minimal set of variables that, when conditioned on, renders x independent of all remaining state attributes.

Definition B.2 : Markov Blanket and Markov Boundary. — *Let $R \in \mathcal{R}$. A* Markov blanket *of node i is any $M \subseteq N \setminus \{i\}$ such that $\omega_i \perp\!\!\!\perp_R \boldsymbol{\omega}_{N\setminus(M\cup\{i\})} | \boldsymbol{\omega}_M$. A* Markov boundary *is a minimal Markov blanket.*

For any DAG, the set $\mathrm{MB}_R(i) = \mathrm{pa}(i) \cup \mathrm{ch}(i) \cup \bigcup_{j\in\mathrm{ch}(i)} \mathrm{pa}(j) \setminus \{i\}$ of parents, children and co-parents of i is a Markov blanket of i (Neapolitan, 2004, Theorem 2.14), and it is the unique Markov boundary whenever the distribution is strictly positive.[7] The position of x determines what $\mathrm{MB}_R(0)$ contains and, through it, what consistency requires of the experiment.

[7] The objective joint μ need not be strictly positive, since π may assign probability zero to some realizations at some states. Nothing below relies on uniqueness: only the blanket property of $\mathrm{MB}_R(0)$ is used.

B.3.1. Terminal Signal: a Conditioning Restriction

For a terminal node, since x has no children, and hence no co-parents, $\mathrm{MB}_R(0) = \mathrm{pa}(0)$.

Lemma B.3 . — *When the signal is evidence, an experiment is consistent with the receiver's causal model if and only if it depends on the state only through the direct causes of the signal. Formally, fix* $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$ *and* $R \in \mathcal{R}_0$ *with* $\mathrm{ch}(0) = \emptyset$*. Then*

$$\mathcal{E}_R = \Big\{\pi \in \mathcal{E} : \ \pi(\cdot|\boldsymbol{\omega}) = \pi(\cdot|\boldsymbol{\omega}_{\mathrm{pa}(0)}) \ \forall\, \boldsymbol{\omega} \in \Omega\Big\}.$$

In the receiver's causal model, x is a direct consequence of its parents. She therefore updates her beliefs about $\boldsymbol{\omega}_{\mathrm{pa}(0)}$ only, and any additional conditioning in the experiment is discarded. The sender need only condition the experiment on the direct parents of x, and any experiment that also conditions on state attributes outside $\mathrm{pa}(0)$ can be replaced by one depending only on $\mathrm{pa}(0)$ that achieves the same payoff-relevant posteriors with the same frequencies (No Confusion Principle).

The lemma restricts the *form* of the experiment, not directly the outcomes it can generate. For a given realization space, the number of free parameters falls from $|\Omega|\,(|X| - 1)$ to $|\Omega_{\mathrm{pa}(0)}|\,(|X| - 1)$, a reduction that does not depend on the cardinality of X; but there is no further condition on the joint distribution of x and $\boldsymbol{\omega}_{\mathrm{pa}(0)}$. The receiver's causal model limits *which* variables the signal can be designed to depend on, not *how* it can depend on them.

B.3.2. Root Signal: Conditioning and Independence Restrictions

For a root node, $\mathrm{MB}_R(0) = \mathrm{ch}(0) \cup \bigcup_{i \in \mathrm{ch}(0)} \mathrm{pa}_\omega(i)$: it includes the direct children of x *and* their state-attribute co-parents, whereas for a terminal node it contained only the direct parents of x. Let $\mathrm{nd}(0) \subseteq N_\omega$ denote the set of state attributes that are not descendants of x in R.

Lemma B.4 . — *When the signal is a cause, a consistent experiment depends on the state only through the effects of the signal and their other causes, and it is independent of every state attribute that is not an effect of the signal, direct or indirect. When the signal has a single effect, these two conditions characterize consistency. Formally, fix* $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$ *and* $R \in \mathcal{R}_0$ *with* $\mathrm{pa}(0) = \emptyset$ *and* $\mathrm{ch}(0) \neq \emptyset$*. Then*

$$\mathcal{E}_R \ \subseteq \ \Big\{\pi \in \mathcal{E} : \ \pi(\cdot|\boldsymbol{\omega}) = \pi(\cdot|\boldsymbol{\omega}_{\mathrm{MB}_R(0)})\ \forall\, \boldsymbol{\omega} \in \Omega, \ \ x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{nd}(0)}\Big\},$$

with equality when x *has a single child. In particular,* $x \perp\!\!\!\perp_\mu \omega_j$ *for every co-parent* $j \in \mathrm{pa}_\omega(i)$*,* $i \in \mathrm{ch}(0)$*, that is not a descendant of* x*.*

Consistent experiments are thus subject to two distinct constraints. First, a *conditioning restriction*: the experiment depends on the state only through the full Markov boundary $\mathrm{MB}_R(0)$, which is larger here than in the terminal case because it includes the co-parents. Second, an *independence restriction*: the signal must be independent, in the objective distribution, of every state attribute that the receiver does not regard as an effect of x; in particular, of the co-parents of its children whenever they are not themselves descendants of x. When x has several children the inclusion is in general strict, because consistency also imposes the independencies R entails among state attributes *given* x, which involve the experiment: for the common-cause DAG $\omega_1 \leftarrow x \rightarrow \omega_2$, consistency requires $\omega_1 \perp\!\!\!\perp_\mu \omega_2 | x$.

This is the structural difference from the terminal case. In Section B.3.1, only the conditioning re-

striction was present: the experiment depends on the state only through $\mathrm{pa}(0)$, with no further conditions. Here, the causal model restricts not just *which* variables the signal depends on, but also *how* it can depend on them. The independence condition $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{nd}(0)}$ is a moment restriction on the data-generating process, not a restriction on the variables the experiment conditions on.

B.4. Markov-Equivalent Causal Models

Two DAGs are *Markov equivalent* when they encode the same conditional-independence relations among their nodes; by Verma and Pearl (1990), this holds if and only if they share the same *skeleton*, that is, the undirected graph obtained by ignoring edge directions, and the same *v-structures*, triples $a \to c \leftarrow b$ in which a and b are not linked. Markov-equivalent DAGs induce identical subjective beliefs in my model. I denote with $[R]$ the class of DAGs that are Markov equivalent to R (see Andersson, Madigan, and Perlman, 1997).

Lemma B.5 . — *Two causal models that imply the same independence relations lead the receiver to the same beliefs. Formally, let $R, R' \in \mathcal{R}$ be Markov equivalent. Then for every $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$ and $\pi \in \mathcal{E}$, $\mu^R = \mu^{R'}$. In particular, $R \in \mathcal{R}_0$ if and only if $R' \in \mathcal{R}_0$, and $\mathcal{E}_R = \mathcal{E}_{R'}$.*

By Lemma B.5, any R that is Markov equivalent to a DAG in which x is terminal reduces to the case in which x is terminal (Section 5.1). The next lemma identifies the conditions under which this happens.

Lemma B.6 . — *A causal model in which the signal is a cause leads to the same beliefs as one in which it is evidence if and only if the signal is not one of two independent causes of a state attribute and the state attributes it causes are linked to each other. Formally, let $R \in \mathcal{R}$ have $\mathrm{pa}(0) = \emptyset$ and $\mathrm{ch}(0) \neq \emptyset$. There exists a DAG R' Markov equivalent to R in which x is terminal if and only if*

- **(i)** *x belongs to no v-structure, that is $\mathrm{pa}_\omega(i) \subseteq \mathrm{ch}(0)$ for every $i \in \mathrm{ch}(0)$; and*
- **(ii)** *the children of x are pairwise linked.*

Both kinds of root DAGs that fail these conditions, those in which x takes part in a v-structure and those in which x is a common cause of two state attributes that are not linked, are consistent with every full-support prior, since neither entails an independence among state attributes once x is marginalized out (see Lemma B.2).

B.5. Proofs

Proof of Lemma B.2

(i) Suppose $R|_{N_\omega}$ is fully connected. A conditional independence $\omega_i \perp\!\!\!\perp_R \omega_j | \boldsymbol{\omega}_M$ obtained by d-separation requires i and j not to be linked, since no conditioning set blocks a path consisting of a single edge. As every pair in N_ω is linked, R entails no conditional independence among the state attributes and consistency of (R, μ_0) holds vacuously for every $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$.

(ii) Consistency of (R, μ_0) concerns only relations among the variables in N_ω, that is, relations entailed by the latent projection $\Pi(R)$. If $\Pi(R)$ is fully connected, the argument in **(i)** applies. Conversely, suppose some pair $i, j \in N_\omega$ is not linked in $\Pi(R)$. Then $\Pi(R)$ entails $\omega_i \perp\!\!\!\perp \omega_j | \boldsymbol{\omega}_M$ for some $M \subseteq N_\omega \setminus \{i, j\}$: taking M to be the set of parents of whichever of the two is not a descen-

dant of the other blocks every path between them. Since μ_0 ranges over the interior of the simplex and $|\Omega_i|, |\Omega_j| \geq 2$, there exists $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ violating this relation, and for such a prior $R \notin \mathcal{R}_0$.

(iii) If $\mathrm{ch}(0) = \emptyset$, marginalizing the terminal node x deletes it without creating any bidirected edge, so $\Pi(R) = R|_{N_\omega}$ and **(i)** and **(ii)** coincide. ■

Proof of Lemma B.3

($\subseteq$) Let $\pi \in \mathcal{E}_R$ and let μ be its objective joint. Since (R, μ) satisfies consistency and $\mathrm{pa}(0)$ is a Markov blanket of node 0, $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{N_\omega \setminus \mathrm{pa}(0)} | \boldsymbol{\omega}_{\mathrm{pa}(0)}$. As μ_0 has full support, every configuration $\boldsymbol{\omega}$ has positive probability, so $\pi(x|\boldsymbol{\omega}) = \mu(x|\boldsymbol{\omega}) = \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})$ for all $\boldsymbol{\omega}$.

($\supseteq$) Let $\pi \in \mathcal{E}$ depend on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_{\mathrm{pa}(0)}$. By (A.2) in the proof of Lemma 3**(i)**, $\mu(\boldsymbol{\omega}, x)$ is a factorization along R, hence (R, μ) satisfies consistency, as in Case 1 of the proof of Proposition 1. ■

Proof of Lemma B.4

Let $\pi \in \mathcal{E}_R$ with objective joint μ. Since $\mathrm{MB}_R(0)$ is a Markov blanket of node 0 and (R, μ) is consistent, $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{N_\omega \setminus \mathrm{MB}_R(0)} | \boldsymbol{\omega}_{\mathrm{MB}_R(0)}$; as μ_0 has full support, $\pi(x|\boldsymbol{\omega}) = \mu(x|\boldsymbol{\omega}) = \mu(x|\boldsymbol{\omega}_{\mathrm{MB}_R(0)})$. For the second property, the local Markov condition at the root x states that x is d-separated from its non-descendants by the empty set: any path leaving x does so along an edge $x \to \omega_i$, and either it is a directed path, which ends at a descendant, or it contains a node at which the direction changes from $\to$ to $\leftarrow$, which is a collider and blocks the path. Consistency then gives $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{nd}(0)}$. A co-parent $j \in \mathrm{pa}_\omega(i)$ with $j \in \mathrm{nd}(0)$ is covered by the joint independence.

For equality when $\mathrm{ch}(0) = \{i^*\}$, let π belong to the right-hand side and let μ be its objective joint. Since x is linked only to ω_{i^*}, no path between two state attributes passes through x, so the independencies that R entails among state attributes are those of $R|_{N_\omega}$, and by Assumption 1 $\mu_0(\boldsymbol{\omega}) = \prod_{j \in N_\omega} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)})$ (as in the proof of Lemma 3**(ii)**). Write $\mathrm{MB}_R(0) = \{i^*\} \cup \mathrm{pa}_\omega(i^*)$ and note that $\mathrm{pa}_\omega(i^*) \subseteq \mathrm{nd}(0)$, since a parent of i^* cannot be a descendant of i^*. Then

$$\mu(\boldsymbol{\omega}, x) = \pi(x | \omega_{i^*}, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})\, \mu_0(\omega_{i^*} | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) \prod_{j \in N_\omega \setminus \{i^*\}} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}).$$

Summing the identity $\mu(\boldsymbol{\omega}, x) = \pi(x|\boldsymbol{\omega}_{\mathrm{MB}_R(0)}) \mu_0(\boldsymbol{\omega})$ over $\boldsymbol{\omega}_{N_\omega \setminus \mathrm{MB}_R(0)}$ shows that $\mu(x, \boldsymbol{\omega}_{\mathrm{MB}_R(0)})$ equals $\pi(x|\boldsymbol{\omega}_{\mathrm{MB}_R(0)}) \mu_0(\boldsymbol{\omega}_{\mathrm{MB}_R(0)})$, so the product of the first two factors equals

$$\mu(x, \omega_{i^*} | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) = \mu(x | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})\, \mu(\omega_{i^*} | x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) = \mu(x)\, \mu(\omega_{i^*} | x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}),$$

where the last equality uses $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}$, a consequence of $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{nd}(0)}$. In particular, $\mu(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$ equals $\mu(x)\mu_0(\boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) > 0$, so $\mu(\cdot | x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$ is a probability distribution on Ω_{i^*} for every $(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)})$. Hence

$$\mu(\boldsymbol{\omega}, x) = \mu(x)\, \mu(\omega_{i^*} | x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i^*)}) \prod_{j \in N_\omega \setminus \{i^*\}} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}),$$

one kernel per node, each conditioning on the node's parents in R: $\mathrm{pa}(0) = \emptyset$, $\mathrm{pa}(i^*) = \{0\} \cup \mathrm{pa}_\omega(i^*)$, and $\mathrm{pa}(j) = \mathrm{pa}_\omega(j)$ for $j \neq i^*$. Thus μ factorizes along R, (R, μ) satisfies consistency (Lauritzen, 1996, Theorem 3.27), and $\pi \in \mathcal{E}_R$. The last display is (3), so in addition $\mu^R = \mu$.

■

Proof of Lemma B.5

Theorem 2 in Chickering (1995) shows that two Markov-equivalent DAGs are connected by a finite sequence of reversals of *covered* edges, that is, edges $i \to j$ with $\mathrm{pa}(j) = \mathrm{pa}(i) \cup \{i\}$, each intermediate DAG being Markov equivalent to R. It therefore suffices to show that (A.1) is invariant to the reversal of a covered edge $i \to j$. Let $S = \mathrm{pa}(i)$, so that $\mathrm{pa}(j) = S \cup \{i\}$, and after reversal $\mathrm{pa}(i)' = S \cup \{j\}$, $\mathrm{pa}(j)' = S$. Only the factors of i and j change, and

$$\mu(\omega_i|\boldsymbol{\omega}_S)\,\mu(\omega_j|\omega_i,\boldsymbol{\omega}_S) = \mu(\omega_i,\omega_j|\boldsymbol{\omega}_S) = \mu(\omega_j|\boldsymbol{\omega}_S)\,\mu(\omega_i|\omega_j,\boldsymbol{\omega}_S),$$

pointwise, including at configurations of measure zero. The remaining claims follow because consistency of (R,μ_0) and of (R,μ) depends only on the independencies R entails, which are shared by R'. ■

Proof of Lemma B.6

Fix an arbitrary $R \in \mathcal{R}$.

Necessity. Let R' be Markov equivalent to R with x terminal. If **(i)** fails, some child i has a parent $j \notin \mathrm{ch}(0)$; since x is a root, j is not a parent of x, so $x \to \omega_i \leftarrow \omega_j$ is a v-structure of R, hence of R', contradicting that x is terminal in R'. If **(ii)** fails, two children i, k of x are not linked; in R' both edges point into x, so $\omega_i \to x \leftarrow \omega_k$ is a v-structure of R' that is not a v-structure of R, a contradiction.

Sufficiency. Under **(ii)**, the subgraph of R induced by $\mathrm{ch}(0)$ is a complete DAG, so its nodes can be ordered $i_1,\dots,i_m$ with $i_k \to i_l$ whenever $k < l$. Reverse the edges $x \to \omega_{i_1},\dots,x \to \omega_{i_m}$ in this order. At step k, $\mathrm{pa}(x) = \{i_1,\dots,i_{k-1}\}$ and, by **(i)** and **(ii)**, $\mathrm{pa}(i_k) = \{0, i_1,\dots,i_{k-1}\} = \mathrm{pa}(x) \cup \{0\}$: the edge $x \to \omega_{i_k}$ is covered, and its reversal yields a Markov-equivalent DAG (Chickering, 1995). After step m, x has no children. ■

C. Appendix: No NDD

This appendix proves the results of Section 6 under its standing assumptions: $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and $R \in \mathcal{R}_0$, where x is a root with the single child i^* and the set $C = \text{pa}_\omega(i^*)$ is non-empty. Throughout, $J \equiv |\Omega_C| \geq 2$ is the number of groups.

C.1. Section 6.2: The Design Space

An experiment is described by the mass $\alpha_x(\boldsymbol{c}) \equiv \mu(\boldsymbol{\omega}_C = \boldsymbol{c}, x)$ of the states with $\boldsymbol{\omega}_C = \boldsymbol{c}$ that generate each realization x and by the within-group conditional $p_x(\boldsymbol{c}) \equiv \mu(\cdot|\boldsymbol{c}, x)$. The probability of realization x is $\alpha_x \equiv \sum_{\boldsymbol{c}} \alpha_x(\boldsymbol{c}) = \mu(x)$. The next lemma characterizes the families that some experiment induces. Since $\mu(x|\boldsymbol{c}) = \alpha_x(\boldsymbol{c})/\mu_0(\boldsymbol{c})$ is the probability of the conditional $p_x(\boldsymbol{c})$ in the experiment of group $\boldsymbol{c}$, its condition is the one of KG11 in each group.

Lemma C.1: Feasible experiments. — *Fix $\mu_0 \in \text{int}\,(\Delta(\Omega))$ and $R \in \mathcal{R}_0$ as in Section 6, and a finite set X. Let $(\alpha_x(\boldsymbol{c}), p_x(\boldsymbol{c}))_{x\in X, \boldsymbol{c}\in\Omega_C}$ satisfy $\alpha_x(\boldsymbol{c}) \geq 0$ and $\alpha_x > 0$ for every x, with $p_x(\boldsymbol{c}) \in \Delta(\Omega_{i^*})$ and $p_x(\boldsymbol{c}) = \mu_0(\cdot|\boldsymbol{c})$ whenever $\alpha_x(\boldsymbol{c}) = 0$. This family is induced by some $\pi \in \mathcal{E}$ if and only if*

$$\sum_{x\in X} \alpha_x(\boldsymbol{c}) p_x(\boldsymbol{c})(\omega_{i^*}) = \mu_0(\omega_{i^*}, \boldsymbol{c}) \qquad \forall \omega_{i^*} \in \Omega_{i^*},\ \forall \boldsymbol{c} \in \Omega_C, \tag{C.1}$$

that is, if and only if, for every group $\boldsymbol{c}$, the distributions $(p_x(\boldsymbol{c}))_{x\in X}$, weighted by $\alpha_x(\boldsymbol{c})/\mu_0(\boldsymbol{c})$, average back to $\mu_0(\cdot|\boldsymbol{c})$. The experiment that induces it can be taken to depend on $\boldsymbol{\omega}$ only through $(\omega_{i^*}, \boldsymbol{\omega}_C)$.

C.2. Section 6.3, Two Actions: Gain, Lowered Bar and Capacity

Let $A = \{a_1, a_2\}$ with $v(a_2) > v(a_1)$. Define the receiver's gain from choosing a_2 rather than a_1 at an arbitrary belief $q \in \Delta(\Omega_P)$ by

$$G(q) \equiv \bar{u}(a_2, q) - \bar{u}(a_1, q),$$

which is affine in q. Since a_1 is the default action, $G(\mu_0) < 0$. The receiver takes a_2 after realization x if and only if $G(\mu_x^R) \geq 0$, where ties are broken in favor of the sender. For any given Blackwell experiment $\pi \in \mathcal{E}$, let $X^+ \equiv \{x \in X : G(\mu_x^R) \geq 0\}$ denote the set of persuading realizations, so that $\mathbb{P}(\pi) = \sum_{x\in X^+} \mu(x)$.

By (4), the receiver's posterior over payoff-relevant attributes after x is

$$\mu_x^R(\boldsymbol{\omega}_P) = \sum_{\boldsymbol{c}} \mu_0(\boldsymbol{c}) \sum_{\omega_{i^*}} \mu(\omega_{i^*}|\boldsymbol{c}, x)\mu_0(\boldsymbol{\omega}_P|\omega_{i^*}, \boldsymbol{c}).$$

Given that G is affine, the receiver's gain at x can be written as $G(\mu_x^R) = \sum_{\boldsymbol{c}} \mu_0(\boldsymbol{c}) G_{\boldsymbol{c}}\big(\mu(\cdot|\boldsymbol{c}, x)\big)$, where, for any $p \in \Delta(\Omega_{i^*})$,

$$G_{\boldsymbol{c}}(p) \equiv \sum_{\omega_{i^*}\in\Omega_{i^*}} p(\omega_{i^*}) G\big(\mu_0(\boldsymbol{\omega}_P|\omega_{i^*}, \boldsymbol{c})\big). \tag{C.2}$$

Thus, $G\big(\mu_0(\boldsymbol{\omega}_P|\tilde{\omega}_{i^*}, \boldsymbol{c})\big)$ is the receiver's gain within group $\boldsymbol{c}$ when she is certain that $\boldsymbol{\omega}_C = \boldsymbol{c}$ and $\omega_1 = \tilde{\omega}_{i^*}$. The function $G_{\boldsymbol{c}}(p)$ is the within-group gain when ω_{i^*} is distributed according to p. With this notation, $\omega_{i^*}^{\boldsymbol{c}} \in \arg\max_{\omega_{i^*}\in\Omega_{i^*}} G\big(\mu_0(\boldsymbol{\omega}_P|\omega_{i^*}, \boldsymbol{c})\big)$, and I denote the maximal value by

$\overline{G}_{\boldsymbol{c}} = G\big(\mu_0(\boldsymbol{\omega}_P|\omega_{i^*}^{\boldsymbol{c}}, \boldsymbol{c})\big)$. Since $G_{\boldsymbol{c}}(p)$ is an average of the values $G\big(\mu_0(\boldsymbol{\omega}_P|\omega_{i^*}, \boldsymbol{c})\big)$, it satisfies $G_{\boldsymbol{c}}(p) \leq \overline{G}_{\boldsymbol{c}}$ for every $p \in \Delta(\Omega_{i^*})$, with equality when p assigns probability one to $\omega_{i^*}^{\boldsymbol{c}}$.

Now consider a realization $x \in X^+$ and fix a group $\boldsymbol{c}$. Since the contribution of every other group $\boldsymbol{c}' \neq \boldsymbol{c}$ is at most $\mu_0(\boldsymbol{c}')\overline{G}_{\boldsymbol{c}'}$, persuasion requires

$$0 \leq \sum_{\boldsymbol{c}'} \mu_0(\boldsymbol{c}') G_{\boldsymbol{c}'}\big(\mu(\cdot|\boldsymbol{c}', x)\big) \leq \mu_0(\boldsymbol{c}) G_{\boldsymbol{c}}\big(\mu(\cdot|\boldsymbol{c}, x)\big) + \sum_{\boldsymbol{c}' \neq \boldsymbol{c}} \mu_0(\boldsymbol{c}')\overline{G}_{\boldsymbol{c}'}.$$

Hence, at every realization $x \in X^+$,

$$G_{\boldsymbol{c}}\big(\mu(\cdot|\boldsymbol{c}, x)\big) \geq t_{\boldsymbol{c}}, \qquad t_{\boldsymbol{c}} \equiv -\frac{1}{\mu_0(\boldsymbol{c})} \sum_{\boldsymbol{c}' \neq \boldsymbol{c}} \mu_0(\boldsymbol{c}')\overline{G}_{\boldsymbol{c}'}.$$

I call $t_{\boldsymbol{c}}$ the *lowered bar* of group $\boldsymbol{c}$: it is the within-group gain at which the receiver is indifferent when every other group is at its most favorable conditional, namely the one that puts probability one on $\omega_{i^*}^{\boldsymbol{c}'}$.

For each group $\boldsymbol{c}$, let $S_{\boldsymbol{c}} \equiv \{p \in \Delta(\Omega_{i^*}) : G_{\boldsymbol{c}}(p) \geq t_{\boldsymbol{c}}\}$, which is a closed convex set because $G_{\boldsymbol{c}}$ is affine. A *splitting* of $p_0 \in \Delta(\Omega_{i^*})$ is a finite family $(\lambda_j, p_j)_j$ of weights $\lambda_j \geq 0$ summing to one and conditionals $p_j \in \Delta(\Omega_{i^*})$ satisfying $\sum_j \lambda_j p_j = p_0$. Let $\varphi_{\boldsymbol{c}}(p_0)$ denote the largest probability that a splitting of p_0 assigns to conditionals in $S_{\boldsymbol{c}}$:

$$\varphi_{\boldsymbol{c}}(p_0) \;\equiv\; \max\Big\{\sum_j \lambda_j \mathbf{1}\{p_j \in S_{\boldsymbol{c}}\} : \sum_j \lambda_j p_j = p_0,\ \lambda_j \geq 0,\ \sum_j \lambda_j = 1\Big\}.$$

By KG11, $\varphi_{\boldsymbol{c}}$ is the concave envelope of the indicator of $S_{\boldsymbol{c}}$. The maximum is attained because the indicator of the closed set $S_{\boldsymbol{c}}$ is upper semicontinuous. Since $\varphi_{\boldsymbol{c}}$ is concave, it is continuous on the relative interior of $\Delta(\Omega_{i^*})$, which contains $\mu_0(\cdot|\boldsymbol{c})$ by full support. The *capacity* of group $\boldsymbol{c}$ is

$$\kappa_{\boldsymbol{c}} \equiv \mu_0(\boldsymbol{c})\varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c})).$$

If $\mu_0(\cdot|\boldsymbol{c}) \in S_{\boldsymbol{c}}$, the unsplit prior itself clears the bar, so $\varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c})) = 1$ and $\kappa_{\boldsymbol{c}} = \mu_0(\boldsymbol{c})$.

The capacity is the largest mass of states with $\boldsymbol{\omega}_C = \boldsymbol{c}$ that can generate realizations at which the within-group conditional clears the bar. To see this, consider any experiment. By Lemma C.1, the conditionals $\mu(\cdot|\boldsymbol{c}, x)$, with weights $\mu(x|\boldsymbol{c})$, form a splitting of $\mu_0(\cdot|\boldsymbol{c})$. Hence,

$$\sum_{x:\mu(\boldsymbol{c},x)>0} \mu(x|\boldsymbol{c})\mathbf{1}\{\mu(\cdot|\boldsymbol{c}, x) \in S_{\boldsymbol{c}}\} \leq \varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c})),$$

so the mass from group $\boldsymbol{c}$ that can generate realizations at which it clears the bar is at most $\mu_0(\boldsymbol{c})\varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c})) = \kappa_{\boldsymbol{c}}$. Conversely, let $(\lambda_j, p_j)_j$ attain $\varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c}))$, with $\lambda_j > 0$ for every j. By Lemma C.1, there is an experiment with realizations $(x_j)_j$ such that $\mu(\boldsymbol{c}', x_j) = \lambda_j \mu_0(\boldsymbol{c}')$ for every group $\boldsymbol{c}'$, with

$$\mu(\cdot|\boldsymbol{c}, x_j) = p_j \quad \text{and} \quad \mu(\cdot|\boldsymbol{c}', x_j) = \mu_0(\cdot|\boldsymbol{c}')$$

for every $\boldsymbol{c}' \neq \boldsymbol{c}$. The mass from group $\boldsymbol{c}$ that generates realizations at which it clears the bar is

$$\mu_0(\boldsymbol{c}) \sum_{j:p_j \in S_{\boldsymbol{c}}} \lambda_j = \kappa_{\boldsymbol{c}}.$$

Thus, $\kappa_{\boldsymbol{c}}$ is exactly the largest mass of group $\boldsymbol{c}$ that can be made to clear its lowered bar.

Since Ω_1 is binary, I identify the within-group conditional with the probability it assigns to ω_1^+: $p_x(\boldsymbol{c}) \equiv \mu(\omega_1^+|\boldsymbol{c}, x)$. Whenever $\mu(\boldsymbol{c}, x) = 0$, this conditional is understood according to the prior convention, so that $p_x(\boldsymbol{c}) = \mu_0(\omega_1^+|\boldsymbol{c})$. Hence, for every x and $\boldsymbol{c}$, $\mu(\boldsymbol{c}, x)p_x(\boldsymbol{c}) = \mu(\omega_1^+, \boldsymbol{c}, x)$.

By (4), the receiver's posterior at x is $\mu_x^R(\omega_1^+) = \sum_{\boldsymbol{c}} \mu_0(\boldsymbol{c})p_x(\boldsymbol{c})$, and therefore the set of persuading signal realizations is $X^+ = \{x \in X : \mu_x^R(\omega_1^+) \geq \theta\}$. The lowered bar $t_{\boldsymbol{c}}(\theta)$ is defined in (5). Whenever $t_{\boldsymbol{c}}(\theta) > 0$, $\mu_0(\boldsymbol{c})t_{\boldsymbol{c}}(\theta) = \theta - 1 + \mu_0(\boldsymbol{c})$ and group $\boldsymbol{c}$ binds if and only if $t_{\boldsymbol{c}}(\theta) > \mu_0(\omega_1^+|\boldsymbol{c})$. Equivalently, group $\boldsymbol{c}$ binds if and only if $\theta > \bar{\theta}_{\boldsymbol{c}}$, where $\bar{\theta}_{\boldsymbol{c}} = 1 - \mu_0(\omega_1^-, \boldsymbol{c})$ is the value of the receiver's threshold defined in Section 6.4.

The next lemma collects the inequalities underlying the upper bound $\overline{\mathbb{P}}(\theta)$ in Proposition 4 and characterizes when they hold with equality. It is used in the proof of Proposition 4**(i)**.

Lemma C.2 . — *Under the assumptions of Proposition 4, every $\pi \in \mathcal{E}$, every group $\boldsymbol{c}$, and every $x \in X^+$ satisfy $p_x(\boldsymbol{c}) \geq t_{\boldsymbol{c}}(\theta)$. Therefore,*

$$t_{\boldsymbol{c}}(\theta) \sum_{x \in X^+} \mu(\boldsymbol{c}, x) \leq \sum_{x \in X^+} \mu(\omega_1^+, \boldsymbol{c}, x) \leq \mu_0(\omega_1^+, \boldsymbol{c}), \qquad \sum_{x \in X^+} \mu(\boldsymbol{c}, x) \leq \mu_0(\boldsymbol{c}). \quad \text{(C.3)}$$

If $\mathbb{P}(\pi) = \overline{\mathbb{P}}(\theta)$, then the inequalities in (C.3) *must bind for every group. In particular, for every group $\boldsymbol{c}$ with $t_{\boldsymbol{c}}(\theta) \geq \mu_0(\omega_1^+|\boldsymbol{c})$, $p_x(\boldsymbol{c}) = t_{\boldsymbol{c}}(\theta)$ at every $x \in X^+$ with $\mu(\boldsymbol{c}, x) > 0$, and $\mu(\omega_1^+, \boldsymbol{c}, x) = 0$ at every $x \notin X^+$. If instead $t_{\boldsymbol{c}}(\theta) \leq \mu_0(\omega_1^+|\boldsymbol{c})$, then $\mu(\boldsymbol{c}, x) = 0$ at every $x \notin X^+$.*

The last result of the binary case is the one stated in Section 6.4. An experiment *replicates* the Bayesian persuasion outcome if the receiver's subjective posterior $\mu_x^R(\omega_1^+)$ equals θ on a set of realizations of total probability $\mu_0(\omega_1^+)/\theta$ and 0 at every other realization, and $\tilde{\theta} = \mu_0(\omega_1^+)/\max_{\boldsymbol{c}} \mu_0(\omega_1^+|\boldsymbol{c})$.

Lemma C.3 : Replication. — *Under the assumptions of Proposition 4, a consistent experiment replicates the Bayesian persuasion outcome if and only if $\theta \leq \tilde{\theta}$.*

C.4. Proofs

Derivation of Equation (4)

Since $\mathrm{pa}(0) = \emptyset$ and $\mathrm{ch}(0) = \{i^*\}$, (3) is

$$\mu^R(\boldsymbol{\omega}, x) = \mu(x)\mu(\omega_{i^*}|\boldsymbol{\omega}_C, x) \prod_{j \in N_\omega \setminus \{i^*\}} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}),$$

with $\mu(\omega_{i^*}|\boldsymbol{\omega}_C, x) = \mu_0(\omega_{i^*}|\boldsymbol{\omega}_C)$ if $\mu(\boldsymbol{\omega}_C, x) = 0$. Since x has a single neighbor, no path between two state attributes passes through x, so the independencies that R entails among the state attributes are those of the DAG $R_{\boldsymbol{\omega}}$ obtained by deleting x, in which the parents of i^* are C. By Assumption 1, μ_0 is Markov with respect to $R_{\boldsymbol{\omega}}$ and factorizes along it (Lauritzen, 1996, Theorem 3.27), as in the proof of Lemma 3**(ii)**:

$$\mu_0(\boldsymbol{\omega}) = \mu_0(\omega_{i^*}|\boldsymbol{\omega}_C) \prod_{j \in N_\omega \setminus \{i^*\}} \mu_0(\omega_j|\boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}).$$

Since μ_0 has full support, the product equals

$$\frac{\mu_0(\boldsymbol{\omega})}{\mu_0(\omega_{i^*}|\boldsymbol{\omega}_C)} = \mu_0(\boldsymbol{\omega}_C)\mu_0\big(\boldsymbol{\omega}_{N_\omega\setminus(C\cup\{i^*\})}\big|\omega_{i^*},\boldsymbol{\omega}_C\big).$$

Substituting into the first display and dividing by $\mu^R(x) = \mu(x) > 0$ (Lemma 1) gives (4).

Summing (4) over the state attributes outside $C \cup \{i^*\}$ gives $\mu^R(\omega_{i^*},\boldsymbol{\omega}_C|x) = \mu_0(\boldsymbol{\omega}_C)\mu(\omega_{i^*}|\boldsymbol{\omega}_C,x)$. Summing this over ω_{i^*} gives $\mu^R(\boldsymbol{\omega}_C|x) = \mu_0(\boldsymbol{\omega}_C)$, since $\mu(\cdot|\boldsymbol{\omega}_C,x)$ is a probability distribution, and summing it over $\boldsymbol{\omega}_C$ instead gives $\mu^R(\omega_{i^*}|x) = \sum_{\boldsymbol{c}}\mu_0(\boldsymbol{c})\mu(\omega_{i^*}|\boldsymbol{c},x)$. ■

Proof of Lemma C.1

(⇒) Let $\pi \in \mathcal{E}$ induce the family. For every group $\boldsymbol{c}$ and every ω_{i^*},

$$\sum_{x\in X}\alpha_x(\boldsymbol{c})p_x(\boldsymbol{c})(\omega_{i^*}) = \sum_{x\in X}\mu(\omega_{i^*},\boldsymbol{c},x) = \mu_0(\omega_{i^*},\boldsymbol{c}),$$

where a realization with $\alpha_x(\boldsymbol{c}) = 0$ contributes zero, whatever value the prior completion of (3) assigns to $p_x(\boldsymbol{c})$.

(⇐) Let $(\alpha_x(\boldsymbol{c}),p_x(\boldsymbol{c}))_{x\in X,\boldsymbol{c}\in\Omega_C}$ satisfy (C.1) and set $\mu(\omega_{i^*},\boldsymbol{c},x) \equiv \alpha_x(\boldsymbol{c})p_x(\boldsymbol{c})(\omega_{i^*})$, which is non-negative and, by (C.1), sums to $\mu_0(\omega_{i^*},\boldsymbol{c})$ over x. Hence

$$\pi(x|\boldsymbol{\omega}) \;\equiv\; \frac{\mu(\omega_{i^*},\boldsymbol{\omega}_C,x)}{\mu_0(\omega_{i^*},\boldsymbol{\omega}_C)}$$

is well defined because μ_0 has full support, non-negative, and sums to one over x. Hence π is a Blackwell experiment, and it depends on $\boldsymbol{\omega}$ only through $(\omega_{i^*},\boldsymbol{\omega}_C)$, which is the last claim. Its objective joint on $(\omega_{i^*},\boldsymbol{\omega}_C,x)$ is $\mu_0(\omega_{i^*},\boldsymbol{\omega}_C)\pi(x|\omega_{i^*},\boldsymbol{\omega}_C) = \mu(\omega_{i^*},\boldsymbol{\omega}_C,x)$, so $\mu(\boldsymbol{c},x) = \alpha_x(\boldsymbol{c})$ and $\mu(x) = \alpha_x > 0$ for every $x \in X$: $\pi \in \mathcal{E}$ with realization space X. When $\alpha_x(\boldsymbol{c}) > 0$, $\mu(\cdot|\boldsymbol{c},x) = p_x(\boldsymbol{c})$ by definition; when $\alpha_x(\boldsymbol{c}) = 0$, both equal $\mu_0(\cdot|\boldsymbol{c})$, by the prior completion and by hypothesis. Thus π induces $(\alpha_x(\boldsymbol{c}),p_x(\boldsymbol{c}))_{x\in X,\boldsymbol{c}\in\Omega_C}$.

Dividing (C.1) by $\mu_0(\boldsymbol{c}) > 0$ gives the splitting formulation, since summing (C.1) over ω_{i^*} gives $\sum_x \alpha_x(\boldsymbol{c}) = \mu_0(\boldsymbol{c})$, so the weights $\alpha_x(\boldsymbol{c})/\mu_0(\boldsymbol{c})$ sum to one. ■

Proof of Proposition 3

I proceed in two steps: first, I establish the upper bound, and second, the number of realizations needed to approach the upper bound.

Step 1. Fix $\pi \in \mathcal{E}$ and a group $\boldsymbol{c}$. At every $x \in X^+$, $\mu(\cdot|\boldsymbol{c},x) \in S_{\boldsymbol{c}}$ (Section C.2). Hence $\sum_{x\in X^+}\mu(\boldsymbol{c},x) \le \sum_{x:\,\mu(\cdot|\boldsymbol{c},x)\in S_c}\mu(\boldsymbol{c},x) \le \kappa_{\boldsymbol{c}}$, because $\kappa_{\boldsymbol{c}}$ is the largest mass of states with $\boldsymbol{\omega}_C = \boldsymbol{c}$ that generate realizations at which the within-group conditional clears the bar (Section C.2). Summing over groups gives $\mathbb{P}(\pi) = \sum_{\boldsymbol{c}}\sum_{x\in X^+}\mu(\boldsymbol{c},x) \le \sum_{\boldsymbol{c}}\kappa_{\boldsymbol{c}}$.

Step 2. Fix $\varepsilon \in (0, 1/(J-1))$ and let the realization space be $X = \{x_{\boldsymbol{c}} : \boldsymbol{c} \in \Omega_C\} \cup \{x_0\}$. For every group $\boldsymbol{c}$ and every $\boldsymbol{c}' \neq \boldsymbol{c}$, let $\pi(x_{\boldsymbol{c}'}|\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c}) = \varepsilon$, that is, $\mu(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c},x_{\boldsymbol{c}'}) = \varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})$, and $\mu(\omega_{i^*},\boldsymbol{c},x_{\boldsymbol{c}'}) = 0$ for $\omega_{i^*} \neq \omega_{i^*}^{\boldsymbol{c}}$. This leaves, in group $\boldsymbol{c}$, a mass $\mu_0(\boldsymbol{c}) - (J-1)\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})$ not

yet assigned to any signal realization, among which ω_{i^*} is distributed according to

$$\tilde{p}_{\boldsymbol{c}}(\omega_{i^*}) \;\equiv\; \frac{\mu_0(\omega_{i^*},\boldsymbol{c})\big[1-(J-1)\varepsilon\mathbf{1}\{\omega_{i^*}=\omega_{i^*}^{\boldsymbol{c}}\}\big]}{\mu_0(\boldsymbol{c})-(J-1)\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})}.$$

Since $(J-1)\varepsilon<1$, $\tilde{p}_{\boldsymbol{c}}$ has full support, and it tends to $\mu_0(\cdot|\boldsymbol{c})$ as $\varepsilon\to 0$. Take a splitting $(\lambda_j,p_j)_j$ of $\tilde{p}_{\boldsymbol{c}}$ that attains $\varphi_{\boldsymbol{c}}(\tilde{p}_{\boldsymbol{c}})$, and divide the unassigned states between $x_{\boldsymbol{c}}$ and x_0 according to it:

$$\mu(\cdot,\boldsymbol{c},x_{\boldsymbol{c}}) = \big[\mu_0(\boldsymbol{c})-(J-1)\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})\big]\sum_{j:p_j\in S_{\boldsymbol{c}}}\lambda_j p_j,$$

$$\mu(\cdot,\boldsymbol{c},x_0) = \big[\mu_0(\boldsymbol{c})-(J-1)\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})\big]\sum_{j:p_j\notin S_{\boldsymbol{c}}}\lambda_j p_j.$$

This defines an experiment since $\sum_j\lambda_j p_j=\tilde{p}_{\boldsymbol{c}}$.

Now, we need to verify that $x_{\boldsymbol{c}}$ leads to the sender's preferred action whenever $\varphi_{\boldsymbol{c}}(\tilde{p}_{\boldsymbol{c}})>0$. Fix $x_{\boldsymbol{c}}$. For any group $\boldsymbol{c}'\neq\boldsymbol{c}$, $\mu(\boldsymbol{c}',x_{\boldsymbol{c}})=\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}'},\boldsymbol{c}')>0$, and $\mu(\omega_{i^*}|\boldsymbol{c}',x_{\boldsymbol{c}})=1$, so that $G_{\boldsymbol{c}'}=\overline{G}_{\boldsymbol{c}'}$. Group $\boldsymbol{c}$ has $\mu(\boldsymbol{c},x_{\boldsymbol{c}})=[\mu_0(\boldsymbol{c})-(J-1)\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})]\varphi_{\boldsymbol{c}}(\tilde{p}_{\boldsymbol{c}})>0$ and is at a convex combination of points of $S_{\boldsymbol{c}}$, which lies in $S_{\boldsymbol{c}}$ because $S_{\boldsymbol{c}}$ is convex. Hence $G(\mu^R_{x_{\boldsymbol{c}}})\geq\mu_0(\boldsymbol{c})t_{\boldsymbol{c}}+\sum_{\boldsymbol{c}'\neq\boldsymbol{c}}\mu_0(\boldsymbol{c}')\overline{G}_{\boldsymbol{c}'}=0$, and $x_{\boldsymbol{c}}\in X^+$. Every $x_{\boldsymbol{c}}$ has positive probability, and since $J\geq 2$ no realization has probability one. If no mass is assigned to x_0, then it is dropped. Therefore

$$\mathbb{P}(\pi)\;\geq\;\sum_{\boldsymbol{c}}\big[\mu_0(\boldsymbol{c})-(J-1)\varepsilon\mu_0(\omega_{i^*}^{\boldsymbol{c}},\boldsymbol{c})\big]\varphi_{\boldsymbol{c}}(\tilde{p}_{\boldsymbol{c}})\;\xrightarrow[\varepsilon\to 0]{}\;\sum_{\boldsymbol{c}}\mu_0(\boldsymbol{c})\varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c}))=\sum_{\boldsymbol{c}}\kappa_{\boldsymbol{c}},$$

by continuity of $\varphi_{\boldsymbol{c}}$ at $\mu_0(\cdot|\boldsymbol{c})$. Together with Step 1 this proves the claim. ■

Proof of Proposition 4

Since $P=\{1\}$ and Ω_1 is binary, I identify an arbitrary belief $q\in\Delta(\Omega_1)$ by $q(\omega_1^+)$. For any q, $G(q)=K(q(\omega_1^+)-\theta)$, where $K\equiv G(\delta_{\omega_1^+})-G(\delta_{\omega_1^-})$, δ_{ω_1} is the belief that puts probability one on ω_1, and $\theta=-G(\delta_{\omega_1^-})/K$ is the threshold of the statement. The constant K is strictly positive because $G(\mu_0)<0\leq G(\delta_{\omega_1^+})$. Moreover, in (C.2), it also holds that $\mu_0(\boldsymbol{\omega}_P|\omega_1,\boldsymbol{c})=\delta_{\omega_1}$. Therefore, $G_{\boldsymbol{c}}(p)=K(p(\omega_1^+)-\theta)$ for every group, $\overline{G}_{\boldsymbol{c}}=K(1-\theta)$, and $t_{\boldsymbol{c}}=-K(1-\theta)(1-\mu_0(\boldsymbol{c}))/\mu_0(\boldsymbol{c})$. Hence $p\in S_{\boldsymbol{c}}$ if and only if

$$p(\omega_1^+)\;\geq\;\theta-\frac{(1-\theta)(1-\mu_0(\boldsymbol{c}))}{\mu_0(\boldsymbol{c})}=\frac{\theta-1+\mu_0(\boldsymbol{c})}{\mu_0(\boldsymbol{c})}=t_{\boldsymbol{c}}(\theta).$$

$\varphi_{\boldsymbol{c}}(\mu_0(\cdot|\boldsymbol{c}))$ is the largest probability that a splitting of $\mu_0(\omega_1^+|\boldsymbol{c})$ assigns to conditionals at least $t_{\boldsymbol{c}}(\theta)$. If $\mu_0(\omega_1^+|\boldsymbol{c})\geq t_{\boldsymbol{c}}(\theta)$, the splitting that leaves $\mu_0(\omega_1^+|\boldsymbol{c})$ unsplit assigns probability one, so $\kappa_{\boldsymbol{c}}=\mu_0(\boldsymbol{c})$. If $\mu_0(\omega_1^+|\boldsymbol{c})<t_{\boldsymbol{c}}(\theta)$, then $t_{\boldsymbol{c}}(\theta)>0$; every splitting $(\lambda_j,p_j)_j$ of $\mu_0(\omega_1^+|\boldsymbol{c})$ satisfies $t_{\boldsymbol{c}}(\theta)\sum_{j:p_j\in S_{\boldsymbol{c}}}\lambda_j\leq\sum_j\lambda_j p_j(\omega_1^+)=\mu_0(\omega_1^+|\boldsymbol{c})$, and the splitting of $\mu_0(\omega_1^+|\boldsymbol{c})$ between the conditionals $t_{\boldsymbol{c}}(\theta)$, with probability $\mu_0(\omega_1^+|\boldsymbol{c})/t_{\boldsymbol{c}}(\theta)$, and 0 attains this bound. Thus, the capacity of group $\boldsymbol{c}$ is $\kappa_{\boldsymbol{c}}=\mu_0(\boldsymbol{c})\mu_0(\omega_1^+|\boldsymbol{c})/t_{\boldsymbol{c}}(\theta)=\mu_0(\omega_1^+,\boldsymbol{c})/t_{\boldsymbol{c}}(\theta)$. In both cases

$$\kappa_{\boldsymbol{c}}=\min\{\mu_0(\boldsymbol{c}),\mu_0(\omega_1^+,\boldsymbol{c})/t_{\boldsymbol{c}}(\theta)\},$$

with the convention for $t_{\boldsymbol{c}}(\theta)=0$, and $\sum_{\boldsymbol{c}}\kappa_{\boldsymbol{c}}=\overline{\mathbb{P}}(\theta)$. The claim follows from Proposition 3.

(i) $\overline{\mathbb{P}}(\theta)=1$ if and only if every term of (6) equals $\mu_0(\boldsymbol{c})$ or, equivalently, if and only if $t_{\boldsymbol{c}}(\theta)\mu_0(\boldsymbol{c})\leq$

$\mu_0(\omega_1^+, \boldsymbol{c})$ for every group. When $t_{\boldsymbol{c}}(\theta) > 0$, this implies $\theta \leq 1 - \mu_0(\omega_1^-, \boldsymbol{c})$; when $t_{\boldsymbol{c}}(\theta) = 0$, it holds, and $\theta \leq 1 - \mu_0(\boldsymbol{c}) \leq 1 - \mu_0(\omega_1^-, \boldsymbol{c})$. Hence $\overline{\mathbb{P}}(\theta) = 1$ if and only if $\theta \leq \min_{\boldsymbol{c}} \bar{\theta}_{\boldsymbol{c}}$. Since μ_0 has full support and $J \geq 2$, $\max_{\boldsymbol{c}} \mu_0(\omega_1^-, \boldsymbol{c}) < \sum_{\boldsymbol{c}} \mu_0(\omega_1^-, \boldsymbol{c}) = 1 - \mu_0(\omega_1^+)$, so this bound is strictly above $\mu_0(\omega_1^+)$.

Let $\theta < \min_{\boldsymbol{c}} \bar{\theta}_{\boldsymbol{c}}$. Then $\mu_0(\omega_1^+|\boldsymbol{c}) > t_{\boldsymbol{c}}(\theta)$ for every group: if $t_{\boldsymbol{c}}(\theta) > 0$, this is $\mu_0(\omega_1^+, \boldsymbol{c}) > \theta - 1 + \mu_0(\boldsymbol{c})$, that is, $\theta < 1 - \mu_0(\omega_1^-, \boldsymbol{c})$, and if $t_{\boldsymbol{c}}(\theta) = 0$, it is $\mu_0(\omega_1^+|\boldsymbol{c}) > 0$. Let $X = \{x_{\boldsymbol{c}} : \boldsymbol{c} \in \Omega_C\}$, let $\varepsilon > 0$ be small, and for every group $\boldsymbol{c}$ let

$$\mu(\omega_1^+, \boldsymbol{c}, x_{\boldsymbol{c}'}) = \varepsilon \qquad \text{for every } \boldsymbol{c}' \neq \boldsymbol{c},$$
$$\mu(\omega_1^+, \boldsymbol{c}, x_{\boldsymbol{c}}) = \mu_0(\omega_1^+, \boldsymbol{c}) - (J-1)\varepsilon, \qquad \mu(\omega_1^-, \boldsymbol{c}, x_{\boldsymbol{c}}) = \mu_0(\omega_1^-, \boldsymbol{c}).$$

For every group and every value of ω_1, the masses sum to $\mu_0(\omega_1, \boldsymbol{c})$ over the realizations, so this defines an experiment. At $x_{\boldsymbol{c}}$, every group $\boldsymbol{c}' \neq \boldsymbol{c}$ is at the conditional 1, and group $\boldsymbol{c}$ is at the conditional $[\mu_0(\omega_1^+, \boldsymbol{c}) - (J-1)\varepsilon]/[\mu_0(\boldsymbol{c}) - (J-1)\varepsilon]$, which tends to $\mu_0(\omega_1^+|\boldsymbol{c}) > t_{\boldsymbol{c}}(\theta)$ as $\varepsilon \to 0$ and is therefore at least $t_{\boldsymbol{c}}(\theta)$ for ε small. Then $\mu^R_{x_{\boldsymbol{c}}}(\omega_1^+) \geq \mu_0(\boldsymbol{c}) t_{\boldsymbol{c}}(\theta) + 1 - \mu_0(\boldsymbol{c}) \geq \theta$, where the last inequality is an equality if $t_{\boldsymbol{c}}(\theta) > 0$ and is the definition of $t_{\boldsymbol{c}}(\theta)$ otherwise. Every realization persuades, so $\mathbb{P}(\pi) = 1$, which equals $\overline{\mathbb{P}}(\theta)$ by the first claim of this part.

Let now $\theta \geq \min_{\boldsymbol{c}} \bar{\theta}_{\boldsymbol{c}}$, and suppose that some $\pi \in \mathcal{E}$ attains $\overline{\mathbb{P}}(\theta)$. Fix a group $\boldsymbol{c}^*$ with $\theta \geq \bar{\theta}_{\boldsymbol{c}^*}$. Then $\theta - 1 + \mu_0(\boldsymbol{c}^*) \geq \mu_0(\omega_1^+, \boldsymbol{c}^*) > 0$, so $t_{\boldsymbol{c}^*}(\theta) = [\theta - 1 + \mu_0(\boldsymbol{c}^*)]/\mu_0(\boldsymbol{c}^*) \geq \mu_0(\omega_1^+|\boldsymbol{c}^*)$: the prior conditional of group $\boldsymbol{c}^*$ does not clear its bar. Fix $x \in X^+$. If $\mu(\boldsymbol{c}^*, x) > 0$, then $p_x(\boldsymbol{c}^*) = t_{\boldsymbol{c}^*}(\theta)$ by Lemma C.2; otherwise $p_x(\boldsymbol{c}^*) = \mu_0(\omega_1^+|\boldsymbol{c}^*) \leq t_{\boldsymbol{c}^*}(\theta)$. In both cases $\mu_0(\boldsymbol{c}^*) p_x(\boldsymbol{c}^*) \leq \theta - 1 + \mu_0(\boldsymbol{c}^*)$, so the persuasion inequality $\sum_{\boldsymbol{c}} \mu_0(\boldsymbol{c}) p_x(\boldsymbol{c}) \geq \theta$ gives $\sum_{\boldsymbol{c} \neq \boldsymbol{c}^*} \mu_0(\boldsymbol{c})[1 - p_x(\boldsymbol{c})] \leq 0$: every other group is at the conditional 1 at x, that is, $\mu(\omega_1^-, \boldsymbol{c}, x) = \mu(\boldsymbol{c}, x)[1 - p_x(\boldsymbol{c})] = 0$ for every $\boldsymbol{c} \neq \boldsymbol{c}^*$. Hence the states with $\omega_1 = \omega_1^-$ of every group $\boldsymbol{c} \neq \boldsymbol{c}^*$ generate non-persuading realizations only, and $\sum_{x \in X^+} \mu(\boldsymbol{c}, x) \leq \mu_0(\omega_1^+, \boldsymbol{c})$. This is below $\kappa_{\boldsymbol{c}}$, because both terms in the minimum of (6) exceed $\mu_0(\omega_1^+, \boldsymbol{c})$: the first by full support, the second because $t_{\boldsymbol{c}}(\theta) < 1$, as $\theta < 1$, and it equals the first when $t_{\boldsymbol{c}}(\theta) = 0$. Since $\sum_{x \in X^+} \mu(\boldsymbol{c}, x) \leq \kappa_{\boldsymbol{c}}$ for every group by Lemma C.2, and some group $\boldsymbol{c} \neq \boldsymbol{c}^*$ exists because $J \geq 2$, $\mathbb{P}(\pi) = \sum_{\boldsymbol{c}} \sum_{x \in X^+} \mu(\boldsymbol{c}, x) < \sum_{\boldsymbol{c}} \kappa_{\boldsymbol{c}} = \overline{\mathbb{P}}(\theta)$. Hence no experiment attains $\overline{\mathbb{P}}(\theta)$.

(ii) Let $\theta \geq \max_{\boldsymbol{c}} \mu_0(\omega_1^+|\boldsymbol{c})$ and fix a group $\boldsymbol{c}$. If $t_{\boldsymbol{c}}(\theta) > 0$, then $t_{\boldsymbol{c}}(\theta) < \theta$, because

$$\frac{\theta - 1 + \mu_0(\boldsymbol{c})}{\mu_0(\boldsymbol{c})} < \theta \qquad \text{is} \qquad (1 - \theta)(1 - \mu_0(\boldsymbol{c})) > 0;$$

hence the second term in the minimum of (6) exceeds $\mu_0(\omega_1^+, \boldsymbol{c})/\theta$. The first term, $\mu_0(\boldsymbol{c})$, is at least $\mu_0(\omega_1^+, \boldsymbol{c})/\theta$ because $\mu_0(\omega_1^+|\boldsymbol{c}) \leq \theta$. Thus every term of (6) is at least $\mu_0(\omega_1^+, \boldsymbol{c})/\theta$, and summing gives $\overline{\mathbb{P}}(\theta) \geq \mu_0(\omega_1^+)/\theta$. The inequality is strict: since $\theta > \mu_0(\omega_1^+) = \sum_{\boldsymbol{c}} \mu_0(\boldsymbol{c}) \mu_0(\omega_1^+|\boldsymbol{c})$, some group has $\mu_0(\omega_1^+|\boldsymbol{c}) < \theta$, and for that group both terms in the minimum exceed $\mu_0(\omega_1^+, \boldsymbol{c})/\theta$ strictly. ■

Proof of Lemma C.2

Fix $x \in X^+$ and a group $\boldsymbol{c}$. Then $\mu(\cdot|\boldsymbol{c}, x) \in S_{\boldsymbol{c}}$ (Section C.2), and $p \in S_{\boldsymbol{c}}$ if and only if $p(\omega_1^+) \geq t_{\boldsymbol{c}}(\theta)$ by the first paragraph of the proof of Proposition 4, so $p_x(\boldsymbol{c}) \geq t_{\boldsymbol{c}}(\theta)$. Multiplying by $\mu(\boldsymbol{c}, x)$

and summing over X^+ gives the first inequality of (C.3), because $\mu(\boldsymbol{c},x)p_x(\boldsymbol{c}) = \mu(\omega_1^+,\boldsymbol{c},x)$; the second and the third hold because the same sums over all of X equal $\mu_0(\omega_1^+,\boldsymbol{c})$ and $\mu_0(\boldsymbol{c})$.

By (C.3), $\sum_{x\in X^+}\mu(\boldsymbol{c},x) \leq \kappa_{\boldsymbol{c}}$ for every group, with $\kappa_{\boldsymbol{c}}$ as in the proof of Proposition 4, and summing over groups gives $\mathbb{P}(\pi) \leq \overline{\mathbb{P}}(\theta)$. If $\mathbb{P}(\pi) = \overline{\mathbb{P}}(\theta)$, then $\sum_{x\in X^+}\mu(\boldsymbol{c},x) = \kappa_{\boldsymbol{c}}$ for every group. For a group with $t_{\boldsymbol{c}}(\theta) \geq \mu_0(\omega_1^+|\boldsymbol{c})$, $t_{\boldsymbol{c}}(\theta) > 0$ and $\kappa_{\boldsymbol{c}} = \mu_0(\omega_1^+,\boldsymbol{c})/t_{\boldsymbol{c}}(\theta)$, so both inequalities in the chain of (C.3) hold with equality: the first requires $p_x(\boldsymbol{c}) = t_{\boldsymbol{c}}(\theta)$ at every $x \in X^+$ with $\mu(\boldsymbol{c},x) > 0$, and the second requires $\sum_{x\in X^+}\mu(\omega_1^+,\boldsymbol{c},x) = \mu_0(\omega_1^+,\boldsymbol{c})$, that is, $\mu(\omega_1^+,\boldsymbol{c},x) = 0$ at every $x \notin X^+$. For a group with $t_{\boldsymbol{c}}(\theta) \leq \mu_0(\omega_1^+|\boldsymbol{c})$, $\kappa_{\boldsymbol{c}} = \mu_0(\boldsymbol{c})$, so $\sum_{x\in X^+}\mu(\boldsymbol{c},x) = \mu_0(\boldsymbol{c})$, that is, $\mu(\boldsymbol{c},x) = 0$ at every $x \notin X^+$. ■

Proof of Lemma C.3

Recall that $\mu_0(\omega_1^+) < \theta$ and that $\mu_0(\omega_1^+|\boldsymbol{c}) \in (0,1)$ for every group.

($\Rightarrow$) Let $\pi \in \mathcal{E}_R$ replicate the outcome: $\mu_x^R(\omega_1^+) = \theta$ for any $x \in X^+$, $\mu_x^R(\omega_1^+) = 0$ for any $x \notin X^+$, and $\mathbb{P}(\pi) = \mu_0(\omega_1^+)/\theta$. Let $x \notin X^+$. Then

$$\sum_{\boldsymbol{c}}\mu_0(\boldsymbol{c})\mu(\omega_1^+|\boldsymbol{c},x) = 0,$$

that is, $\mu(\omega_1^+|\boldsymbol{c},x) = 0$ for every group $\boldsymbol{c} \in \Omega_C$. Since $\mu(\omega_1^+|\boldsymbol{c},x) = \mu_0(\omega_1^+|\boldsymbol{c}) > 0$ when $\mu(\boldsymbol{c},x) = 0$, it must be the case that $\mu(\boldsymbol{c},x) > 0$ and $\mu(\omega_1^+,\boldsymbol{c},x) = \mu(\omega_1^+|\boldsymbol{c},x)\mu(\boldsymbol{c},x) = 0$. Therefore, every $\boldsymbol{\omega}$ such that $\omega_1 = \omega_1^+$ generates a realization $x \in X^+$: $\sum_{x\in X^+}\mu(\omega_1^+,\boldsymbol{c},x) = \mu_0(\omega_1^+,\boldsymbol{c})$ for every group.

By Lemma B.4, $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_C$, so $\sum_{x\in X^+}\mu(\boldsymbol{c},x) = \mu_0(\boldsymbol{c})\sum_{x\in X^+}\mu(x) = \mu_0(\boldsymbol{c})\mu_0(\omega_1^+)/\theta$. This sum is at least $\sum_{x\in X^+}\mu(\omega_1^+,\boldsymbol{c},x) = \mu_0(\omega_1^+,\boldsymbol{c})$. Thus $\mu_0(\omega_1^+|\boldsymbol{c}) \leq \mu_0(\omega_1^+)/\theta$ for every group, that is, $\theta \leq \tilde{\theta}$.

($\Leftarrow$) Let $\theta \leq \tilde{\theta}$ and $X = \{H,L\}$. For every group $\boldsymbol{c}$, consider

$$\mu(\omega_1^+,\boldsymbol{c},H) = \mu_0(\omega_1^+,\boldsymbol{c}), \qquad \mu(\omega_1^-,\boldsymbol{c},H) = \mu_0(\boldsymbol{c})\frac{\mu_0(\omega_1^+)}{\theta} - \mu_0(\omega_1^+,\boldsymbol{c}),$$
$$\mu(\omega_1^+,\boldsymbol{c},L) = 0, \qquad \mu(\omega_1^-,\boldsymbol{c},L) = \mu_0(\boldsymbol{c})\left(1 - \frac{\mu_0(\omega_1^+)}{\theta}\right).$$

The second probability is non-negative because $\mu_0(\omega_1^+|\boldsymbol{c}) \leq \max_{\boldsymbol{c}'}\mu_0(\omega_1^+|\boldsymbol{c}') \leq \mu_0(\omega_1^+)/\theta$, and the fourth is positive because $\mu_0(\omega_1^+) < \theta$. This corresponds to an experiment $\pi \in \mathcal{E}$ with $\mu(H) = \mu_0(\omega_1^+)/\theta$ that depends on $\boldsymbol{\omega}$ only through $(\omega_1,\boldsymbol{\omega}_C)$. Since the parents of 1 are the state attributes in C, we obtain

$$\mu(H|\boldsymbol{\omega}_C) = \sum_{\omega_1}\pi(H|\omega_1,\boldsymbol{\omega}_C)\mu_0(\omega_1|\boldsymbol{\omega}_C) = \mu_0(\omega_1^+)/\theta = \mu(H),$$

that is, $x \perp\!\!\!\perp_\mu \boldsymbol{\omega}_{\mathrm{nd}(0)}$. Thus, $\pi \in \mathcal{E}_R$ by Lemma B.4. Finally, $\mu(\omega_1^+|\boldsymbol{c},H) = \theta\mu_0(\omega_1^+|\boldsymbol{c})/\mu_0(\omega_1^+)$, so $\mu_H^R(\omega_1^+) = \theta$, and $p_L(\boldsymbol{c}) = 0$ for every group, so $\mu_L^R(\omega_1^+) = 0$. Thus π replicates the outcome. ■

D. Appendix: Irrelevance

In this appendix, I separate the two channels of Definition 5 when x is terminal, characterize the causal models that are Markov equivalent to a terminal one with $\mathrm{pa}(0) = P$ and give the results behind the converse when x is a root. The last subsection collects the proofs.

D.1. The Terminal Case

When x is terminal, the two channels of Definition 5 separate, each pinned down by one inclusion between P and the Markov boundary $\mathrm{MB}_R(0) = \mathrm{pa}(0)$ of the signal.

Lemma D.1 . — *Fix $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$ and $R \in \mathcal{R}_0$ with $\mathrm{ch}(0) = \emptyset$. Then*

- **(i)** *R is belief-irrelevant if and only if $P \subseteq \mathrm{MB}_R(0)$; equivalently, $\mathcal{B}_R = \mathcal{B}$;*
- **(ii)** *R is design-irrelevant if and only if $\mathrm{MB}_R(0) \subseteq P$.*

Hence R is irrelevant if and only if $\mathrm{MB}_R(0) = P$.

Part **(i)** is the belief channel. It imposes no condition on how R arranges the state attributes among themselves: that arrangement is immaterial. Since $\mathcal{B}_R \subseteq \mathcal{B}$ always (Lemma 4), the equivalence with $\mathcal{B}_R = \mathcal{B}$ says that the receiver reaches the entire feasible set exactly when no posterior is distorted, and that $\mathcal{B}_R \subsetneq \mathcal{B}$ precisely when a payoff-relevant state attribute lies outside $\mathrm{pa}(0)$.

Part **(ii)** is the design channel, governed by the reverse inclusion. By Lemma B.3, $\mathcal{E}_R$ is exactly the set of experiments measurable with respect to $\omega_{\mathrm{MB}_R(0)}$, so the sender can restrict attention to the reduced state space Ω_P if and only if $\mathrm{MB}_R(0) \subseteq P$. This inclusion is silent about beliefs: it constrains the *form* of the instrument, not the posteriors it can induce.

D.2. Terminal Equivalents

The next lemma states the condition of Proposition 5 in terms of the edges of R. A set of nodes is a *clique* in R if every two distinct nodes in it are linked, that is, if $i \to_R j$ or $j \to_R i$ for any two of them.

Lemma D.2 . — *Let $R \in \mathcal{R}$ with x as a root node. Then R is Markov equivalent to a DAG in which x is terminal and $\mathrm{pa}(0) = P$ if and only if*

- **(a)** *$\mathrm{MB}_R(0) = P$, and*
- **(b)** *$P \cup \{0\}$ is a clique.*

The two conditions require that the children of x are the payoff-relevant state attributes, that none of them has a parent outside P, and that every two of them are linked. With a single child i^* they reduce to $\mathrm{pa}_\omega(i^*) = \emptyset$ and $P = \{i^*\}$.

D.3. The Converse when x is a Root

When x is a root, the next lemma gives what the converse of Proposition 5 needs. Its first part holds for every root DAG. The second covers a single child, the class of Section 6.

Lemma D.3 . — *Fix $\mu_0 \in \mathrm{int}\,(\Delta(\Omega))$ and $R \in \mathcal{R}_0$ with x as a root node.*

(i) *If R is belief-irrelevant, then $P \subseteq \mathrm{ch}(0)$.*

(ii) *Suppose that x has a single child i^*, and let $C = \mathrm{pa}_\omega(i^*)$. Then R is belief-irrelevant if and only if $C = \emptyset$ and $P = \{i^*\}$.*

Hence, when x has a single child and R is irrelevant, conditions **(a)** *and* **(b)** *of Lemma D.2 hold.*

Part **(i)** is the root counterpart of the inclusion $P \subseteq \mathrm{pa}(0)$ in Lemma D.1, and it is proved in the same way: an attribute that the receiver does not interpret as an effect of x enters her belief only through prior conditionals, so her posterior on it has full support at every realization, while the sender can make the objective one degenerate. Part **(ii)** says that, with a single child, belief-irrelevance leaves no room for a co-parent, payoff-relevant or not: by (4), the receiver weights the co-parent groups by their prior shares at every realization, and an experiment that carries a single group at some realization has an objective posterior concentrated on that group.

Remark D.1 : The Fully Connected Model. — *Let $R \in \mathcal{R}_0$ be fully connected. Then* (3) *is the chain-rule factorization of μ along a causal order of R, so $\mu^R = \mu$ for every $\pi \in \mathcal{E}$. Consequently $\mathcal{E}_R = \mathcal{E}$, NDD holds automatically, $\mu^R_x = \mu_x$, and $\mathcal{B}_R = \mathcal{B}$: the receiver is indistinguishable from the KG11 receiver. This holds both when x is terminal, with $\mathrm{pa}(0) = N_\omega$, and when x is a root, with $\mathrm{ch}(0) = N_\omega$; and it holds for each of the $n!$ fully connected DAGs on N_ω, which are Markov equivalent.*

D.4. Proofs

Proof of Lemma D.1

Throughout, $\mathrm{ch}(0) = \emptyset$, so $\mathrm{MB}_R(0) = \mathrm{pa}(0)$ and, for every $\pi \in \mathcal{E}$,

$$\mu^R(\boldsymbol{\omega}, x) = \mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\,\mu_0(\boldsymbol{\omega}), \quad \mu^R(\boldsymbol{\omega}) = \mu_0(\boldsymbol{\omega}), \quad \mu^R(x) = \mu(x), \tag{D.1}$$

by Lemma 1 and Lemma 3**(i)**. Marginalizing (D.1) over Ω_{-P} gives, for every $\pi \in \mathcal{E}$,

$$\mu^R_x(\boldsymbol{\omega}_P) = \frac{\mu_0(\boldsymbol{\omega}_P)\,\mathbb{E}_{\mu_0}\left[\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) \,\middle|\, \boldsymbol{\omega}_P\right]}{\mu(x)}. \tag{D.2}$$

Preliminary. Let $i \in P \setminus \mathrm{pa}(0)$, and fix $\pi \in \mathcal{E}$ and $x \in X$. By (D.1), $\mu^R(\boldsymbol{\omega}_P, x)$ is the sum of $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})\,\mu_0(\boldsymbol{\omega})$ over Ω_{-P}, and $\mu_0(\boldsymbol{\omega}) > 0$ for every $\boldsymbol{\omega}$. Hence $\mu^R_x(\boldsymbol{\omega}_P) > 0$ if and only if $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) > 0$ for some value of the state attributes outside P. Since $i \notin \mathrm{pa}(0)$, this condition does not involve ω_i. Therefore, if $\mu^R_x(\boldsymbol{\omega}_P) > 0$, then $\mu^R_x(\boldsymbol{\omega}'_P) > 0$ for every $\boldsymbol{\omega}'_P$ that differs from $\boldsymbol{\omega}_P$ only in ω_i.

(i) I prove separately belief-irrelevance and the equivalence $\mathcal{B}_R = \mathcal{B}$.

Belief-irrelevance.

Sufficiency. Suppose $P \subseteq \mathrm{pa}(0)$. Since $\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)}) = \mathbb{E}_{\mu_0}[\pi(x|\boldsymbol{\omega})|\boldsymbol{\omega}_{\mathrm{pa}(0)}]$ and $P \subseteq \mathrm{pa}(0)$, by the law of iterated expectations $\mathbb{E}_{\mu_0}[\mu(x|\boldsymbol{\omega}_{\mathrm{pa}(0)})|\boldsymbol{\omega}_P] = \mathbb{E}_{\mu_0}[\pi(x|\boldsymbol{\omega})|\boldsymbol{\omega}_P] = \mu(x|\boldsymbol{\omega}_P)$. By (D.2)

$$\mu^R_x(\boldsymbol{\omega}_P) = \frac{\mu(x|\boldsymbol{\omega}_P)\mu_0(\boldsymbol{\omega}_P)}{\mu(x)} = \mu_x(\boldsymbol{\omega}_P). \tag{D.3}$$

Necessity. Suppose $P \not\subseteq \mathrm{pa}(0)$ and pick $i \in P \setminus \mathrm{pa}(0)$ and $\bar{\omega}_i \in \Omega_i$. Let $X = \{x', x''\}$ and

$\pi(x'|\boldsymbol{\omega}) = \mathbf{1}\{\omega_i = \bar{\omega}_i\}$, $\pi(x''|\boldsymbol{\omega}) = 1 - \pi(x'|\boldsymbol{\omega})$; then $\mu(x') = \mu_0(\omega_i = \bar{\omega}_i) \in (0,1)$, so $\pi \in \mathcal{E}$. The objective posterior is $\mu_{x'}(\boldsymbol{\omega}_P) = \mathbf{1}\{\omega_i = \bar{\omega}_i\}\mu_0(\boldsymbol{\omega}_P)/\mu(x')$, so $\operatorname{supp}\mu_{x'} = \{\boldsymbol{\omega}_P : \omega_i = \bar{\omega}_i\}$. Take $\boldsymbol{\omega}_P$ with $\omega_i = \bar{\omega}_i$ and $\boldsymbol{\omega}'_P$ that differs from it only in ω_i, which exists since $|\Omega_i| \geq 2$. Then $\mu_{x'}(\boldsymbol{\omega}_P) > 0 = \mu_{x'}(\boldsymbol{\omega}'_P)$, whereas by the preliminary step $\mu^R_{x'}$ is positive at both or at neither. Hence $\mu^R_{x'} \neq \mu_{x'}$, and R is not belief-irrelevant.

Equivalence with $\mathcal{B}_R = \mathcal{B}$.

Sufficiency. Let $P \subseteq \mathrm{pa}(0)$. By Lemma 4, $\mathcal{B}_R \subseteq \mathcal{B}$. For the reverse inclusion, take $\tau = (\alpha_x, \mu_x)_{x\in X} \in \mathcal{B}$, so that $\alpha \in \operatorname{int}(\Delta(X))$ and $\mu_0(\boldsymbol{\omega}_P) = \sum_x \alpha_x \mu_x(\boldsymbol{\omega}_P)$, and define $\pi^\alpha(x|\boldsymbol{\omega}) = \mu_x(\boldsymbol{\omega}_P)\alpha_x/\mu_0(\boldsymbol{\omega}_P)$, which lies in $[0,1]$ and sums to one over x; its objective marginal is $\mu(x) = \sum_{\Omega_P} \pi^\alpha(x|\boldsymbol{\omega}_P)\mu_0(\boldsymbol{\omega}_P) = \alpha_x \in (0,1)$, so $\pi^\alpha \in \mathcal{E}$. Since π^α depends on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_P$, and $P \subseteq \mathrm{pa}(0)$, Lemma B.3 gives $\pi^\alpha \in \mathcal{E}_R$. By (D.3), $\mu^R_x = \mu_x$ and $\mu(x) = \alpha_x$, so $\tau \in \mathcal{B}_R$.

Necessity. Suppose $P \not\subseteq \mathrm{pa}(0)$, and let π be the experiment of the previous step, with $i \in P \setminus \mathrm{pa}(0)$ and $\bar{\omega}_i \in \Omega_i$. Its objective weighted vector $\tau = (\mu(x), \mu_x)_{x\in\{x',x''\}}$ is Bayes-consistent, hence in $\mathcal{B}$. If τ were in $\mathcal{B}_R$, some $\pi' \in \mathcal{E}_R \subseteq \mathcal{E}$ with realization space $\{x', x''\}$ would induce the subjective posterior $\mu'^R_{x'} = \mu_{x'}$. By the preliminary step applied to π', $\mu'^R_{x'}$ is positive at both $\boldsymbol{\omega}_P$ and $\boldsymbol{\omega}'_P$ of the previous step or at neither, whereas $\mu_{x'}$ is positive at the first and zero at the second. Hence $\tau \notin \mathcal{B}_R$ and $\mathcal{B}_R \subsetneq \mathcal{B}$.

(ii) By Lemma B.3, $\mathcal{E}_R = \{\pi \in \mathcal{E} : \pi(\cdot|\boldsymbol{\omega}) = \pi(\cdot|\boldsymbol{\omega}_{\mathrm{pa}(0)})\}$.

Sufficiency. If $\mathrm{pa}(0) \subseteq P$, every such π depends on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_P$, so R is design-irrelevant.

Necessity. If some $j \in \mathrm{pa}(0) \setminus P$, pick $\bar{\omega}_j \in \Omega_j$ and let $\pi(x'|\boldsymbol{\omega}) = \mathbf{1}\{\omega_j = \bar{\omega}_j\}$, $\pi(x''|\boldsymbol{\omega}) = 1 - \pi(x'|\boldsymbol{\omega})$ on $X = \{x', x''\}$. Then $\pi \in \mathcal{E}_R$, since it is measurable in $\boldsymbol{\omega}_{\mathrm{pa}(0)}$ and $\mu(x') = \mu_0(\omega_j = \bar{\omega}_j) \in (0,1)$, but $\pi(\cdot|\boldsymbol{\omega})$ is not a function of $\boldsymbol{\omega}_P$, because changing ω_j while holding $\boldsymbol{\omega}_P$ fixed changes $\pi(x'|\boldsymbol{\omega})$. Hence R is not design-irrelevant.

Since $\mathrm{MB}_R(0) = \mathrm{pa}(0)$, (i) and (ii) together give: R is irrelevant if and only if $\mathrm{MB}_R(0) = P$. ■

Proof of Lemma D.2

Let $P_0 = P \cup \{0\}$. Since x is a root, $\mathrm{MB}_R(0) = \mathrm{ch}(0) \cup \bigcup_{i\in\mathrm{ch}(0)} \mathrm{pa}_\omega(i)$.

Sufficiency. Let **(a)** and **(b)** hold. Since P_0 is a clique, x is linked to every $i \in P$, so $P \subseteq \mathrm{ch}(0)$. By **(a)**, $\mathrm{ch}(0) \subseteq \mathrm{MB}_R(0) = P$. Hence $\mathrm{ch}(0) = P$, and, again by **(a)**, $\mathrm{pa}_\omega(i) \subseteq P = \mathrm{ch}(0)$ for every child i. Since P is a clique, the children of x are pairwise linked. By Lemma B.6, there is a DAG R' Markov equivalent to R in which x is terminal. In the construction in the proof of that lemma only the edges $x \to \omega_i$, $i \in \mathrm{ch}(0)$, are reversed, so $\mathrm{pa}(0)$ in R' equals $\mathrm{ch}(0) = P$.

Necessity. Let R' be Markov equivalent to R, with x terminal and $\mathrm{pa}(0) = P$ in R'. The two DAGs have the same skeleton, so the neighbors of x in R are the attributes in P, that is, $\mathrm{ch}(0) = P$. Since such an R' exists, Lemma B.6 gives $\mathrm{pa}_\omega(i) \subseteq \mathrm{ch}(0)$ for every child i, and the children of x are pairwise linked. The first property gives $\mathrm{MB}_R(0) = \mathrm{ch}(0) = P$, which is **(a)**. The second, together with the fact that x is linked to each of its children, makes $P_0 = \mathrm{ch}(0) \cup \{0\}$ a clique, which is **(b)**. ■

Proof of Lemma D.3

Since $\mathrm{pa}(0) = \emptyset$, recall that by (3), for every $\pi \in \mathcal{E}$,

$$\mu^R(\boldsymbol{\omega}, x) = \mu(x) \prod_{i \in \mathrm{ch}(0)} \mu(\omega_i | x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) \prod_{j \in N_\omega \setminus \mathrm{ch}(0)} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}),$$

with the prior completion $\mu(\omega_i | x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = \mu_0(\omega_i | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$ whenever $\mu(x, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = 0$, and $\mu^R(x) = \mu(x)$ by Lemma 1.

(i) I prove the statement by contraposition. Let $k \in P$ be such that $k \notin \mathrm{ch}(0)$. Fix a realization space $X = \{x', x''\}$ and a value $\omega^* \in \Omega_k$. Consider a Blackwell experiment that assigns $\pi(x' | \boldsymbol{\omega}) = 1$ if $\omega_k = \omega^*$, and 0 otherwise. Then $\mu(x') = \mu_0(\omega_k = \omega^*) \in (0, 1)$, thus $\pi \in \mathcal{E}$. Since $\mu_{x'}(\boldsymbol{\omega}) = 0$ whenever $\omega_k \neq \omega^*$, $\mu_{x'}(\boldsymbol{\omega}_P) = 0$ at every $\boldsymbol{\omega}_P$ with $\omega_k \neq \omega^*$.

I claim that $\mu^R(\boldsymbol{\omega}, x') \in \mathrm{int}\,(\Delta(\Omega \times X))$, or equivalently, that each factor of $\mu^R(\boldsymbol{\omega}, x')$ is strictly positive. First, $\mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) > 0$ for $j \in N_\omega \setminus \mathrm{ch}(0)$ by full support. Fix now $i \in \mathrm{ch}(0)$, with $i \neq k$. Then there are two cases.

- If $k \notin \mathrm{pa}_\omega(i)$, then $\mu(\omega_i | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}, x') = \mu_0(\omega_i | \omega_k = \omega^*, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) > 0$, given that $\mu(\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}, x') = \mu_0(\omega_k = \omega^*, \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) > 0$.
- If $k \in \mathrm{pa}_\omega(i)$, then $\mu(x', \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) = \mu_0(\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}) > 0$ when $\omega_k = \omega^*$ in $\boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}$, in which case $\mu(\omega_i | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)}, x') = \mu_0(\omega_i | \boldsymbol{\omega}_{\mathrm{pa}_\omega(i)})$, and it is zero otherwise, in which case the factor is the prior conditional by completion; either way it is positive.

Hence $\mu^R(\boldsymbol{\omega}, x') > 0$ for every $\boldsymbol{\omega} \in \Omega$, and $\mu^R_{x'}(\boldsymbol{\omega}_P) = \sum_{\Omega_{-P}} \mu^R(\boldsymbol{\omega}, x') / \mu(x') > 0$ for every $\boldsymbol{\omega}_P$. At any $\boldsymbol{\omega}_P$ with $\omega_k \neq \omega^*$, $\mu^R_{x'}(\boldsymbol{\omega}_P) > 0 = \mu_{x'}(\boldsymbol{\omega}_P)$, so R is not belief-irrelevant.

(ii) Since x has the single child i^*, no path between two state attributes passes through x, so the independencies that R entails among the state attributes are those of the DAG R_ω obtained by deleting x, in which the parents of i^* are C; by Assumption 1, μ_0 is Markov with respect to R_ω and factorizes as

$$\mu_0(\boldsymbol{\omega}) = \mu_0(\omega_{i^*} | \boldsymbol{\omega}_C) \prod_{j \neq i^*} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}). \tag{D.4}$$

Sufficiency. Let $C = \emptyset$ and $P = \{i^*\}$, fix $\pi \in \mathcal{E}$ and $x \in X$. In this case, (3) becomes

$$\mu^R(\boldsymbol{\omega}, x) = \mu(x)\, \mu(\omega_{i^*} | x) \prod_{j \neq i^*} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}),$$

since $\mu(x) > 0$. When $C = \emptyset$, (D.4) gives $\prod_{j \neq i^*} \mu_0(\omega_j | \boldsymbol{\omega}_{\mathrm{pa}_\omega(j)}) = \mu_0(\boldsymbol{\omega}_{N_\omega \setminus \{i^*\}} | \omega_{i^*})$, which sums to one over $\boldsymbol{\omega}_{N_\omega \setminus \{i^*\}}$. Hence $\mu^R(\omega_{i^*}, x) = \mu(x)\mu(\omega_{i^*} | x) = \mu(\omega_{i^*}, x)$, and dividing by $\mu^R(x) = \mu(x)$ gives $\mu^R_x(\omega_{i^*}) = \mu_x(\omega_{i^*})$. Thus R is belief-irrelevant.

Necessity. I prove this statement by contradiction. Let R be belief-irrelevant. By **(i)**, $P \subseteq \{i^*\}$, and P is non-empty, so $P = \{i^*\}$. Suppose that $C \neq \emptyset$. Fix $c_0 \in \Omega_C$, $\bar{\omega}_{i^*} \in \Omega_{i^*}$, and let $X = \{x', x''\}$. Consider a Blackwell experiment such that $\pi(x' | \boldsymbol{\omega}) = 1$ if $\omega_{i^*} = \bar{\omega}_{i^*}$, $\boldsymbol{\omega}_C = c_0$, and 0 otherwise. Then $\mu(x') = \mu_0(\bar{\omega}_{i^*}, c_0) \in (0, 1)$, so $\pi \in \mathcal{E}$. The objective posterior $\mu_{x'}(\omega_{i^*}) = 0$ if $\omega_{i^*} \neq \bar{\omega}_{i^*}$.

Turning to the subjective posterior, $\mu(c_0, x') = \mu_0(\bar{\omega}_{i^*}, c_0) > 0$ and $\mu(\omega_{i^*}, x', c_0) = \mu_0(\bar{\omega}_{i^*}, c_0)$ if $\omega_{i^*} = \bar{\omega}_{i^*}$, and 0 otherwise. Therefore, $\mu(\cdot | c_0, x')$ is the point mass at $\bar{\omega}_{i^*}$. For every $c \neq c_0$,

$\mu(x', c) = 0$ and $\mu(\cdot|c, x') = \mu_0(\cdot|c)$ by completion. By (4),

$$\mu^R_{x'}(\omega_{i^*}) = \sum_{c \in \Omega_C} \mu_0(c)\, \mu(\omega_{i^*}|c, x') = \mu_0(c_0)\, \mathbf{1}\{\omega_{i^*} = \bar{\omega}_{i^*}\} + \sum_{c \neq c_0} \mu_0(c)\, \mu_0(\omega_{i^*}|c).$$

Since $C \neq \emptyset$ and $|\Omega_j| \geq 2$ for $j \in C$, the last sum has at least one term, and every term is strictly positive by full support. Hence $\mu^R_{x'}(\omega_{i^*}) > 0$ at every ω_{i^*}, in particular at some $\omega_{i^*} \neq \bar{\omega}_{i^*}$, where $\mu_{x'}(\omega_{i^*}) = 0$. This contradicts belief-irrelevance, so $C = \emptyset$.

Finally, with a single child, **(b)** requires every payoff-relevant state attribute to be linked to x, so $P = \{i^*\}$, and **(a)** then reads $\{i^*\} \cup C = \{i^*\}$. Hence **(a)** and **(b)** hold if and only if $C = \emptyset$ and $P = \{i^*\}$. If R is irrelevant, it is belief-irrelevant: by **(ii)**, $C = \emptyset$ and $P = \{i^*\}$. ■

Proof of Proposition 5

Sufficiency. Let R' be Markov equivalent to R, with x terminal and $\mathrm{pa}(0) = P$ in R'. By Lemma B.5, $R' \in \mathcal{R}_0$, $\mu^{R'} = \mu^R$ for every $\pi \in \mathcal{E}$, and $\mathcal{E}_{R'} = \mathcal{E}_R$. Both halves of Definition 5 are stated in terms of the subjective posteriors and of the consistent experiments only, so R is irrelevant if and only if R' is. In R' the Markov boundary of x is $\mathrm{pa}(0) = P$, so R' is irrelevant by Lemma D.1.

Converse. Let R be irrelevant. If x is terminal, $\mathrm{pa}(0) = \mathrm{MB}_R(0) = P$ by Lemma D.1, and R itself is the required DAG. If x has a single child, conditions **(a)** and **(b)** hold by Lemma D.3, and Lemma D.2 provides the required DAG.

■

Proof of Corollary 2

By hypothesis, there is a DAG R' Markov equivalent to R in which x is terminal and $\mathrm{pa}(0) = P$. By Lemma B.5, $\mathcal{E}_R = \mathcal{E}_{R'}$ and $\mu^R_x = \mu^{R'}_x$ for every $\pi \in \mathcal{E}$ and $x \in X$, so $\mathcal{B}_R = \mathcal{B}_{R'}$ by the definition of R-Bayes consistency. By Lemma B.3 applied to R', $\mathcal{E}_{R'}$ is the set of experiments that depend on $\boldsymbol{\omega}$ only through $\boldsymbol{\omega}_{\mathrm{pa}(0)} = \boldsymbol{\omega}_P$, which is the first claim. By Lemma D.1(i) applied to R', in which $P \subseteq \mathrm{MB}_R(0) = P$, $\mathcal{B}_{R'} = \mathcal{B}$, which is the second; and its proof implements every $\tau \in \mathcal{B}$ by the experiment $\pi^\alpha(x|\boldsymbol{\omega}) = \mu_x(\boldsymbol{\omega}_P)\alpha_x / \mu_0(\boldsymbol{\omega}_P)$, which is defined on Ω_P and belongs to $\mathcal{E}_{R'} = \mathcal{E}_R$. ■

E. Appendix: Welfare

E.1. More than two actions

Let $\Omega_P = \Omega_1$ be binary and, as in the proof of Proposition 4, identify a belief $q \in \Delta(\Omega_1)$ with $q(\omega_1^+)$. The receiver has $|A|$ actions. Let $\mu_0(\omega_1^+) < \theta_1 < \cdots < \theta_{|A|-1}$ be the kinks of $\hat{u}$, with actions labeled so that at θ_k the receiver is indifferent between a_k and a_{k+1}, and a_1 is the default.

Lemma E.1 : More than two actions. — *Fix $\mu_0 \in \operatorname{int}(\Delta(\Omega))$ and $R \in \mathcal{R}_0$, and let $\pi \in \mathcal{E}$ satisfy $\mu_x^R(\boldsymbol{\omega}_P) = \mu_x(\boldsymbol{\omega}_P)$ for every $x \in X$. Then $U(\pi) > \hat{u}(\mu_0)$ if and only if $\mu_x(\omega_1^+) > \theta_1$ at some $x \in X$. If, moreover, every posterior above θ_1 is one of the kinks, then*

$$U(\pi) - \hat{u}(\mu_0) = \sum_{k=2}^{|A|-1} \left[\hat{u}(\theta_k) - \bar{u}(a_1, \theta_k)\right] \sum_{x:\, \mu_x(\omega_1^+)=\theta_k} \mu(x), \tag{E.1}$$

and each coefficient $\hat{u}(\theta_k) - \bar{u}(a_1, \theta_k)$ is strictly positive and does not depend on R.

With two actions the sum in (E.1) is empty, and the lemma gives Proposition 6**(ii)** when Ω_P is binary.

E.2. Proofs

Proof of Proposition 6

(i) The receiver's expected payoff from an action depends on her belief only through its marginal on Ω_P, so μ_x^R and μ_x give the same expected payoff to every action and the same set of optimal actions; since the tie-breaking rule selects from that set, $\hat{a}(\mu_x^R) = \hat{a}(\mu_x)$ for every x. Hence $U(\pi) = \sum_x \mu(x)\, \hat{u}(\mu_x) = U^{\mathrm{BP}}(\pi)$. Since the objective posteriors average to the prior and $\hat{u}$ is convex, Jensen's inequality gives $U^{\mathrm{BP}}(\pi) \geq \hat{u}(\sum_x \mu(x)\mu_x) = \hat{u}(\mu_0)$.

(ii) At a realization at which the receiver takes a_2, obedience binding means she is indifferent at μ_x, so $\hat{u}(\mu_x) = \bar{u}(a_1, \mu_x)$. At every other realization she takes a_1, which is optimal at $\mu_x^R = \mu_x$, so again $\hat{u}(\mu_x) = \bar{u}(a_1, \mu_x)$. By (i), $U(\pi) = \sum_x \mu(x)\, \bar{u}(a_1, \mu_x) = \bar{u}(a_1, \mu_0)$, because the map $q \mapsto \bar{u}(a_1, q)$ is affine and the objective posteriors average to the prior. Since a_1 is the default, the last expression is $\hat{u}(\mu_0)$. These last two steps do not use the hypothesis on π: whenever a_1 is the default, $\sum_x \mu(x)\, \bar{u}(a_1, \mu_x) = \hat{u}(\mu_0)$ at every $\pi \in \mathcal{E}$. The proofs below use this identity.

■

Proof of Lemma E.1

By Proposition 6**(i)**, $U(\pi) = \sum_x \mu(x)\, \hat{u}(\mu_x)$, and by Proposition 6**(ii)**, $\hat{u}(\mu_0) = \sum_x \mu(x)\, \bar{u}(a_1, \mu_x)$, so $U(\pi) - \hat{u}(\mu_0) = \sum_x \mu(x)\, [\hat{u}(\mu_x) - \bar{u}(a_1, \mu_x)]$. The mapping $q \mapsto \hat{u}(q) - \bar{u}(a_1, q)$ is convex and non-negative, and it is zero on $[0, \theta_1]$, where a_1 is optimal. Hence the bracket is zero at a posterior with $\mu_x(\omega_1^+) \leq \theta_1$ and strictly positive at every other posterior, which gives the first claim. If every posterior above θ_1 is one of $\theta_2, \ldots, \theta_{|A|-1}$, the sum is the right-hand side of (E.1). Its coefficients are strictly positive by the same argument, and they depend only on the receiver's decision problem. ■

Proof of Proposition 7

Let $X^+ = \{x \in X : \mu_x^R(\omega_1^+) \geq \theta\}$ be the set of realizations at which the receiver takes a_2, so that

$\mathbb{P}(\pi) = \sum_{x \in X^+} \mu(x)$, and $G(q) = K(q(\omega_1^+) - \theta)$.

I first derive (8). By the identity that closes the proof of Proposition 6(ii), $\hat{u}(\mu_0) = \sum_x \mu(x)\, \bar{u}(a_1, \mu_x)$. At each realization the receiver's payoff is the payoff from a_1 plus, if $x \in X^+$, the gain $G(\mu_x)$, so

$$U(\pi) = \sum_x \mu(x)\, \bar{u}(a_1, \mu_x) + \sum_{x \in X^+} \mu(x)\, K\big(\mu_x(\omega_1^+) - \theta\big),$$

and subtracting $\hat{u}(\mu_0)$ gives (8).

(i) Since $\sum_x \mu(x)\, \mu_x(\omega_1^+) = \mu_0(\omega_1^+)$ at every experiment,

$$\begin{aligned} U(\pi) - \hat{u}(\mu_0) &= K\Big[\mu_0(\omega_1^+) - \sum_{x \notin X^+} \mu(x)\, \mu_x(\omega_1^+)\Big] - K\theta\, \mathbb{P}(\pi) \\ &= K\theta\Big[\frac{\mu_0(\omega_1^+)}{\theta} - \mathbb{P}(\pi)\Big] - K \sum_{x \notin X^+} \mu(x)\, \mu_x(\omega_1^+), \end{aligned}$$

which is (9), since $\mathbb{P}^{\mathrm{BP}} = \mu_0(\omega_1^+)/\theta$ (Section 6.4) and $x \notin X^+$ if and only if $\mu_x^R(\omega_1^+) < \theta$. The last sum is non-negative, so $\hat{u}(\mu_0) - U(\pi) \geq K\theta\, [\mathbb{P}(\pi) - \mathbb{P}^{\mathrm{BP}}]$, and the right-hand side is strictly positive if $\mathbb{P}(\pi) > \mathbb{P}^{\mathrm{BP}}$.

(ii) By the martingale property of the subjective posteriors (Section 4.1) and Lemma 1, $\sum_x \mu(x)\, \mu_x^R = \mu^R(\boldsymbol{\omega}_P)$. Every persuading realization satisfies $\mu_x^R(\omega_1^+) \geq \theta$, so $\mu^R(\omega_1^+) \geq \theta \mathbb{P}(\pi) > \mu_0(\omega_1^+)$.

Next, $\hat{u}(q) \geq \bar{u}(a_1, q)$ at every belief, with equality if and only if $q(\omega_1^+) \leq \theta$. Since the expected payoff from a_1 is affine,

$$U^{\mathrm{R}}(\pi) \;\geq\; \sum_x \mu(x)\, \bar{u}(a_1, \mu_x^R),$$

with equality if and only if $\mu_x^R(\omega_1^+) \leq \theta$ at every x, that is, if and only if every persuading subjective posterior equals θ. Since a_1 is the default, $\hat{u}(\mu_0) = \bar{u}(a_1, \mu_0)$, and $\bar{u}(a_1, q) = u(a_1, \omega_1^-) + q(\omega_1^+)[u(a_1, \omega_1^+) - u(a_1, \omega_1^-)]$ gives (10).

Finally, if $u(a_1, \omega_1^+) \geq u(a_1, \omega_1^-)$, the right-hand side of (10) is non-negative by the first step, so $U^{\mathrm{R}}(\pi) \geq \hat{u}(\mu_0)$, and $\hat{u}(\mu_0) > U(\pi)$ by **(i)**.

■